Extended methods, results and tables

Necessity versus sufficiency in last-layer class activation maps: a structural check of the classification head

This document belongs to the code-and-data archive of the article. Equation, figure, table and section numbers without an S refer to the main text. Tables S1–S13 report every architecture, contrast and cell behind the counts of the main text. Note S1 gives the details needed to reproduce the study that the main text does not state, with the design of the study (Fig. S1) and of every experiment (Table S14). Note S2 lists, as numbered paragraphs, the results that the main text cites and that no table carries, with the full-width figures of every analysis (Figs. S2–S6) and Tables S15–S19 (normalisation edit, versions of the check, channel-wise pooling, heads trained end to end, retrained ISIC classifiers). The README of the code-and-data archive maps each table and figure to the script and output that produce it.

Table S1. Structural verdict and measured label for each of the 148 architectures of the development and held-out sets. The verdict is the structural check's (Table 1 of the main text), read from the modules the classification head executes and never from the measurement it is scored against: degenerate, the readout is degenerate as distributed; norm.-blocked, normalisation-blocked; nonlin.-blocked, nonlinearity-blocked; class-token, a readout that reads a single class token rather than pooling spatially. The measured label (additive, vacuous, norm.-only, hard-blocked) is built from direct double-precision evaluation, as set out in Section 2.4 of the main text and Notes S1.2 and S1.6. Development architectures were used to write the check; held-out architectures were not used in its development. crossvit_9_240 could not be measured; volo_d1_224 is the one disagreement (Note S2.6).

Architecture	Set	Verdict	Blocking operation	Measured label	Agree
beitv2_base_patch16_224	development	norm.-blocked	fc_norm(LayerNorm)	norm.-only	yes
caformer_s18	development	nonlin.-blocked	head.fc.act.relu(ReLU)	hard-blocked	yes
cait_xxs24_224	development	class-token		vacuous	yes
coatnet_0_rw_224	development	degenerate		additive	yes
convformer_s18	development	nonlin.-blocked	head.fc.act.relu(ReLU)	hard-blocked	yes
convnext_base	development	norm.-blocked	head.norm(LayerNorm2d)	norm.-only	yes
convnext_tiny	development	norm.-blocked	head.norm(LayerNorm2d)	norm.-only	yes
convnextv2_tiny	development	norm.-blocked	head.norm(LayerNorm2d)	norm.-only	yes
crossvit_9_240	development	class-token		not measured	
davit_tiny	development	norm.-blocked	head.norm(LayerNorm2d)	norm.-only	yes
deit3_small_patch16_224	development	class-token		vacuous	yes
densenet121	development	degenerate		additive	yes
dpn68	development	degenerate		additive	yes
edgenext_small	development	norm.-blocked	head.norm(LayerNorm2d)	norm.-only	yes
efficientformer_l1	development	degenerate		additive	yes
efficientformerv2_s0	development	degenerate		additive	yes
efficientnet_b0	development	degenerate		additive	yes
eva02_tiny_patch14_336	development	norm.-blocked	fc_norm(LayerNorm)	norm.-only	yes
fastvit_t8	development	degenerate		additive	yes
focalnet_tiny_srf	development	degenerate		additive	yes
gcvit_tiny	development	degenerate		additive	yes
ghostnet_100	development	nonlin.-blocked	act2(ReLU)	hard-blocked	yes
hardcorenas_a	development	nonlin.-blocked	act2(Hardswish)	hard-blocked	yes
hgnetv2_b0	development	nonlin.-blocked	head.last_conv.1(ReLU)	hard-blocked	yes
hiera_tiny_224	development	norm.-blocked	head.norm(LayerNorm)	norm.-only	yes
inception_v3	development	degenerate		additive	yes
inception_v4	development	degenerate		additive	yes
lcnet_050	development	nonlin.-blocked	act2(Hardswish)	hard-blocked	yes
levit_128	development	degenerate		additive	yes
mambaout_tiny	development	nonlin.-blocked	head.pre_logits.act(GELU)	hard-blocked	yes
maxvit_tiny_tf_224	development	nonlin.-blocked	head.pre_logits.act(Tanh)	hard-blocked	yes
mnasnet_100	development	degenerate		additive	yes
mobilenetv3_large_100	development	nonlin.-blocked	act2(Hardswish)	hard-blocked	yes
mobilenetv3_small_100	development	nonlin.-blocked	act2(Hardswish)	hard-blocked	yes
mobilenetv4_conv_small	development	nonlin.-blocked	norm_head.act(ReLU)	hard-blocked	yes
mobilevit_s	development	degenerate		additive	yes
mobilevitv2_050	development	degenerate		additive	yes
naflexvit_base_patch16_gap	development	norm.-blocked	fc_norm(LayerNorm)	norm.-only	yes
nest_tiny_jx	development	degenerate		additive	yes
nextvit_small	development	degenerate		additive	yes
pit_s_224	development	class-token		vacuous	yes
poolformer_s12	development	norm.-blocked	head.norm(LayerNorm2d)	norm.-only	yes
poolformerv2_s12	development	norm.-blocked	head.norm(LayerNorm2d)	norm.-only	yes
pvt_v2_b0	development	degenerate		additive	yes
rdnet_tiny	development	norm.-blocked	head.norm(LayerNorm2d)	norm.-only	yes
regnetx_032	development	degenerate		additive	yes
regnety_032	development	degenerate		additive	yes
repghostnet_100	development	nonlin.-blocked	act2(ReLU)	hard-blocked	yes
repvgg_a2	development	degenerate		additive	yes
res2net50_26w_4s	development	degenerate		additive	yes
resnet18	development	degenerate		additive	yes
resnet50	development	degenerate		additive	yes

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Table S1 (continued)

Architecture	Set	Verdict	Blocking operation	Measured label	Agree
resnext50_32x4d	development	degenerate		additive	yes
swin_tiny_patch4_window7_224	development	degenerate		additive	yes
swinv2_tiny_window8_256	development	degenerate		additive	yes
tf_efficientnetv2_s	development	degenerate		additive	yes
tiny_vit_5m_224	development	norm.-blocked	head.norm(LayerNorm2d)	norm.-only	yes
tinynet_a	development	degenerate		additive	yes
twins_svt_small	development	degenerate		additive	yes
vgg16	development	nonlin.-blocked	pre_logits.act1(ReLU)	hard-blocked	yes
vit_base_patch16_224	development	class-token		vacuous	yes
vit_base_patch16_rope_reg1_gap_256	development	norm.-blocked	fc_norm(LayerNorm)	norm.-only	yes
vit_betwixt_patch16_reg4_gap_256	development	norm.-blocked	fc_norm(LayerNorm)	norm.-only	yes
vitamin_small_224	development	norm.-blocked	fc_norm(LayerNorm)	norm.-only	yes
wide_resnet50_2	development	degenerate		additive	yes
xception41	development	degenerate		additive	yes
beit_base_patch16_224	held out	norm.-blocked	fc_norm(LayerNorm)	norm.-only	yes
coat_lite_tiny	held out	class-token		vacuous	yes
coatnet_1_rw_224	held out	degenerate		additive	yes
coatnext_nano_rw_224	held out	degenerate		additive	yes
convformer_s36	held out	nonlin.-blocked	head.fc.act.relu(ReLU)	hard-blocked	yes
convit_tiny	held out	class-token		vacuous	yes
convmixer_768_32	held out	degenerate		additive	yes
cspresnet50	held out	degenerate		additive	yes
davit_small	held out	norm.-blocked	head.norm(LayerNorm2d)	norm.-only	yes
densenet169	held out	degenerate		additive	yes
densenetblur121d	held out	degenerate		additive	yes
dla34	held out	degenerate		additive	yes
dm_nfnet_f0	held out	degenerate		additive	yes
eca_nfnet_l0	held out	degenerate		additive	yes
ecaresnet50d	held out	degenerate		additive	yes
efficientformer_l3	held out	degenerate		additive	yes
efficientformerv2_s1	held out	degenerate		additive	yes
efficientnet_b3	held out	degenerate		additive	yes
efficientnet_lite0	held out	degenerate		additive	yes
efficientnetv2_rw_t	held out	degenerate		additive	yes
efficientvit_b1	held out	nonlin.-blocked	head.classifier.2(Hardswish)	hard-blocked	yes
efficientvit_m1	held out	degenerate		additive	yes
ese_vovnet39b	held out	degenerate		additive	yes
fasternet_t0	held out	nonlin.-blocked	act(GELU)	hard-blocked	yes
fastvit_s12	held out	degenerate		additive	yes
fastvit_sa12	held out	degenerate		additive	yes
flexivit_small	held out	class-token		vacuous	yes
ghostnetv2_100	held out	nonlin.-blocked	act2(ReLU)	hard-blocked	yes
gmixer_24_224	held out	degenerate		additive	yes
gmlp_s16_224	held out	degenerate		additive	yes
hgnet_tiny	held out	nonlin.-blocked	head.last_conv.1(ReLU)	hard-blocked	yes
hiera_base_224	held out	norm.-blocked	head.norm(LayerNorm)	norm.-only	yes
hiera_small_224	held out	norm.-blocked	head.norm(LayerNorm)	norm.-only	yes
hrnet_w18_small	held out	degenerate		additive	yes
inception_next_tiny	held out	nonlin.-blocked	head.act(GELU)	hard-blocked	yes
legacy_seresnet18	held out	degenerate		additive	yes
legacy_xception	held out	degenerate		additive	yes
levit_192	held out	degenerate		additive	yes
levit_conv_128s	held out	degenerate		additive	yes
mambaout_femto	held out	nonlin.-blocked	head.pre_logits.act(GELU)	hard-blocked	yes
mambaout_kobe	held out	nonlin.-blocked	head.pre_logits.act(GELU)	hard-blocked	yes
mambaout_small	held out	nonlin.-blocked	head.pre_logits.act(GELU)	hard-blocked	yes
mixer_b16_224	held out	degenerate		additive	yes
mixnet_s	held out	degenerate		additive	yes
mobilevit_xs	held out	degenerate		additive	yes
mobilevitv2_100	held out	degenerate		additive	yes
mvitv2_tiny	held out	degenerate		additive	yes
poolformer_s24	held out	norm.-blocked	head.norm(LayerNorm2d)	norm.-only	yes
poolformerv2_s24	held out	norm.-blocked	head.norm(LayerNorm2d)	norm.-only	yes
pvt_v2_b1	held out	degenerate		additive	yes
pvt_v2_b2	held out	degenerate		additive	yes
regnetx_016	held out	degenerate		additive	yes
regnety_040	held out	degenerate		additive	yes
regnetz_b16	held out	degenerate		additive	yes
regnetz_c16	held out	degenerate		additive	yes
repvit_m1_0	held out	degenerate		additive	yes
resmlp_12_224	held out	degenerate		additive	yes
resnest14d	held out	degenerate		additive	yes

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Table S1 (continued)

Architecture	Set	Verdict	Blocking operation	Measured label	Agree
resnest50d	held out	degenerate		additive	yes
resnetv2_50	held out	degenerate		additive	yes
rexnet_100	held out	degenerate		additive	yes
selecsls42b	held out	degenerate		additive	yes
sequencer2d_s	held out	degenerate		additive	yes
seresnet50	held out	degenerate		additive	yes
seresnext26d_32x4d	held out	degenerate		additive	yes
shvit_s1	held out	degenerate		additive	yes
skresnet18	held out	degenerate		additive	yes
starnet_s1	held out	degenerate		additive	yes
swiftformer_s	held out	degenerate		additive	yes
tf_efficientnetv2_b0	held out	degenerate		additive	yes
tiny_vit_11m_224	held out	norm.-blocked	head.norm(LayerNorm2d)	norm.-only	yes
tiny_vit_21m_224	held out	norm.-blocked	head.norm(LayerNorm2d)	norm.-only	yes
tnt_s_patch16_224	held out	class-token		vacuous	yes
tresnet_m	held out	degenerate		additive	yes
twins_pcpvt_small	held out	degenerate		additive	yes
twins_svt_base	held out	degenerate		additive	yes
vgg19	held out	nonlin.-blocked	pre_logits.act1(ReLU)	hard-blocked	yes
visformer_small	held out	degenerate		additive	yes
vit_small_patch16_224	held out	class-token		vacuous	yes
volo_d1_224	held out	class-token		hard-blocked	no
xception65	held out	degenerate		additive	yes
xcit_tiny_12_p16_224	held out	class-token		vacuous	yes

Table S2. The 192 primary localisation contrasts, each with its effect size, followed by the 96 contrasts of the same separated maps against plain CAM. Of the 96 primary contrasts against the collapsed map, 72 are in the written protocol, and likewise 72 of the 96 against the centre prior; the other 24 of each (best IoU and both pointing games on CUB) were added to the protocol's metrics (Table S6, deviation (i)). Datasets: cub, CUB-200-2011 (test split, 5,794 images); voc12, PASCAL VOC 2012 with boxes (train/val, 8,856 image–class records); voc12seg, PASCAL VOC 2012 with masks (train/segmentation-val, 2,147 records); coco17, MS-COCO val2017 (7,339 records). Backbones: resnet50, ResNet-50; vgg16gap, VGG-16 with a GAP head; vitb16, ViT-B/16. Arms: *replaced aggregator* is the aggregator arm of Note S2.8 and *relocated normalisation* the relocation arm. diff is the separated map's score minus its comparator's in the benchmark's own metric, with a 95% bootstrap interval. P is Holm-corrected within each (dataset, backbone, metric) family, whose size is four for a convolutional backbone and eight for the transformer, and the verdict applies $\alpha = 0.05$ to that corrected P. A Wilcoxon or McNemar verdict can differ from whether the bootstrap interval of the mean difference spans zero. The last block (comparator added after the primary outcome; Table S6) replaces the collapsed map by plain CAM, the collapsed map without subtraction of the reference's per-cell logit; its Holm family is the same cell with plain CAM in place of the collapsed map. Against plain CAM the separated maps win 18, lose 33 and tie 45 of the 96 contrasts (Fig. S6(a)); of the 39 wins over the collapsed map, 16 are also wins over plain CAM, 19 are ties and 4 are losses.

Dataset	Backbone	Metric	Arm	Contrast	n	Test	Separated map	Comparator	diff [95% CI]	P_{Holm}	Verdict
coco17	resnet50	peak-mask IoU	replaced aggregator	necessity vs centre prior	7,339	Wilcoxon	0.1956	0.1736	+0.0220 [+0.0181, +0.0259]	3.8e-07	win
coco17	resnet50	peak-mask IoU	replaced aggregator	necessity vs collapsed	7,339	Wilcoxon	0.1956	0.2937	−0.0980 [−0.1013, −0.0950]	<1e-300	loss
coco17	resnet50	peak-mask IoU	replaced aggregator	sufficiency vs centre prior	7,339	Wilcoxon	0.2903	0.1736	+0.1167 [+0.1126, +0.1210]	<1e-300	win
coco17	resnet50	peak-mask IoU	replaced aggregator	sufficiency vs collapsed	7,339	Wilcoxon	0.2903	0.2937	−0.0034 [−0.0042, −0.0026]	8.4e-24	loss
coco17	resnet50	pointing, 15 px	replaced aggregator	necessity vs centre prior	7,339	exact McNemar	0.7085	0.3796	+0.3289 [+0.3158, +0.3426]	<1e-300	win
coco17	resnet50	pointing, 15 px	replaced aggregator	necessity vs collapsed	7,339	exact McNemar	0.7085	0.7083	+0.0003 [−0.0045, +0.0049]	0.987	tie
coco17	resnet50	pointing, 15 px	replaced aggregator	sufficiency vs centre prior	7,339	exact McNemar	0.7100	0.3796	+0.3304 [+0.3175, +0.3439]	<1e-300	win
coco17	resnet50	pointing, 15 px	replaced aggregator	sufficiency vs collapsed	7,339	exact McNemar	0.7100	0.7083	+0.0018 [−0.0030, +0.0065]	0.987	tie
coco17	resnet50	pointing, strict	replaced aggregator	necessity vs centre prior	7,339	exact McNemar	0.4368	0.2002	+0.2367 [+0.2247, +0.2492]	4.5e-294	win
coco17	resnet50	pointing, strict	replaced aggregator	necessity vs collapsed	7,339	exact McNemar	0.4368	0.4332	+0.0037 [−0.0018, +0.0093]	0.423	tie
coco17	resnet50	pointing, strict	replaced aggregator	sufficiency vs centre prior	7,339	exact McNemar	0.4338	0.2002	+0.2337 [+0.2214, +0.2464]	3.1e-286	win
coco17	resnet50	pointing, strict	replaced aggregator	sufficiency vs collapsed	7,339	exact McNemar	0.4338	0.4332	+0.0007 [−0.0048, +0.0061]	0.842	tie
coco17	vgg16gap	peak-mask IoU	replaced aggregator	necessity vs centre prior	7,339	Wilcoxon	0.1847	0.1736	+0.0110 [+0.0071, +0.0149]	0.0581	tie
coco17	vgg16gap	peak-mask IoU	replaced aggregator	necessity vs collapsed	7,339	Wilcoxon	0.1847	0.2772	−0.0925 [−0.0956, −0.0893]	<1e-300	loss
coco17	vgg16gap	peak-mask IoU	replaced aggregator	sufficiency vs centre prior	7,339	Wilcoxon	0.2611	0.1736	+0.0875 [+0.0833, +0.0917]	<1e-300	win
coco17	vgg16gap	peak-mask IoU	replaced aggregator	sufficiency vs collapsed	7,339	Wilcoxon	0.2611	0.2772	−0.0161 [−0.0173, −0.0148]	5.3e-145	loss
coco17	vgg16gap	pointing, 15 px	replaced aggregator	necessity vs centre prior	7,339	exact McNemar	0.6495	0.3796	+0.2699 [+0.2573, +0.2831]	<1e-300	win

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Table S2 (continued)

Dataset	Backbone	Metric	Arm	Contrast	n	Test	Separated map	Comparator	diff [95% CI]	P_{Holm}	Verdict
coco17	vgg16gap	pointing, 15 px	replaced aggregator	necessity vs collapsed	7,339	exact McNemar	0.6495	0.6491	+0.0004 [−0.0040, +0.0046]	1	tie
coco17	vgg16gap	pointing, 15 px	replaced aggregator	sufficiency vs centre prior	7,339	exact McNemar	0.6498	0.3796	+0.2702 [+0.2577, +0.2833]	<1e-300	win
coco17	vgg16gap	pointing, 15 px	replaced aggregator	sufficiency vs collapsed	7,339	exact McNemar	0.6498	0.6491	+0.0007 [−0.0037, +0.0052]	1	tie
coco17	vgg16gap	pointing, strict	replaced aggregator	necessity vs centre prior	7,339	exact McNemar	0.4175	0.2002	+0.2173 [+0.2064, +0.2300]	4.9e-275	win
coco17	vgg16gap	pointing, strict	replaced aggregator	necessity vs collapsed	7,339	exact McNemar	0.4175	0.4217	−0.0042 [−0.0091, +0.0007]	0.202	tie
coco17	vgg16gap	pointing, strict	replaced aggregator	sufficiency vs centre prior	7,339	exact McNemar	0.4219	0.2002	+0.2217 [+0.2105, +0.2341]	2.1e-282	win
coco17	vgg16gap	pointing, strict	replaced aggregator	sufficiency vs collapsed	7,339	exact McNemar	0.4219	0.4217	+0.0001 [−0.0045, +0.0048]	1	tie
coco17	vitb16	peak-mask IoU	relocated normalisation	necessity vs centre prior	7,339	Wilcoxon	0.2647	0.1736	+0.0910 [+0.0867, +0.0956]	1.7e-294	win
coco17	vitb16	peak-mask IoU	relocated normalisation	necessity vs collapsed	7,339	Wilcoxon	0.2647	0.2559	+0.0088 [+0.0077, +0.0098]	6.7e-56	win
coco17	vitb16	peak-mask IoU	relocated normalisation	sufficiency vs centre prior	7,339	Wilcoxon	0.2634	0.1736	+0.0898 [+0.0856, +0.0942]	7.0e-289	win
coco17	vitb16	peak-mask IoU	relocated normalisation	sufficiency vs collapsed	7,339	Wilcoxon	0.2634	0.2559	+0.0075 [+0.0062, +0.0088]	5.4e-18	win
coco17	vitb16	peak-mask IoU	replaced aggregator	necessity vs centre prior	7,339	Wilcoxon	0.2101	0.1736	+0.0364 [+0.0323, +0.0407]	4.6e-35	win
coco17	vitb16	peak-mask IoU	replaced aggregator	necessity vs collapsed	7,339	Wilcoxon	0.2101	0.2559	−0.0458 [−0.0486, −0.0434]	3.5e-257	loss
coco17	vitb16	peak-mask IoU	replaced aggregator	sufficiency vs centre prior	7,339	Wilcoxon	0.2666	0.1736	+0.0930 [+0.0885, +0.0976]	1.5e-288	win
coco17	vitb16	peak-mask IoU	replaced aggregator	sufficiency vs collapsed	7,339	Wilcoxon	0.2666	0.2559	+0.0107 [+0.0098, +0.0117]	2.3e-79	win
coco17	vitb16	pointing, 15 px	relocated normalisation	necessity vs centre prior	7,339	exact McNemar	0.5775	0.3796	+0.1978 [+0.1839, +0.2115]	1.8e-165	win
coco17	vitb16	pointing, 15 px	relocated normalisation	necessity vs collapsed	7,339	exact McNemar	0.5775	0.5835	−0.0060 [−0.0140, +0.0018]	0.152	tie
coco17	vitb16	pointing, 15 px	relocated normalisation	sufficiency vs centre prior	7,339	exact McNemar	0.5687	0.3796	+0.1891 [+0.1756, +0.2025]	4.2e-152	win
coco17	vitb16	pointing, 15 px	relocated normalisation	sufficiency vs collapsed	7,339	exact McNemar	0.5687	0.5835	−0.0147 [−0.0236, −0.0057]	0.00215	loss
coco17	vitb16	pointing, 15 px	replaced aggregator	necessity vs centre prior	7,339	exact McNemar	0.5953	0.3796	+0.2157 [+0.2019, +0.2293]	1.4e-192	win
coco17	vitb16	pointing, 15 px	replaced aggregator	necessity vs collapsed	7,339	exact McNemar	0.5953	0.5835	+0.0119 [+0.0053, +0.0183]	0.00151	win
coco17	vitb16	pointing, 15 px	replaced aggregator	sufficiency vs centre prior	7,339	exact McNemar	0.5990	0.3796	+0.2194 [+0.2057, +0.2330]	4.1e-198	win
coco17	vitb16	pointing, 15 px	replaced aggregator	sufficiency vs collapsed	7,339	exact McNemar	0.5990	0.5835	+0.0155 [+0.0087, +0.0221]	2.4e-05	win
coco17	vitb16	pointing, strict	relocated normalisation	necessity vs centre prior	7,339	exact McNemar	0.3728	0.2002	+0.1726 [+0.1613, +0.1852]	2.8e-162	win
coco17	vitb16	pointing, strict	relocated normalisation	necessity vs collapsed	7,339	exact McNemar	0.3728	0.3540	+0.0188 [+0.0112, +0.0266]	1.0e-05	win
coco17	vitb16	pointing, strict	relocated normalisation	sufficiency vs centre prior	7,339	exact McNemar	0.3603	0.2002	+0.1601 [+0.1485, +0.1720]	1.1e-143	win
coco17	vitb16	pointing, strict	relocated normalisation	sufficiency vs collapsed	7,339	exact McNemar	0.3603	0.3540	+0.0063 [−0.0020, +0.0147]	0.159	tie
coco17	vitb16	pointing, strict	replaced aggregator	necessity vs centre prior	7,339	exact McNemar	0.3656	0.2002	+0.1654 [+0.1530, +0.1780]	5.7e-146	win
coco17	vitb16	pointing, strict	replaced aggregator	necessity vs collapsed	7,339	exact McNemar	0.3656	0.3540	+0.0116 [+0.0054, +0.0180]	0.0015	win
coco17	vitb16	pointing, strict	replaced aggregator	sufficiency vs centre prior	7,339	exact McNemar	0.3687	0.2002	+0.1686 [+0.1557, +0.1812]	4.1e-151	win
coco17	vitb16	pointing, strict	replaced aggregator	sufficiency vs collapsed	7,339	exact McNemar	0.3687	0.3540	+0.0147 [+0.0084, +0.0211]	4.5e-05	win
cub	resnet50	MaxBoxAccV2	replaced aggregator	necessity vs centre prior	5,794	paired bootstrap	0.2509	0.5636	−0.3126 [−0.3194, −0.3074]	0.004	loss
cub	resnet50	MaxBoxAccV2	replaced aggregator	necessity vs collapsed	5,794	paired bootstrap	0.2509	0.5295	−0.2786 [−0.2887, −0.2710]	0.004	loss
cub	resnet50	MaxBoxAccV2	replaced aggregator	sufficiency vs centre prior	5,794	paired bootstrap	0.5645	0.5636	+0.0010 [−0.0087, +0.0093]	0.88	tie
cub	resnet50	MaxBoxAccV2	replaced aggregator	sufficiency vs collapsed	5,794	paired bootstrap	0.5645	0.5295	+0.0350 [+0.0270, +0.0415]	0.004	win
cub	resnet50	best IoU	replaced aggregator	necessity vs centre prior	5,794	Wilcoxon	0.3516	0.6251	−0.2736 [−0.2773, −0.2700]	<1e-300	loss
cub	resnet50	best IoU	replaced aggregator	necessity vs collapsed	5,794	Wilcoxon	0.3516	0.6608	−0.3092 [−0.3131, −0.3054]	<1e-300	loss
cub	resnet50	best IoU	replaced aggregator	sufficiency vs centre prior	5,794	Wilcoxon	0.6466	0.6251	+0.0215 [+0.0166, +0.0259]	1.5e-15	win
cub	resnet50	best IoU	replaced aggregator	sufficiency vs collapsed	5,794	Wilcoxon	0.6466	0.6608	−0.0142 [−0.0164, −0.0119]	1.1e-28	loss
cub	resnet50	pointing, 15 px	replaced aggregator	necessity vs centre prior	5,794	exact McNemar	0.9924	0.9867	+0.0057 [+0.0021, +0.0095]	0.00955	win
cub	resnet50	pointing, 15 px	replaced aggregator	necessity vs collapsed	5,794	exact McNemar	0.9924	0.9952	−0.0028 [−0.0047, −0.0009]	0.0181	loss
cub	resnet50	pointing, 15 px	replaced aggregator	sufficiency vs centre prior	5,794	exact McNemar	0.9934	0.9867	+0.0067 [+0.0031, +0.0104]	0.00124	win
cub	resnet50	pointing, 15 px	replaced aggregator	sufficiency vs collapsed	5,794	exact McNemar	0.9934	0.9952	−0.0017 [−0.0036, +0.0000]	0.0872	tie

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Table S2 (continued)

Dataset	Backbone	Metric	Arm	Contrast	n	Test	Separated map	Comparator	diff [95% CI]	P_{Holm}	Verdict
cub	resnet50	pointing, strict	replaced aggregator	necessity vs centre prior	5,794	exact McNemar	0.9581	0.9446	+0.0135 [+0.0057, +0.0212]	0.00201	win
cub	resnet50	pointing, strict	replaced aggregator	necessity vs collapsed	5,794	exact McNemar	0.9581	0.9650	−0.0069 [−0.0110, −0.0028]	0.00274	loss
cub	resnet50	pointing, strict	replaced aggregator	sufficiency vs centre prior	5,794	exact McNemar	0.9600	0.9446	+0.0154 [+0.0076, +0.0228]	0.000346	win
cub	resnet50	pointing, strict	replaced aggregator	sufficiency vs collapsed	5,794	exact McNemar	0.9600	0.9650	−0.0050 [−0.0091, −0.0009]	0.0181	loss
cub	vgg16gap	MaxBoxAccV2	replaced aggregator	necessity vs centre prior	5,794	paired bootstrap	0.2509	0.5636	−0.3126 [−0.3194, −0.3074]	0.004	loss
cub	vgg16gap	MaxBoxAccV2	replaced aggregator	necessity vs collapsed	5,794	paired bootstrap	0.2509	0.4054	−0.1544 [−0.1631, −0.1475]	0.004	loss
cub	vgg16gap	MaxBoxAccV2	replaced aggregator	sufficiency vs centre prior	5,794	paired bootstrap	0.3786	0.5636	−0.1850 [−0.1954, −0.1758]	0.004	loss
cub	vgg16gap	MaxBoxAccV2	replaced aggregator	sufficiency vs collapsed	5,794	paired bootstrap	0.3786	0.4054	−0.0268 [−0.0357, −0.0193]	0.004	loss
cub	vgg16gap	best IoU	replaced aggregator	necessity vs centre prior	5,794	Wilcoxon	0.3496	0.6251	−0.2755 [−0.2791, −0.2720]	<1e-300	loss
cub	vgg16gap	best IoU	replaced aggregator	necessity vs collapsed	5,794	Wilcoxon	0.3496	0.6243	−0.2747 [−0.2786, −0.2709]	<1e-300	loss
cub	vgg16gap	best IoU	replaced aggregator	sufficiency vs centre prior	5,794	Wilcoxon	0.5650	0.6251	−0.0601 [−0.0652, −0.0549]	1.3e-114	loss
cub	vgg16gap	best IoU	replaced aggregator	sufficiency vs collapsed	5,794	Wilcoxon	0.5650	0.6243	−0.0593 [−0.0630, −0.0558]	3.7e-197	loss
cub	vgg16gap	pointing, 15 px	replaced aggregator	necessity vs centre prior	5,794	exact McNemar	0.9865	0.9867	−0.0002 [−0.0047, +0.0040]	1	tie
cub	vgg16gap	pointing, 15 px	replaced aggregator	necessity vs collapsed	5,794	exact McNemar	0.9865	0.9917	−0.0052 [−0.0083, −0.0024]	0.00421	loss
cub	vgg16gap	pointing, 15 px	replaced aggregator	sufficiency vs centre prior	5,794	exact McNemar	0.9877	0.9867	+0.0010 [−0.0033, +0.0048]	1	tie
cub	vgg16gap	pointing, 15 px	replaced aggregator	sufficiency vs collapsed	5,794	exact McNemar	0.9877	0.9917	−0.0040 [−0.0069, −0.0014]	0.0286	loss
cub	vgg16gap	pointing, strict	replaced aggregator	necessity vs centre prior	5,794	exact McNemar	0.9767	0.9446	+0.0321 [+0.0249, +0.0385]	3.7e-19	win
cub	vgg16gap	pointing, strict	replaced aggregator	necessity vs collapsed	5,794	exact McNemar	0.9767	0.9808	−0.0041 [−0.0079, −0.0005]	0.0833	tie
cub	vgg16gap	pointing, strict	replaced aggregator	sufficiency vs centre prior	5,794	exact McNemar	0.9786	0.9446	+0.0340 [+0.0271, +0.0406]	1.1e-21	win
cub	vgg16gap	pointing, strict	replaced aggregator	sufficiency vs collapsed	5,794	exact McNemar	0.9786	0.9808	−0.0022 [−0.0057, +0.0012]	0.271	tie
cub	vitb16	MaxBoxAccV2	relocated normalisation	necessity vs centre prior	5,794	paired bootstrap	0.3996	0.5636	−0.1640 [−0.1729, −0.1561]	0.008	loss
cub	vitb16	MaxBoxAccV2	relocated normalisation	necessity vs collapsed	5,794	paired bootstrap	0.3996	0.3337	+0.0659 [+0.0592, +0.0713]	0.008	win
cub	vitb16	MaxBoxAccV2	relocated normalisation	sufficiency vs centre prior	5,794	paired bootstrap	0.4769	0.5636	−0.0866 [−0.0950, −0.0773]	0.008	loss
cub	vitb16	MaxBoxAccV2	relocated normalisation	sufficiency vs collapsed	5,794	paired bootstrap	0.4769	0.3337	+0.1433 [+0.1357, +0.1511]	0.008	win
cub	vitb16	MaxBoxAccV2	replaced aggregator	necessity vs centre prior	5,794	paired bootstrap	0.2691	0.5636	−0.2944 [−0.3020, −0.2889]	0.008	loss
cub	vitb16	MaxBoxAccV2	replaced aggregator	necessity vs collapsed	5,794	paired bootstrap	0.2691	0.3337	−0.0645 [−0.0723, −0.0591]	0.008	loss
cub	vitb16	MaxBoxAccV2	replaced aggregator	sufficiency vs centre prior	5,794	paired bootstrap	0.3235	0.5636	−0.2401 [−0.2483, −0.2316]	0.008	loss
cub	vitb16	MaxBoxAccV2	replaced aggregator	sufficiency vs collapsed	5,794	paired bootstrap	0.3235	0.3337	−0.0102 [−0.0146, −0.0049]	0.008	loss
cub	vitb16	best IoU	relocated normalisation	necessity vs centre prior	5,794	Wilcoxon	0.5970	0.6251	−0.0281 [−0.0326, −0.0235]	1.2e-42	loss
cub	vitb16	best IoU	relocated normalisation	necessity vs collapsed	5,794	Wilcoxon	0.5970	0.5406	+0.0565 [+0.0536, +0.0594]	7.7e-292	win
cub	vitb16	best IoU	relocated normalisation	sufficiency vs centre prior	5,794	Wilcoxon	0.6376	0.6251	+0.0125 [+0.0076, +0.0173]	9.2e-06	win
cub	vitb16	best IoU	relocated normalisation	sufficiency vs collapsed	5,794	Wilcoxon	0.6376	0.5406	+0.0970 [+0.0937, +0.1003]	<1e-300	win
cub	vitb16	best IoU	replaced aggregator	necessity vs centre prior	5,794	Wilcoxon	0.4511	0.6251	−0.1740 [−0.1790, −0.1692]	<1e-300	loss
cub	vitb16	best IoU	replaced aggregator	necessity vs collapsed	5,794	Wilcoxon	0.4511	0.5406	−0.0895 [−0.0926, −0.0863]	<1e-300	loss
cub	vitb16	best IoU	replaced aggregator	sufficiency vs centre prior	5,794	Wilcoxon	0.5476	0.6251	−0.0775 [−0.0826, −0.0722]	1.0e-177	loss
cub	vitb16	best IoU	replaced aggregator	sufficiency vs collapsed	5,794	Wilcoxon	0.5476	0.5406	+0.0071 [+0.0049, +0.0094]	3.3e-08	win
cub	vitb16	pointing, 15 px	relocated normalisation	necessity vs centre prior	5,794	exact McNemar	0.7654	0.9867	−0.2213 [−0.2321, −0.2100]	<1e-300	loss
cub	vitb16	pointing, 15 px	relocated normalisation	necessity vs collapsed	5,794	exact McNemar	0.7654	0.5644	+0.2011 [+0.1873, +0.2145]	1.6e-178	win
cub	vitb16	pointing, 15 px	relocated normalisation	sufficiency vs centre prior	5,794	exact McNemar	0.8386	0.9867	−0.1481 [−0.1574, −0.1386]	1.8e-203	loss
cub	vitb16	pointing, 15 px	relocated normalisation	sufficiency vs collapsed	5,794	exact McNemar	0.8386	0.5644	+0.2742 [+0.2603, +0.2875]	<1e-300	win
cub	vitb16	pointing, 15 px	replaced aggregator	necessity vs centre prior	5,794	exact McNemar	0.5853	0.9867	−0.4014 [−0.4147, −0.3889]	<1e-300	loss
cub	vitb16	pointing, 15 px	replaced aggregator	necessity vs collapsed	5,794	exact McNemar	0.5853	0.5644	+0.0209 [+0.0085, +0.0324]	0.00127	win
cub	vitb16	pointing, 15 px	replaced aggregator	sufficiency vs centre prior	5,794	exact McNemar	0.5851	0.9867	−0.4016 [−0.4139, −0.3895]	<1e-300	loss

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Table S2 (continued)

Dataset	Backbone	Metric	Arm	Contrast	n	Test	Separated map	Comparator	diff [95% CI]	P_{Holm}	Verdict
cub	vitb16	pointing, 15 px	replaced aggregator	sufficiency vs collapsed	5,794	exact McNemar	0.5851	0.5644	+0.0207 [+0.0086, +0.0323]	0.00127	win
cub	vitb16	pointing, strict	relocated normalisation	necessity vs centre prior	5,794	exact McNemar	0.6545	0.9446	−0.2901 [−0.3036, −0.2774]	<1e-300	loss
cub	vitb16	pointing, strict	relocated normalisation	necessity vs collapsed	5,794	exact McNemar	0.6545	0.4220	+0.2325 [+0.2190, +0.2466]	8.5e-224	win
cub	vitb16	pointing, strict	relocated normalisation	sufficiency vs centre prior	5,794	exact McNemar	0.7478	0.9446	−0.1968 [−0.2092, −0.1843]	3.6e-209	loss
cub	vitb16	pointing, strict	relocated normalisation	sufficiency vs collapsed	5,794	exact McNemar	0.7478	0.4220	+0.3259 [+0.3120, +0.3402]	<1e-300	win
cub	vitb16	pointing, strict	replaced aggregator	necessity vs centre prior	5,794	exact McNemar	0.4537	0.9446	−0.4909 [−0.5043, −0.4774]	<1e-300	loss
cub	vitb16	pointing, strict	replaced aggregator	necessity vs collapsed	5,794	exact McNemar	0.4537	0.4220	+0.0318 [+0.0202, +0.0435]	1.4e-07	win
cub	vitb16	pointing, strict	replaced aggregator	sufficiency vs centre prior	5,794	exact McNemar	0.4534	0.9446	−0.4912 [−0.5048, −0.4770]	<1e-300	loss
cub	vitb16	pointing, strict	replaced aggregator	sufficiency vs collapsed	5,794	exact McNemar	0.4534	0.4220	+0.0314 [+0.0200, +0.0430]	1.4e-07	win
voc12	resnet50	pointing, 15 px	replaced aggregator	necessity vs centre prior	8,856	exact McNemar	0.9163	0.7053	+0.2110 [+0.2010, +0.2210]	<1e-300	win
voc12	resnet50	pointing, 15 px	replaced aggregator	necessity vs collapsed	8,856	exact McNemar	0.9163	0.9177	−0.0014 [−0.0040, +0.0014]	0.503	tie
voc12	resnet50	pointing, 15 px	replaced aggregator	sufficiency vs centre prior	8,856	exact McNemar	0.9194	0.7053	+0.2141 [+0.2044, +0.2239]	<1e-300	win
voc12	resnet50	pointing, 15 px	replaced aggregator	sufficiency vs collapsed	8,856	exact McNemar	0.9194	0.9177	+0.0017 [−0.0010, +0.0045]	0.503	tie
voc12	resnet50	pointing, strict	replaced aggregator	necessity vs centre prior	8,856	exact McNemar	0.8398	0.5911	+0.2486 [+0.2380, +0.2590]	<1e-300	win
voc12	resnet50	pointing, strict	replaced aggregator	necessity vs collapsed	8,856	exact McNemar	0.8398	0.8368	+0.0029 [−0.0005, +0.0063]	0.108	tie
voc12	resnet50	pointing, strict	replaced aggregator	sufficiency vs centre prior	8,856	exact McNemar	0.8434	0.5911	+0.2523 [+0.2420, +0.2625]	<1e-300	win
voc12	resnet50	pointing, strict	replaced aggregator	sufficiency vs collapsed	8,856	exact McNemar	0.8434	0.8368	+0.0065 [+0.0033, +0.0098]	0.000277	win
voc12	vgg16gap	pointing, 15 px	replaced aggregator	necessity vs centre prior	8,856	exact McNemar	0.8945	0.7053	+0.1893 [+0.1799, +0.1987]	<1e-300	win
voc12	vgg16gap	pointing, 15 px	replaced aggregator	necessity vs collapsed	8,856	exact McNemar	0.8945	0.8957	−0.0011 [−0.0040, +0.0015]	0.931	tie
voc12	vgg16gap	pointing, 15 px	replaced aggregator	sufficiency vs centre prior	8,856	exact McNemar	0.8967	0.7053	+0.1914 [+0.1820, +0.2005]	<1e-300	win
voc12	vgg16gap	pointing, 15 px	replaced aggregator	sufficiency vs collapsed	8,856	exact McNemar	0.8967	0.8957	+0.0010 [−0.0017, +0.0037]	0.931	tie
voc12	vgg16gap	pointing, strict	replaced aggregator	necessity vs centre prior	8,856	exact McNemar	0.8093	0.5911	+0.2182 [+0.2075, +0.2288]	<1e-300	win
voc12	vgg16gap	pointing, strict	replaced aggregator	necessity vs collapsed	8,856	exact McNemar	0.8093	0.8089	+0.0003 [−0.0032, +0.0037]	0.899	tie
voc12	vgg16gap	pointing, strict	replaced aggregator	sufficiency vs centre prior	8,856	exact McNemar	0.8145	0.5911	+0.2234 [+0.2130, +0.2340]	<1e-300	win
voc12	vgg16gap	pointing, strict	replaced aggregator	sufficiency vs collapsed	8,856	exact McNemar	0.8145	0.8089	+0.0055 [+0.0023, +0.0088]	0.00239	win
voc12	vitb16	pointing, 15 px	relocated normalisation	necessity vs centre prior	8,856	exact McNemar	0.8285	0.7053	+0.1232 [+0.1121, +0.1338]	4.4e-107	win
voc12	vitb16	pointing, 15 px	relocated normalisation	necessity vs collapsed	8,856	exact McNemar	0.8285	0.8199	+0.0086 [+0.0024, +0.0141]	0.0102	win
voc12	vitb16	pointing, 15 px	relocated normalisation	sufficiency vs centre prior	8,856	exact McNemar	0.8476	0.7053	+0.1423 [+0.1317, +0.1524]	1.2e-156	win
voc12	vitb16	pointing, 15 px	relocated normalisation	sufficiency vs collapsed	8,856	exact McNemar	0.8476	0.8199	+0.0277 [+0.0218, +0.0333]	3.0e-20	win
voc12	vitb16	pointing, 15 px	replaced aggregator	necessity vs centre prior	8,856	exact McNemar	0.8234	0.7053	+0.1181 [+0.1067, +0.1288]	5.7e-96	win
voc12	vitb16	pointing, 15 px	replaced aggregator	necessity vs collapsed	8,856	exact McNemar	0.8234	0.8199	+0.0035 [−0.0012, +0.0082]	0.166	tie
voc12	vitb16	pointing, 15 px	replaced aggregator	sufficiency vs centre prior	8,856	exact McNemar	0.8385	0.7053	+0.1332 [+0.1217, +0.1440]	1.2e-124	win
voc12	vitb16	pointing, 15 px	replaced aggregator	sufficiency vs collapsed	8,856	exact McNemar	0.8385	0.8199	+0.0186 [+0.0136, +0.0236]	4.3e-13	win
voc12	vitb16	pointing, strict	relocated normalisation	necessity vs centre prior	8,856	exact McNemar	0.7072	0.5911	+0.1161 [+0.1037, +0.1283]	5.9e-76	win
voc12	vitb16	pointing, strict	relocated normalisation	necessity vs collapsed	8,856	exact McNemar	0.7072	0.6834	+0.0238 [+0.0165, +0.0311]	3.6e-10	win
voc12	vitb16	pointing, strict	relocated normalisation	sufficiency vs centre prior	8,856	exact McNemar	0.7311	0.5911	+0.1400 [+0.1288, +0.1513]	5.6e-123	win
voc12	vitb16	pointing, strict	relocated normalisation	sufficiency vs collapsed	8,856	exact McNemar	0.7311	0.6834	+0.0478 [+0.0405, +0.0554]	2.7e-36	win
voc12	vitb16	pointing, strict	replaced aggregator	necessity vs centre prior	8,856	exact McNemar	0.6990	0.5911	+0.1078 [+0.0954, +0.1201]	9.5e-66	win
voc12	vitb16	pointing, strict	replaced aggregator	necessity vs collapsed	8,856	exact McNemar	0.6990	0.6834	+0.0156 [+0.0095, +0.0213]	3.4e-07	win
voc12	vitb16	pointing, strict	replaced aggregator	sufficiency vs centre prior	8,856	exact McNemar	0.7093	0.5911	+0.1182 [+0.1057, +0.1304]	2.1e-78	win
voc12	vitb16	pointing, strict	replaced aggregator	sufficiency vs collapsed	8,856	exact McNemar	0.7093	0.6834	+0.0260 [+0.0200, +0.0317]	1.1e-16	win
voc12seg	resnet50	peak-mask IoU	replaced aggregator	necessity vs centre prior	2,147	Wilcoxon	0.2622	0.3209	−0.0587 [−0.0678, −0.0498]	2.8e-55	loss
voc12seg	resnet50	peak-mask IoU	replaced aggregator	necessity vs collapsed	2,147	Wilcoxon	0.2622	0.4790	−0.2167 [−0.2241, −0.2095]	<1e-300	loss

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Table S2 (continued)

Dataset	Backbone	Metric	Arm	Contrast	n	Test	Separated map	Comparator	diff [95% CI]	P_{Holm}	Verdict
voc12seg	resnet50	peak-mask IoU	replaced aggregator	sufficiency vs centre prior	2,147	Wilcoxon	0.4785	0.3209	+0.1576 [+0.1486, +0.1661]	2.0e-197	win
voc12seg	resnet50	peak-mask IoU	replaced aggregator	sufficiency vs collapsed	2,147	Wilcoxon	0.4785	0.4790	−0.0005 [−0.0021, +0.0012]	0.00514	loss
voc12seg	resnet50	pointing, 15 px	replaced aggregator	necessity vs centre prior	2,147	exact McNemar	0.9082	0.6530	+0.2552 [+0.2347, +0.2748]	1.0e-119	win
voc12seg	resnet50	pointing, 15 px	replaced aggregator	necessity vs collapsed	2,147	exact McNemar	0.9082	0.9054	+0.0028 [−0.0033, +0.0084]	0.441	tie
voc12seg	resnet50	pointing, 15 px	replaced aggregator	sufficiency vs centre prior	2,147	exact McNemar	0.9106	0.6530	+0.2576 [+0.2371, +0.2771]	2.2e-120	win
voc12seg	resnet50	pointing, 15 px	replaced aggregator	sufficiency vs collapsed	2,147	exact McNemar	0.9106	0.9054	+0.0051 [−0.0009, +0.0112]	0.235	tie
voc12seg	resnet50	pointing, strict	replaced aggregator	necessity vs centre prior	2,147	exact McNemar	0.6670	0.4099	+0.2571 [+0.2329, +0.2813]	2.4e-81	win
voc12seg	resnet50	pointing, strict	replaced aggregator	necessity vs collapsed	2,147	exact McNemar	0.6670	0.6591	+0.0079 [−0.0019, +0.0182]	0.145	tie
voc12seg	resnet50	pointing, strict	replaced aggregator	sufficiency vs centre prior	2,147	exact McNemar	0.6712	0.4099	+0.2613 [+0.2366, +0.2864]	5.6e-84	win
voc12seg	resnet50	pointing, strict	replaced aggregator	sufficiency vs collapsed	2,147	exact McNemar	0.6712	0.6591	+0.0121 [+0.0028, +0.0214]	0.0207	win
voc12seg	vgg16gap	peak-mask IoU	replaced aggregator	necessity vs centre prior	2,147	Wilcoxon	0.2603	0.3209	−0.0606 [−0.0692, −0.0518]	4.4e-57	loss
voc12seg	vgg16gap	peak-mask IoU	replaced aggregator	necessity vs collapsed	2,147	Wilcoxon	0.2603	0.4569	−0.1966 [−0.2041, −0.1895]	<1e-300	loss
voc12seg	vgg16gap	peak-mask IoU	replaced aggregator	sufficiency vs centre prior	2,147	Wilcoxon	0.4478	0.3209	+0.1269 [+0.1179, +0.1355]	6.0e-138	win
voc12seg	vgg16gap	peak-mask IoU	replaced aggregator	sufficiency vs collapsed	2,147	Wilcoxon	0.4478	0.4569	−0.0091 [−0.0112, −0.0072]	8.3e-23	loss
voc12seg	vgg16gap	pointing, 15 px	replaced aggregator	necessity vs centre prior	2,147	exact McNemar	0.8887	0.6530	+0.2357 [+0.2152, +0.2557]	3.4e-104	win
voc12seg	vgg16gap	pointing, 15 px	replaced aggregator	necessity vs collapsed	2,147	exact McNemar	0.8887	0.8919	−0.0033 [−0.0098, +0.0023]	0.721	tie
voc12seg	vgg16gap	pointing, 15 px	replaced aggregator	sufficiency vs centre prior	2,147	exact McNemar	0.8915	0.6530	+0.2385 [+0.2180, +0.2585]	5.0e-108	win
voc12seg	vgg16gap	pointing, 15 px	replaced aggregator	sufficiency vs collapsed	2,147	exact McNemar	0.8915	0.8919	−0.0005 [−0.0070, +0.0056]	1	tie
voc12seg	vgg16gap	pointing, strict	replaced aggregator	necessity vs centre prior	2,147	exact McNemar	0.6642	0.4099	+0.2543 [+0.2296, +0.2781]	2.0e-86	win
voc12seg	vgg16gap	pointing, strict	replaced aggregator	necessity vs collapsed	2,147	exact McNemar	0.6642	0.6646	−0.0005 [−0.0112, +0.0093]	1	tie
voc12seg	vgg16gap	pointing, strict	replaced aggregator	sufficiency vs centre prior	2,147	exact McNemar	0.6702	0.4099	+0.2604 [+0.2357, +0.2841]	5.5e-91	win
voc12seg	vgg16gap	pointing, strict	replaced aggregator	sufficiency vs collapsed	2,147	exact McNemar	0.6702	0.6646	+0.0056 [−0.0033, +0.0149]	0.523	tie
voc12seg	vitb16	peak-mask IoU	relocated normalisation	necessity vs centre prior	2,147	Wilcoxon	0.4261	0.3209	+0.1052 [+0.0945, +0.1155]	4.4e-70	win
voc12seg	vitb16	peak-mask IoU	relocated normalisation	necessity vs collapsed	2,147	Wilcoxon	0.4261	0.4107	+0.0154 [+0.0129, +0.0179]	1.7e-34	win
voc12seg	vitb16	peak-mask IoU	relocated normalisation	sufficiency vs centre prior	2,147	Wilcoxon	0.4375	0.3209	+0.1166 [+0.1065, +0.1264]	5.0e-99	win
voc12seg	vitb16	peak-mask IoU	relocated normalisation	sufficiency vs collapsed	2,147	Wilcoxon	0.4375	0.4107	+0.0268 [+0.0235, +0.0301]	1.6e-55	win
voc12seg	vitb16	peak-mask IoU	replaced aggregator	necessity vs centre prior	2,147	Wilcoxon	0.2777	0.3209	−0.0432 [−0.0524, −0.0333]	1.4e-30	loss
voc12seg	vitb16	peak-mask IoU	replaced aggregator	necessity vs collapsed	2,147	Wilcoxon	0.2777	0.4107	−0.1330 [−0.1402, −0.1261]	3.3e-246	loss
voc12seg	vitb16	peak-mask IoU	replaced aggregator	sufficiency vs centre prior	2,147	Wilcoxon	0.4317	0.3209	+0.1108 [+0.0997, +0.1215]	2.6e-73	win
voc12seg	vitb16	peak-mask IoU	replaced aggregator	sufficiency vs collapsed	2,147	Wilcoxon	0.4317	0.4107	+0.0210 [+0.0188, +0.0232]	1.8e-61	win
voc12seg	vitb16	pointing, 15 px	relocated normalisation	necessity vs centre prior	2,147	exact McNemar	0.7857	0.6530	+0.1327 [+0.1062, +0.1560]	1.8e-24	win
voc12seg	vitb16	pointing, 15 px	relocated normalisation	necessity vs collapsed	2,147	exact McNemar	0.7857	0.7848	+0.0009 [−0.0135, +0.0144]	1	tie
voc12seg	vitb16	pointing, 15 px	relocated normalisation	sufficiency vs centre prior	2,147	exact McNemar	0.8272	0.6530	+0.1742 [+0.1509, +0.1956]	2.3e-48	win
voc12seg	vitb16	pointing, 15 px	relocated normalisation	sufficiency vs collapsed	2,147	exact McNemar	0.8272	0.7848	+0.0424 [+0.0289, +0.0545]	8.1e-10	win
voc12seg	vitb16	pointing, 15 px	replaced aggregator	necessity vs centre prior	2,147	exact McNemar	0.7885	0.6530	+0.1355 [+0.1108, +0.1588]	6.9e-25	win
voc12seg	vitb16	pointing, 15 px	replaced aggregator	necessity vs collapsed	2,147	exact McNemar	0.7885	0.7848	+0.0037 [−0.0070, +0.0144]	1	tie
voc12seg	vitb16	pointing, 15 px	replaced aggregator	sufficiency vs centre prior	2,147	exact McNemar	0.8062	0.6530	+0.1532 [+0.1290, +0.1775]	9.6e-32	win
voc12seg	vitb16	pointing, 15 px	replaced aggregator	sufficiency vs collapsed	2,147	exact McNemar	0.8062	0.7848	+0.0214 [+0.0112, +0.0321]	0.000231	win
voc12seg	vitb16	pointing, strict	relocated normalisation	necessity vs centre prior	2,147	exact McNemar	0.5268	0.4099	+0.1169 [+0.0894, +0.1449]	1.5e-15	win
voc12seg	vitb16	pointing, strict	relocated normalisation	necessity vs collapsed	2,147	exact McNemar	0.5268	0.5105	+0.0163 [−0.0009, +0.0331]	0.126	tie
voc12seg	vitb16	pointing, strict	relocated normalisation	sufficiency vs centre prior	2,147	exact McNemar	0.5571	0.4099	+0.1472 [+0.1220, +0.1728]	3.1e-26	win
voc12seg	vitb16	pointing, strict	relocated normalisation	sufficiency vs collapsed	2,147	exact McNemar	0.5571	0.5105	+0.0466 [+0.0284, +0.0629]	7.7e-07	win
voc12seg	vitb16	pointing, strict	replaced aggregator	necessity vs centre prior	2,147	exact McNemar	0.5207	0.4099	+0.1109 [+0.0829, +0.1379]	5.4e-14	win

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Table S2 (continued)

Dataset	Backbone	Metric	Arm	Contrast	n	Test	Separated map	Comparator	diff [95% CI]	P_{Holm}	Verdict
voc12seg	vitb16	pointing, strict	replaced aggregator	necessity vs collapsed	2,147	exact McNemar	0.5207	0.5105	+0.0102 [−0.0037, +0.0242]	0.164	tie
voc12seg	vitb16	pointing, strict	replaced aggregator	sufficiency vs centre prior	2,147	exact McNemar	0.5291	0.4099	+0.1192 [+0.0917, +0.1476]	7.5e-16	win
voc12seg	vitb16	pointing, strict	replaced aggregator	sufficiency vs collapsed	2,147	exact McNemar	0.5291	0.5105	+0.0186 [+0.0051, +0.0326]	0.0218	win
The same separated maps against plain CAM (96 contrasts)											
coco17	resnet50	peak-mask IoU	replaced aggregator	necessity vs plain CAM	7,339	Wilcoxon	0.1956	0.2979	−0.1022 [−0.1056, −0.0992]	<1e-300	loss
coco17	resnet50	peak-mask IoU	replaced aggregator	sufficiency vs plain CAM	7,339	Wilcoxon	0.2903	0.2979	−0.0076 [−0.0082, −0.0069]	1.2e-171	loss
coco17	resnet50	pointing, 15 px	replaced aggregator	necessity vs plain CAM	7,339	exact McNemar	0.7085	0.7100	−0.0015 [−0.0035, +0.0005]	0.37	tie
coco17	resnet50	pointing, 15 px	replaced aggregator	sufficiency vs plain CAM	7,339	exact McNemar	0.7100	0.7100	+0.0000 [−0.0012, +0.0012]	1	tie
coco17	resnet50	pointing, strict	replaced aggregator	necessity vs plain CAM	7,339	exact McNemar	0.4368	0.4333	+0.0035 [+0.0010, +0.0063]	0.0277	win
coco17	resnet50	pointing, strict	replaced aggregator	sufficiency vs plain CAM	7,339	exact McNemar	0.4338	0.4333	+0.0005 [−0.0004, +0.0016]	0.454	tie
coco17	vgg16gap	peak-mask IoU	replaced aggregator	necessity vs plain CAM	7,339	Wilcoxon	0.1847	0.2819	−0.0973 [−0.1004, −0.0940]	<1e-300	loss
coco17	vgg16gap	peak-mask IoU	replaced aggregator	sufficiency vs plain CAM	7,339	Wilcoxon	0.2611	0.2819	−0.0208 [−0.0221, −0.0195]	2.1e-230	loss
coco17	vgg16gap	pointing, 15 px	replaced aggregator	necessity vs plain CAM	7,339	exact McNemar	0.6495	0.6494	+0.0001 [−0.0016, +0.0019]	1	tie
coco17	vgg16gap	pointing, 15 px	replaced aggregator	sufficiency vs plain CAM	7,339	exact McNemar	0.6498	0.6494	+0.0004 [−0.0008, +0.0018]	1	tie
coco17	vgg16gap	pointing, strict	replaced aggregator	necessity vs plain CAM	7,339	exact McNemar	0.4175	0.4199	−0.0025 [−0.0053, +0.0004]	0.111	tie
coco17	vgg16gap	pointing, strict	replaced aggregator	sufficiency vs plain CAM	7,339	exact McNemar	0.4219	0.4199	+0.0019 [+0.0004, +0.0035]	0.0486	win
coco17	vitb16	peak-mask IoU	relocated normalisation	necessity vs plain CAM	7,339	Wilcoxon	0.2647	0.2646	+0.0001 [−0.0011, +0.0013]	0.312	tie
coco17	vitb16	peak-mask IoU	relocated normalisation	sufficiency vs plain CAM	7,339	Wilcoxon	0.2634	0.2646	−0.0011 [−0.0026, +0.0003]	2.1e-06	loss
coco17	vitb16	peak-mask IoU	replaced aggregator	necessity vs plain CAM	7,339	Wilcoxon	0.2101	0.2646	−0.0545 [−0.0573, −0.0520]	<1e-300	loss
coco17	vitb16	peak-mask IoU	replaced aggregator	sufficiency vs plain CAM	7,339	Wilcoxon	0.2666	0.2646	+0.0021 [+0.0013, +0.0028]	0.92	tie
coco17	vitb16	pointing, 15 px	relocated normalisation	necessity vs plain CAM	7,339	exact McNemar	0.5775	0.5989	−0.0214 [−0.0298, −0.0129]	4.5e-06	loss
coco17	vitb16	pointing, 15 px	relocated normalisation	sufficiency vs plain CAM	7,339	exact McNemar	0.5687	0.5989	−0.0301 [−0.0391, −0.0210]	7.5e-10	loss
coco17	vitb16	pointing, 15 px	replaced aggregator	necessity vs plain CAM	7,339	exact McNemar	0.5953	0.5989	−0.0035 [−0.0071, −0.0001]	0.11	tie
coco17	vitb16	pointing, 15 px	replaced aggregator	sufficiency vs plain CAM	7,339	exact McNemar	0.5990	0.5989	+0.0001 [−0.0018, +0.0020]	1	tie
coco17	vitb16	pointing, strict	relocated normalisation	necessity vs plain CAM	7,339	exact McNemar	0.3728	0.3690	+0.0038 [−0.0046, +0.0120]	0.776	tie
coco17	vitb16	pointing, strict	relocated normalisation	sufficiency vs plain CAM	7,339	exact McNemar	0.3603	0.3690	−0.0087 [−0.0177, +0.0000]	0.234	tie
coco17	vitb16	pointing, strict	replaced aggregator	necessity vs plain CAM	7,339	exact McNemar	0.3656	0.3690	−0.0034 [−0.0069, +0.0003]	0.234	tie
coco17	vitb16	pointing, strict	replaced aggregator	sufficiency vs plain CAM	7,339	exact McNemar	0.3687	0.3690	−0.0003 [−0.0019, +0.0012]	0.868	tie
cub	resnet50	MaxBoxAccV2	replaced aggregator	necessity vs plain CAM	5,794	paired bootstrap	0.2509	0.5039	−0.2529 [−0.2631, −0.2441]	0.004	loss
cub	resnet50	MaxBoxAccV2	replaced aggregator	sufficiency vs plain CAM	5,794	paired bootstrap	0.5645	0.5039	+0.0607 [+0.0540, +0.0681]	0.004	win
cub	resnet50	best IoU	replaced aggregator	necessity vs plain CAM	5,794	Wilcoxon	0.3516	0.6598	−0.3083 [−0.3126, −0.3040]	<1e-300	loss
cub	resnet50	best IoU	replaced aggregator	sufficiency vs plain CAM	5,794	Wilcoxon	0.6466	0.6598	−0.0132 [−0.0143, −0.0121]	8.0e-137	loss
cub	resnet50	pointing, 15 px	replaced aggregator	necessity vs plain CAM	5,794	exact McNemar	0.9924	0.9934	−0.0010 [−0.0019, −0.0003]	0.0625	tie
cub	resnet50	pointing, 15 px	replaced aggregator	sufficiency vs plain CAM	5,794	exact McNemar	0.9934	0.9934	+0.0000 [+0.0000, +0.0000]	1	tie
cub	resnet50	pointing, strict	replaced aggregator	necessity vs plain CAM	5,794	exact McNemar	0.9581	0.9600	−0.0019 [−0.0035, −0.0003]	0.0532	tie
cub	resnet50	pointing, strict	replaced aggregator	sufficiency vs plain CAM	5,794	exact McNemar	0.9600	0.9600	+0.0000 [+0.0000, +0.0000]	1	tie
cub	vgg16gap	MaxBoxAccV2	replaced aggregator	necessity vs plain CAM	5,794	paired bootstrap	0.2509	0.3723	−0.1214 [−0.1310, −0.1152]	0.004	loss
cub	vgg16gap	MaxBoxAccV2	replaced aggregator	sufficiency vs plain CAM	5,794	paired bootstrap	0.3786	0.3723	+0.0063 [−0.0031, +0.0128]	0.222	tie
cub	vgg16gap	best IoU	replaced aggregator	necessity vs plain CAM	5,794	Wilcoxon	0.3496	0.6284	−0.2788 [−0.2835, −0.2743]	<1e-300	loss
cub	vgg16gap	best IoU	replaced aggregator	sufficiency vs plain CAM	5,794	Wilcoxon	0.5650	0.6284	−0.0634 [−0.0662, −0.0608]	<1e-300	loss
cub	vgg16gap	pointing, 15 px	replaced aggregator	necessity vs plain CAM	5,794	exact McNemar	0.9865	0.9877	−0.0012 [−0.0024, −0.0002]	0.262	tie
cub	vgg16gap	pointing, 15 px	replaced aggregator	sufficiency vs plain CAM	5,794	exact McNemar	0.9877	0.9877	+0.0000 [−0.0005, +0.0005]	1	tie
cub	vgg16gap	pointing, strict	replaced aggregator	necessity vs plain CAM	5,794	exact McNemar	0.9767	0.9788	−0.0021 [−0.0035, −0.0007]	0.0151	loss

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Table S2 (continued)

Dataset	Backbone	Metric	Arm	Contrast	n	Test	Separated map	Comparator	diff [95% CI]	P_{Holm}	Verdict
cub	vgg16gap	pointing, strict	replaced aggregator	sufficiency vs plain CAM	5,794	exact McNemar	0.9786	0.9788	−0.0002 [−0.0009, +0.0003]	1	tie
cub	vitb16	MaxBoxAccV2	relocated normalisation	necessity vs plain CAM	5,794	paired bootstrap	0.3996	0.3415	+0.0580 [+0.0515, +0.0639]	0.008	win
cub	vitb16	MaxBoxAccV2	relocated normalisation	sufficiency vs plain CAM	5,794	paired bootstrap	0.4769	0.3415	+0.1354 [+0.1278, +0.1435]	0.008	win
cub	vitb16	MaxBoxAccV2	replaced aggregator	necessity vs plain CAM	5,794	paired bootstrap	0.2691	0.3415	−0.0724 [−0.0796, −0.0666]	0.008	loss
cub	vitb16	MaxBoxAccV2	replaced aggregator	sufficiency vs plain CAM	5,794	paired bootstrap	0.3235	0.3415	−0.0180 [−0.0223, −0.0121]	0.008	loss
cub	vitb16	best IoU	relocated normalisation	necessity vs plain CAM	5,794	Wilcoxon	0.5970	0.5465	+0.0506 [+0.0475, +0.0535]	1.2e-217	win
cub	vitb16	best IoU	relocated normalisation	sufficiency vs plain CAM	5,794	Wilcoxon	0.6376	0.5465	+0.0911 [+0.0876, +0.0946]	<1e-300	win
cub	vitb16	best IoU	replaced aggregator	necessity vs plain CAM	5,794	Wilcoxon	0.4511	0.5465	−0.0953 [−0.0985, −0.0922]	<1e-300	loss
cub	vitb16	best IoU	replaced aggregator	sufficiency vs plain CAM	5,794	Wilcoxon	0.5476	0.5465	+0.0012 [+0.0006, +0.0018]	3.9e-12	win
cub	vitb16	pointing, 15 px	relocated normalisation	necessity vs plain CAM	5,794	exact McNemar	0.7654	0.5851	+0.1804 [+0.1672, +0.1947]	1.5e-141	win
cub	vitb16	pointing, 15 px	relocated normalisation	sufficiency vs plain CAM	5,794	exact McNemar	0.8386	0.5851	+0.2535 [+0.2399, +0.2673]	2.1e-266	win
cub	vitb16	pointing, 15 px	replaced aggregator	necessity vs plain CAM	5,794	exact McNemar	0.5853	0.5851	+0.0002 [−0.0041, +0.0043]	1	tie
cub	vitb16	pointing, 15 px	replaced aggregator	sufficiency vs plain CAM	5,794	exact McNemar	0.5851	0.5851	+0.0000 [−0.0010, +0.0010]	1	tie
cub	vitb16	pointing, strict	relocated normalisation	necessity vs plain CAM	5,794	exact McNemar	0.6545	0.4539	+0.2006 [+0.1871, +0.2138]	5.7e-171	win
cub	vitb16	pointing, strict	relocated normalisation	sufficiency vs plain CAM	5,794	exact McNemar	0.7478	0.4539	+0.2939 [+0.2798, +0.3077]	<1e-300	win
cub	vitb16	pointing, strict	replaced aggregator	necessity vs plain CAM	5,794	exact McNemar	0.4537	0.4539	−0.0002 [−0.0041, +0.0036]	1	tie
cub	vitb16	pointing, strict	replaced aggregator	sufficiency vs plain CAM	5,794	exact McNemar	0.4534	0.4539	−0.0005 [−0.0014, +0.0002]	0.75	tie
voc12	resnet50	pointing, 15 px	replaced aggregator	necessity vs plain CAM	8,856	exact McNemar	0.9163	0.9196	−0.0033 [−0.0053, −0.0014]	0.00496	loss
voc12	resnet50	pointing, 15 px	replaced aggregator	sufficiency vs plain CAM	8,856	exact McNemar	0.9194	0.9196	−0.0002 [−0.0008, +0.0002]	0.688	tie
voc12	resnet50	pointing, strict	replaced aggregator	necessity vs plain CAM	8,856	exact McNemar	0.8398	0.8432	−0.0034 [−0.0054, −0.0012]	0.00576	loss
voc12	resnet50	pointing, strict	replaced aggregator	sufficiency vs plain CAM	8,856	exact McNemar	0.8434	0.8432	+0.0002 [−0.0003, +0.0008]	0.727	tie
voc12	vgg16gap	pointing, 15 px	replaced aggregator	necessity vs plain CAM	8,856	exact McNemar	0.8945	0.8974	−0.0028 [−0.0047, −0.0010]	0.0117	loss
voc12	vgg16gap	pointing, 15 px	replaced aggregator	sufficiency vs plain CAM	8,856	exact McNemar	0.8967	0.8974	−0.0007 [−0.0016, +0.0000]	0.146	tie
voc12	vgg16gap	pointing, strict	replaced aggregator	necessity vs plain CAM	8,856	exact McNemar	0.8093	0.8147	−0.0054 [−0.0077, −0.0033]	3.3e-06	loss
voc12	vgg16gap	pointing, strict	replaced aggregator	sufficiency vs plain CAM	8,856	exact McNemar	0.8145	0.8147	−0.0002 [−0.0010, +0.0006]	0.774	tie
voc12	vitb16	pointing, 15 px	relocated normalisation	necessity vs plain CAM	8,856	exact McNemar	0.8285	0.8390	−0.0105 [−0.0169, −0.0044]	0.00332	loss
voc12	vitb16	pointing, 15 px	relocated normalisation	sufficiency vs plain CAM	8,856	exact McNemar	0.8476	0.8390	+0.0086 [+0.0026, +0.0145]	0.0141	win
voc12	vitb16	pointing, 15 px	replaced aggregator	necessity vs plain CAM	8,856	exact McNemar	0.8234	0.8390	−0.0156 [−0.0190, −0.0123]	9.9e-21	loss
voc12	vitb16	pointing, 15 px	replaced aggregator	sufficiency vs plain CAM	8,856	exact McNemar	0.8385	0.8390	−0.0005 [−0.0014, +0.0005]	0.454	tie
voc12	vitb16	pointing, strict	relocated normalisation	necessity vs plain CAM	8,856	exact McNemar	0.7072	0.7092	−0.0020 [−0.0091, +0.0060]	1	tie
voc12	vitb16	pointing, strict	relocated normalisation	sufficiency vs plain CAM	8,856	exact McNemar	0.7311	0.7092	+0.0219 [+0.0145, +0.0297]	7.3e-08	win
voc12	vitb16	pointing, strict	replaced aggregator	necessity vs plain CAM	8,856	exact McNemar	0.6990	0.7092	−0.0103 [−0.0138, −0.0068]	2.8e-08	loss
voc12	vitb16	pointing, strict	replaced aggregator	sufficiency vs plain CAM	8,856	exact McNemar	0.7093	0.7092	+0.0001 [−0.0007, +0.0009]	1	tie
voc12seg	resnet50	peak-mask IoU	replaced aggregator	necessity vs plain CAM	2,147	Wilcoxon	0.2622	0.4880	−0.2258 [−0.2329, −0.2185]	<1e-300	loss
voc12seg	resnet50	peak-mask IoU	replaced aggregator	sufficiency vs plain CAM	2,147	Wilcoxon	0.4785	0.4880	−0.0095 [−0.0106, −0.0085]	4.3e-117	loss
voc12seg	resnet50	pointing, 15 px	replaced aggregator	necessity vs plain CAM	2,147	exact McNemar	0.9082	0.9106	−0.0023 [−0.0065, +0.0019]	0.81	tie
voc12seg	resnet50	pointing, 15 px	replaced aggregator	sufficiency vs plain CAM	2,147	exact McNemar	0.9106	0.9106	+0.0000 [−0.0014, +0.0014]	1	tie
voc12seg	resnet50	pointing, strict	replaced aggregator	necessity vs plain CAM	2,147	exact McNemar	0.6670	0.6712	−0.0042 [−0.0102, +0.0023]	0.486	tie
voc12seg	resnet50	pointing, strict	replaced aggregator	sufficiency vs plain CAM	2,147	exact McNemar	0.6712	0.6712	+0.0000 [+0.0000, +0.0000]	1	tie
voc12seg	vgg16gap	peak-mask IoU	replaced aggregator	necessity vs plain CAM	2,147	Wilcoxon	0.2603	0.4706	−0.2103 [−0.2179, −0.2028]	<1e-300	loss
voc12seg	vgg16gap	peak-mask IoU	replaced aggregator	sufficiency vs plain CAM	2,147	Wilcoxon	0.4478	0.4706	−0.0228 [−0.0247, −0.0208]	2.6e-129	loss
voc12seg	vgg16gap	pointing, 15 px	replaced aggregator	necessity vs plain CAM	2,147	exact McNemar	0.8887	0.8929	−0.0042 [−0.0084, −0.0005]	0.127	tie
voc12seg	vgg16gap	pointing, 15 px	replaced aggregator	sufficiency vs plain CAM	2,147	exact McNemar	0.8915	0.8929	−0.0014 [−0.0033, +0.0005]	0.375	tie

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Table S2 (continued)

Dataset	Backbone	Metric	Arm	Contrast	n	Test	Separated map	Comparator	diff [95% CI]	P_{Holm}	Verdict
voc12seg	vgg16gap	pointing, strict	replaced aggregator	necessity vs plain CAM	2,147	exact McNemar	0.6642	0.6712	−0.0070 [−0.0140, −0.0005]	0.125	tie
voc12seg	vgg16gap	pointing, strict	replaced aggregator	sufficiency vs plain CAM	2,147	exact McNemar	0.6702	0.6712	−0.0009 [−0.0028, +0.0005]	0.625	tie
voc12seg	vitb16	peak-mask IoU	relocated normalisation	necessity vs plain CAM	2,147	Wilcoxon	0.4261	0.4304	−0.0043 [−0.0071, −0.0017]	6.9e-06	loss
voc12seg	vitb16	peak-mask IoU	relocated normalisation	sufficiency vs plain CAM	2,147	Wilcoxon	0.4375	0.4304	+0.0071 [+0.0033, +0.0109]	0.00181	win
voc12seg	vitb16	peak-mask IoU	replaced aggregator	necessity vs plain CAM	2,147	Wilcoxon	0.2777	0.4304	−0.1527 [−0.1601, −0.1454]	1.8e-268	loss
voc12seg	vitb16	peak-mask IoU	replaced aggregator	sufficiency vs plain CAM	2,147	Wilcoxon	0.4317	0.4304	+0.0013 [+0.0003, +0.0022]	4.4e-05	win
voc12seg	vitb16	pointing, 15 px	relocated normalisation	necessity vs plain CAM	2,147	exact McNemar	0.7857	0.8072	−0.0214 [−0.0349, −0.0079]	0.0103	loss
voc12seg	vitb16	pointing, 15 px	relocated normalisation	sufficiency vs plain CAM	2,147	exact McNemar	0.8272	0.8072	+0.0200 [+0.0070, +0.0331]	0.0103	win
voc12seg	vitb16	pointing, 15 px	replaced aggregator	necessity vs plain CAM	2,147	exact McNemar	0.7885	0.8072	−0.0186 [−0.0261, −0.0116]	1.8e-06	loss
voc12seg	vitb16	pointing, 15 px	replaced aggregator	sufficiency vs plain CAM	2,147	exact McNemar	0.8062	0.8072	−0.0009 [−0.0028, +0.0005]	0.625	tie
voc12seg	vitb16	pointing, strict	relocated normalisation	necessity vs plain CAM	2,147	exact McNemar	0.5268	0.5300	−0.0033 [−0.0210, +0.0144]	1	tie
voc12seg	vitb16	pointing, strict	relocated normalisation	sufficiency vs plain CAM	2,147	exact McNemar	0.5571	0.5300	+0.0270 [+0.0088, +0.0447]	0.0139	win
voc12seg	vitb16	pointing, strict	replaced aggregator	necessity vs plain CAM	2,147	exact McNemar	0.5207	0.5300	−0.0093 [−0.0168, −0.0019]	0.0616	tie
voc12seg	vitb16	pointing, strict	replaced aggregator	sufficiency vs plain CAM	2,147	exact McNemar	0.5291	0.5300	−0.0009 [−0.0023, +0.0000]	1	tie

The necessity map's MaxBoxAccV2 on CUB with ResNet-50 and VGG-16 (0.2509) is the score of a box covering the whole image (Note S2.8), which the metric's threshold sweep always includes. The collapsed map itself, compared with plain CAM by the same tests, with Holm correction over the metrics of each dataset–backbone cell, is significantly weaker in 20 of the 36 dataset–backbone–metric cells (all 12 on ViT-B/16), better in 3 and tied in 13 (Note S2.8); its scores are the comparator columns of the two blocks above.

Table S3. The same 192 primary contrasts, and the 96 against plain CAM, by arm, comparator and backbone, with the size of the wins set against the size of the losses. Each magnitude column gives the median paired difference over the wins (respectively the losses), ties excluded, with the interquartile range in brackets; a single value is given where there is a single win or loss. A tally alone cannot distinguish a design in which every win is small and every loss is large from its mirror image; these columns can. Arms, comparators and backbones are named as in Table S2 (replaced aggregator, the aggregator arm; relocated normalisation, the relocation arm, which exists on ViT-B/16 only). The plain-CAM rows summarise the last block of Table S2.

Arm	Comparator	Backbone	n	Win	Loss	Tie	Median diff, wins	Median diff, losses
relocated normalisation	collapsed	vitb16	24	19	1	4	+0.047 [+0.021, +0.120]	−0.015
relocated normalisation	centre prior	vitb16	24	17	7	0	+0.133 [+0.116, +0.160]	−0.164 [−0.209, −0.117]
relocated normalisation	plain CAM	vitb16	24	13	6	5	+0.058 [+0.022, +0.180]	−0.016 [−0.021, −0.006]
replaced aggregator	collapsed	all	72	20	25	27	+0.017 [+0.012, +0.021]	−0.046 [−0.133, −0.007]
		resnet50	24	3	10	11	+0.012 [+0.009, +0.024]	−0.011 [−0.187, −0.004]
		vgg16gap	24	1	10	13	+0.006	−0.043 [−0.139, −0.011]
		vitb16	24	16	5	3	+0.019 [+0.014, +0.021]	−0.065 [−0.089, −0.046]
replaced aggregator	centre prior	all	72	51	17	4	+0.189 [+0.111, +0.238]	−0.274 [−0.313, −0.077]
		resnet50	24	20	3	1	+0.224 [+0.022, +0.256]	−0.274 [−0.293, −0.166]
		vgg16gap	24	16	5	3	+0.220 [+0.174, +0.242]	−0.185 [−0.276, −0.061]
		vitb16	24	15	9	0	+0.119 [+0.111, +0.159]	−0.294 [−0.402, −0.174]
replaced aggregator	plain CAM	all	72	5	27	40	+0.002 [+0.001, +0.004]	−0.023 [−0.112, −0.010]
		resnet50	24	2	9	13	+0.032 [+0.018, +0.046]	−0.013 [−0.226, −0.008]
		vgg16gap	24	1	10	13	+0.002	−0.043 [−0.115, −0.009]
		vitb16	24	2	8	14	+0.001 [+0.001, +0.001]	−0.037 [−0.078, −0.017]

Restricted to the metrics the localisation protocol names (MaxBoxAccV2 on CUB; the two pointing games and peak-mask IoU on VOC and COCO), the tally against the collapsed map is 28 wins, 16 losses and 28 ties of 72; the 24 contrasts on the further CUB metrics (best IoU and both pointing games) contribute 11 wins, 10 losses and 3 ties (Table S6, deviation (i)).

Table S4. Hit rate against radiologist-drawn boxes on chest radiographs (NIH with the distributed (reused) head), by finding. gt frac is the share of the image the annotated box covers, which is the rate a map that points anywhere would achieve. *Best model-derived map* is the best of the 15 model-derived maps of the designated-in-advance grid (the necessity, sufficiency and collapsed maps at both temperatures and both units, and Grad-CAM under the three readouts: the averaged readout and log-sum-exp pooling at $T = -8$ and $T = +8$), with its hit rate in *Hit*. *Plain CAM* is the class-score map without subtraction of the reference's per-cell logit. *Average box* is the average-annotation box, which is image-independent but built per finding: the mean box mask of the same finding over the patients of the other four of five patient-level folds (cross-fitted), scored at full resolution (224×224), the version behind the overall 0.385 of the main text. The leave-one-patient-out average box at full resolution (overall 0.397; 0.371 when pooled to the maps' 7×7 resolution, reported in Note S2.9) gives the same per-finding values except on Atelectasis (0.189), Effusion (0.333), Nodule (0.013) and Pneumonia (0.417); these per-finding values are at full resolution. The centre prior is class-independent and image-independent, and *Random* is a random map. A map is named by its short key: N and S are the necessity and the sufficiency estimate; collapsed and collapsedS are the same two on the averaged readout, which is degenerate, so that collapsedS equals collapsed; lse-8.0 and lse8.0 denote the pooling temperatures $T = -8$ and $T = +8$; and 1x1 and 3x3 are the unit the map is read at.

Finding	n	gt frac	Best model-derived map	Hit	Plain CAM	Average box	Centre prior	Random
Atelectasis	180	0.032	N_lse-8.0_1x1	0.122	0.094	0.178	0.056	0.011
Cardiomegaly	146	0.176	N_lse-8.0_1x1	0.911	0.699	1.000	0.973	0.164
Effusion	153	0.069	S_lse-8.0_1x1	0.105	0.065	0.275	0.026	0.039
Infiltration	123	0.097	N_lse-8.0_1x1	0.358	0.260	0.455	0.228	0.089
Mass	85	0.042	collapsed_1x1 (= collapsedS)	0.259	0.176	0.282	0.106	0.047
Nodule	79	0.005	N_lse-8.0_1x1	0.038	0.025	0.051	0.000	0.013
Pneumonia	120	0.094	N_lse-8.0_3x3	0.283	0.017	0.383	0.142	0.083
Pneumothorax	98	0.052	N_lse8.0_1x1	0.071	0.071	0.296	0.010	0.041

The cross-fitted average box has the highest hit rate in every row, on all 8 findings. The centre prior is above every model-derived map on 1 of the 8 findings, Cardiomegaly (0.973 against 0.911), where the heart occupies the middle of the frame; the average box reaches 1.000 there. Plain CAM is never above the best model-derived map and equals it on Pneumothorax. On Effusion and Pneumothorax the best model-derived map (0.105, 0.071) stays within a factor of two of the hit rate expected from a uniformly random point, which is the box area fraction in the third column (0.069, 0.052). Every number here is a point estimate on the sample size in its own row; the paired tests in the main text run over all 984 boxes together.

Table S5. Correlation between the necessity and sufficiency estimates under twelve reference settings (eleven distinct references; the widest interpolation coincides with the bank), for the ordering that places the normalisation after the spatial mean. *Interpolation, s* uses the reference $b_s = a + s\,(b - a)$, where a is the probed input's activation and b the bank reference, so small s means a reference close to the input. The sweep uses its own evaluation set and reference bank, so its bank row differs from the value in Table S15; the table shows how the correlation depends on the reference. Along the interpolation series the correlation rises toward 1 as s decreases, monotonically except for one reversal of 0.015 (ConvNeXt-T, $s = 1$ to 0.5); across the other kinds of reference the ordering by distance does not hold. Correlations reported for distributed models are therefore properties of their tails at this reference scale. For the model marked †, which ships the normalisation inside the mean and is degenerate as distributed, the column is the relocated model.

Reference	convnext_tiny.fb_in1k	swin_tiny_patch4_window7_224.ms_in1k†	vit_betwixt_patch16_reg4_gap_256.sbb_in1k
interpolation, $s = 1.0$	0.8960	0.8439	0.5355
interpolation, $s = 0.5$	0.8808	0.9642	0.7884
interpolation, $s = 0.25$	0.9451	0.9918	0.9294
interpolation, $s = 0.1$	0.9901	0.9988	0.9872
interpolation, $s = 0.05$	0.9976	0.9997	0.9967
interpolation, $s = 0.02$	0.9996	1.0000	0.9995
interpolation, $s = 0.01$	0.9999	1.0000	0.9999
bank of sixteen, per image (this experiment's own reference)	0.8960	0.8439	0.5355
mean activation over the dataset	0.8881	0.9232	0.7257
Gaussian noise	0.8964	0.9430	0.7163
spatial shuffle of the probed input	0.9840	0.9998	0.9995
all-zero activation	0.8402	0.9695	0.7338

The spatial shuffle of the input, whose pooled vector equals the probed input's, is excluded from the main analysis because it is off the data manifold, not because it degenerates: it is well behaved, and it is the reference least favourable to the reading of Table S15, which is why it is reported here. The same experiment also evaluates the degenerate ordering of each model (the normalisation inside the mean) under three of these references, the all-zero activation, the mean activation over the dataset and Gaussian noise; each of those returns 1.000000, as it must by construction.

Table S6. Status of every analysis reported. *Designated in advance*: the contrast, the map, the operator and temperature, the metric and the test were fixed in writing before the outcome was computed (Note S1.11; see the note on timing below the table). *Primary*: the 192 localisation contrasts of the primary set; 72 of the 96 against the collapsed map and 72 of the 96 against the centre prior are in the written protocol, and best IoU and both pointing games on CUB were added (deviation (i)). *Post hoc*: designed or run after related outcomes were seen. *Exploratory*: run while the phenomenon was being characterised, with no contrast, test or decision rule fixed in advance; its numbers describe and are not confirmatory tests. *Development/held-out split by family*: the structural check was written on the development architectures and then run unchanged on the held-out architectures; both sets were drawn from the same library family by family, so a held-out architecture can have a sibling in the development set (Table S1), and agreement on held-out head traces absent from development is reported separately (Note S2.6).

Analysis	Status	Reported in
Closed-form recomputation of the log-sum-exp necessity and sufficiency maps compared by rank or shown cell by cell, and the estimator checks	post hoc	Eq. (2) of the main text; Figs. S1(c), S2(a,b), S3(b,c); Notes S1.2, S2.2, S2.3 (faithfulness), S2.4
Rank correlation of reference-averaged maps, and of Grad-CAM with them	post hoc	Notes S1.7.2, S2.7, S2.8
Operator sweeps (log-sum-exp, k-of-N, learnable aggregator)	exploratory	Notes S2.2–S2.3
Softmax contrasts, locus by region size and reference	post hoc	Note S2.2
Rank separation of the readouts	post hoc	Note S2.2
Faithfulness: random-ordering baseline across temperatures	post hoc	Note S2.3
Learnable aggregator: all 45 runs	exploratory	Note S2.3
1,000-class experiment with the shipped classifier	exploratory	Note S2.4
Normalisation edit, ten models, costs	exploratory	Note S2.5, Table S15
Exact McNemar tests of the edit costs	post hoc	Table S15
Reconciliation with the self-supervised ablation	post hoc	Note S2.5
Rank correlation of the normalisation lever	post hoc	Note S2.5
Structural check on the 82 held-out architectures	development/held-out split by family	Note S2.6
Ablation of the check (tail predicate only)	post hoc	code-and-data archive (not reported in the text)
Trace novelty, nearest-trace and majority baselines	post hoc	Note S2.6, Fig. S5
Independent identification of normalisations for the labels	post hoc	Note S2.6
Census over 895 timm architectures (audited census; the measured census of Note S2.6 was designated in advance)	post hoc	Note S2.6, Table 2 of the main text
Verdict on random weights against the verdict on trained weights (the 144 of the 148 checked architectures that belong to the census population)	post hoc	Note S2.6
Operator-level variant of the check (a post hoc variant added after the VOLO-D1 disagreement; not part of the check; the audited census rests on the VOLO-D1 measurement), and the census verdicts it changes	post hoc	Note S2.6
Accuracy cost of the 21 costed nonlinearity-blocked architectures (cheapest edit that makes the readout degenerate)	post hoc	Note S2.6, Fig. S5(c)
Audit of the census architectures never measured, and the corrected census	post hoc	Note S2.6, Table 2 of the main text, Table S12
Structural check on 32 constructed heads (error envelope)	post hoc	Note S2.6, Table S11
Grad-CAM weight under shift-equivariant pooling (float64)	exploratory; extended post hoc	Note S2.7
Localisation: 192 primary contrasts at $T = +1$ (metric set differs from the protocol; deviation (i))	primary contrasts; 72 of each 96 in the written protocol	Note S2.8, Fig. S6, Table S2
Localisation: PxAP on VOC-seg and COCO	designated-in-advance metric; secondary	Table S8
Localisation: tally restricted to the protocol's metrics	post hoc	Note S2.8; note to Table S3
Localisation: effect sizes by arm	post hoc	Table S3
Localisation: pre-normalisation control and its paired tests	post hoc	Note S2.8, Fig. S6
Localisation: other temperatures	post hoc	Note S2.8
Localisation: image-level re-analysis	post hoc	Note S2.8
Localisation: plain CAM as comparator	post hoc	Note S2.8, Fig. S6, Tables S2–S3
Localisation: separation of the aggregator arm against its net wins over the twelve dataset–backbone cells	post hoc	Note S2.8
Localisation: refitted log-sum-exp head (ResNet-50)	post hoc	Note S2.8
Localisation: within-model necessity versus sufficiency	post hoc	Note S2.8, Table S9
Localisation: token control at the 3×3 unit	post hoc	Note S2.8
Localisation: participation ratio of the separated maps	post hoc	Note S2.8

Table S6 (continued)

Analysis	Status	Reported in
Localisation: separated maps against activation-magnitude and random controls; dataset-mean reference	required by the protocol, analysed post hoc (deviation (iv))	Note S2.8, Table S10
Medical: three primaries (necessity map, $T = +8$, 1×1; RSNA and ISIC fixed in script headers, deviation (ii))	designated in advance	Note S2.9, Fig. S6, Table S7
Medical: the remaining cells of the grid	designated in advance (per-cell, uncorrected)	Note S2.9, Table S7
Medical: NIH refitted-head arm	post hoc	Note S2.9, Fig. S6, Table S7
Medical: arm with the NIH model trained in this study	post hoc	Note S2.9, Fig. S6, Table S7
Medical: activation-magnitude baseline and its paired test	post hoc	Note S2.9, Fig. S6
Medical: Holm correction over the 48 H1 cells	post hoc	Note S2.9, Table S7
Medical: disjunctive sufficiency against the collapsed map	post hoc	Note S2.9, Table S7
Medical: second training runs of the refitted heads, the NIH model trained in this study and ISIC	post hoc	Note S2.9, Table S7
Medical: cross-fitted disjoint RSNA models	post hoc	Note S2.9, Fig. S6, Table S7
Medical: training-overlap subsets (164 NIH images; 283 RSNA studies)	post hoc	Note S2.9
Medical: average-box baseline	post hoc	Note S2.9, Fig. S6, Table S4
Medical: per-finding plain CAM and average box	post hoc	Table S4
Medical: preprocessing check	post hoc	Note S2.9
Medical: H2 summary over the 32 designated-in-advance cells	designated in advance (per-cell, uncorrected)	Note S2.9, Table S7
Medical: conjunctive sufficiency against the collapsed map in every arm	post hoc	Note S2.9, Fig. S6, Table S13
Medical: closed-form recomputation of the RSNA primary, and of the ISIC primary on a retrained model	post hoc	Note S2.9
Medical: image-level cluster bootstrap of the NIH primary; Grad-CAM with all-zero maps rescored	post hoc	Note S2.9
Conservative check on the constructed heads, the 147 architectures, the census and torchvision	designated in advance (recorded protocol); in-sample except on torchvision	Note S2.6, Table S16
Post hoc variant of the conservative check (two operator-level conditions)	post hoc	Note S2.6, Table S16
Measured census (float64 labels of the 895 entries at random weights); unit-scale label where the two labels differ	designated in advance (recorded protocol); unit-scale rule post hoc	Note S2.6, Table 2 of the main text
Pooling each channel before the linear layer	designated in advance (recorded protocol)	Note S2.7, Table S17
Heads trained end to end: six primary contrasts	designated in advance (recorded protocol)	Note S2.8, Table S18
Heads trained end to end: each head's own maps, the pairing with the sufficiency map, within-image ranks	post hoc	Note S2.8
Three retrained ISIC classifiers (seeds 0–2)	designated in advance (recorded protocol)	Note S2.9, Table S19
Reference sensitivity (Table S5)	post hoc	Note S2.10, Table S5

Deviations from the protocol documents. (i) *Localisation metrics.* The localisation protocol names MaxBoxAccV2 for CUB, and for VOC and COCO the pointing game (strict and with a 15-px tolerance) plus PxAP and peak-mask IoU where masks exist. The 192 reported contrasts add best IoU and both pointing games on CUB (24 of the 96 contrasts against the collapsed map, and 24 of the 96 against the centre prior) and omit PxAP, which is reported as a secondary outcome in Table S8. Restricted to the protocol's metrics, the tally against the collapsed map is 28 wins, 16 losses and 28 ties of 72. (ii) *Medical coverage.* The medical protocol document covers NIH only. The RSNA primary (pointing hit rate, 1×1, $T = +8$, with the NIH comparators) is fixed in the header of the RSNA analysis script, and the ISIC primary metric (Dice at a cross-tuned threshold, with a lesion mask as ground truth) in the header of the ISIC analysis script, which states that it departs from the NIH pre-specification in these two respects; neither is in the protocol document. (iii) *Medical hypothesis.* The medical protocol states the hypothesis as the necessity map beating both the collapsed map and the sufficiency map (H1 and H2) and lists 1×1 and 3×3 units as both reported; the single primary contrast (H1, 1×1, $T = +8$) was named only in the analysis script. The protocol's choice of $T = +8$ and its motivation rested on float32 sweeps whose apparent difference in cell ordering between the two maps is a cancellation artefact (Note S1.7.2). (iv) *Localisation rows.* The localisation protocol asks for every map at a 3×3 perturbation scale, for comparisons with the activation-magnitude and random controls, and for a dataset-mean reference as a robustness row. These rows were analysed after the primary outcome and are in Table S10 (the mean reference was run for ResNet-50 on CUB and VOC masks only). The 3×3 log-sum-exp maps were computed with a block summation of exponentials whose accuracy was characterised only on a stand-in (the shipped ResNet-50 head on 7×7 ImageNet activations at $T = +1$: the maximum located as by the closed form in 223 of 224 necessity and 224 of 224 sufficiency maps, largest normalised error 0.30 and 0.94); those

rows are indicative and were not recomputed. *Note on timing.* The protocol documents carry no time stamp, so their timing cannot be verified independently (Note S1.11); the same holds for the source record of the structural check (Note S1.6).

Table S7. Every cell of the medical grids. Each arm's header gives its status (Table S6), the metrics it carries and the number of scored items: NIH with the distributed (reused) head carries hit, AP and IoU; NIH with the refitted head, the NIH model trained in this study, the distributed RSNA model and the disjoint RSNA models carry hit and AP; ISIC carries hit, AP and Dice. *N vs collapsed* is H1, the necessity map minus the collapsed map; *N vs S* is H2, the necessity map minus the sufficiency map at the same temperature and unit; *S vs collapsed* is the sufficiency map at $T = +8$ (the disjunctive sufficiency map) minus the collapsed map, a contrast that is not part of any designated-in-advance grid and was added for every arm. T is the log-sum-exp temperature and *Unit* the cell block perturbed. Hit is the pointing hit rate, AP the pixel average precision of one item, IoU and Dice are taken at a two-fold cross-tuned top-q threshold. diff is the first map's mean minus the second's with a 95% paired bootstrap interval (10,000 resamples); P is exact McNemar for hit and Wilcoxon signed-rank otherwise, uncorrected; for the hit cells of the first ISIC run, whose per-image outcomes were not stored, the exact McNemar P is computed from the discordant counts, which the stored signed-rank P and mean difference of a binary outcome determine. P_{Holm} is Holm-adjusted over the 48 H1 cells of the five arms of the first runs (bold where below 0.05); the disjoint RSNA models are outside that family. A test can reject where the bootstrap interval of the mean difference spans zero, and the reverse; † marks every cell in which the bootstrap interval and the uncorrected test disagree. *Src* gives where the value comes from: O, the designated-in-advance grid of the first run; M, the stored maps of the first run; R, maps regenerated from the same distributed RSNA weights (2 of 1,385 items differ, from ties in the arg-max); P, a second, independent training run that refitted the NIH heads and retrained the NIH model of this study and the ISIC model; D, the single run of the cross-fitted disjoint models. *Replicate* is the second run's difference for the same cell where one exists. The conjunctive ($T = -8$) necessity cells move most on retraining (ISIC 1×1: hit −0.219 to −0.322, Dice +0.002 to −0.038; the NIH model of this study 1×1 hit +0.005 to −0.043). *Counts.* Of the 48 H1 cells, 6 favour the necessity map (interval above zero), all at $T = -8$, and 4 of these survive Holm; all 24 cells at $T = +8$ lie below zero; the 24 conjunctive cells at $T = -8$ are 6 above zero, 13 spanning zero and 5 below. The three designated-in-advance arms carry 32 of the 48 H1 cells, of which 3 favour the necessity map and 1 survives Holm (RSNA 1×1 hit); the two post hoc arms carry 16, of which 3 favour it and all 3 survive (NIH refitted head). The sufficiency-versus-collapsed cells lie below zero in 25 of 28; the three exceptions are the ISIC 1×1 cells (hit and AP spanning zero, Dice above).

Contrast	T	Unit	Metric	diff [95% CI]	P	P_{Holm}	Src	Replicate
NIH, distributed (reused) head (designated in advance; hit, AP, IoU; n = 984)								
N vs collapsed	+8	1×1	hit	−0.090 [−0.116, −0.066]	3.3e-13	**9e-12**	O	
N vs collapsed	+8	1×1	AP	−0.106 [−0.120, −0.093]	4e-61	**1.8e-59**	O	
N vs collapsed	+8	1×1	IoU	−0.096 [−0.107, −0.086]	4.2e-77	**1.9e-75**	O	
N vs collapsed	+8	3×3	hit	−0.076 [−0.100, −0.054]	6e-11	**1.5e-09**	O	
N vs collapsed	+8	3×3	AP	−0.072 [−0.084, −0.061]	1e-43	**3.7e-42**	O	
N vs collapsed	+8	3×3	IoU	−0.063 [−0.072, −0.055]	6e-48	**2.3e-46**	O	
N vs collapsed	−8	1×1	hit	+0.005 [−0.017, +0.027]	0.72	1	O	
N vs collapsed	−8	1×1	AP	+0.004 [−0.006, +0.015]	0.12	1	O	
N vs collapsed	−8	1×1	IoU	+0.008 [+0.002, +0.014]†	0.087	0.96	O	
N vs collapsed	−8	3×3	hit	−0.002 [−0.023, +0.019]	0.92	1	O	
N vs collapsed	−8	3×3	AP	+0.001 [−0.009, +0.011]	0.11	1	O	
N vs collapsed	−8	3×3	IoU	+0.007 [+0.002, +0.012]	0.021	0.27	O	
N vs S	+8	1×1	hit	−0.050 [−0.065, −0.036]	5.9e-12		O	
N vs S	+8	1×1	AP	−0.074 [−0.085, −0.063]	5.9e-43		O	
N vs S	+8	1×1	IoU	−0.066 [−0.075, −0.057]	1.1e-49		O	
N vs S	+8	3×3	hit	+0.037 [+0.015, +0.058]	0.00096		O	
N vs S	+8	3×3	AP	−0.016 [−0.023, −0.010]	1.2e-16		O	
N vs S	+8	3×3	IoU	−0.026 [−0.031, −0.021]	4.1e-23		O	
N vs S	−8	1×1	hit	+0.048 [+0.026, +0.069]	1.7e-05		O	
N vs S	−8	1×1	AP	+0.069 [+0.060, +0.078]	7.6e-61		O	
N vs S	−8	1×1	IoU	+0.055 [+0.049, +0.060]	3.8e-70		O	
N vs S	−8	3×3	hit	+0.010 [−0.008, +0.028]	0.31		O	
N vs S	−8	3×3	AP	+0.029 [+0.022, +0.035]	7.9e-38		O	
N vs S	−8	3×3	IoU	+0.026 [+0.022, +0.030]	1e-34		O	
S vs collapsed	+8	1×1	hit	−0.041 [−0.061, −0.020]	0.00015		M	
S vs collapsed	+8	1×1	AP	−0.032 [−0.041, −0.023]	3.5e-13		M	
S vs collapsed	+8	1×1	IoU	−0.030 [−0.037, −0.025]	2.8e-20		M	
S vs collapsed	+8	3×3	hit	−0.113 [−0.140, −0.086]	3.4e-16		M	
S vs collapsed	+8	3×3	AP	−0.056 [−0.067, −0.045]	3.2e-25		M	
S vs collapsed	+8	3×3	IoU	−0.037 [−0.044, −0.031]	2.1e-23		M	
RSNA, distributed model (designated in advance; hit, AP; n = 1,385)								
N vs collapsed	+8	1×1	hit	−0.226 [−0.253, −0.200]	5.5e-59	**2.3e-57**	O	
N vs collapsed	+8	1×1	AP	−0.206 [−0.216, −0.195]	3.5e-187	**1.7e-185**	O	
N vs collapsed	+8	3×3	hit	−0.102 [−0.127, −0.077]	1.2e-14	**3.3e-13**	O	
N vs collapsed	+8	3×3	AP	−0.151 [−0.160, −0.141]	2.3e-162	**1.1e-160**	O	
N vs collapsed	−8	1×1	hit	+0.042 [+0.015, +0.069]	0.0029	**0.047**	O	
N vs collapsed	−8	1×1	AP	+0.002 [−0.009, +0.014]†	0.003	**0.047**	O	
N vs collapsed	−8	3×3	hit	−0.211 [−0.239, −0.184]	8.3e-49	**3.2e-47**	O	
N vs collapsed	−8	3×3	AP	−0.049 [−0.059, −0.039]	2.2e-22	**7.2e-21**	O	
N vs S	+8	1×1	hit	+0.001 [−0.007, +0.010]	0.87		O	
N vs S	+8	1×1	AP	−0.067 [−0.072, −0.061]	1.8e-131		O	
N vs S	+8	3×3	hit	+0.084 [+0.058, +0.110]	5.5e-10		O	
N vs S	+8	3×3	AP	−0.014 [−0.019, −0.009]	2.7e-24		O	
N vs S	−8	1×1	hit	−0.002 [−0.022, +0.017]	0.88		O	
N vs S	−8	1×1	AP	+0.067 [+0.057, +0.077]	3.2e-65		O	
N vs S	−8	3×3	hit	+0.011 [−0.006, +0.027]	0.24		O	
N vs S	−8	3×3	AP	+0.036 [+0.029, +0.043]	2.5e-45		O	
S vs collapsed	+8	1×1	hit	−0.227 [−0.253, −0.200]	2.5e-60		R	
S vs collapsed	+8	1×1	AP	−0.139 [−0.148, −0.130]	4.5e-166		R	
S vs collapsed	+8	3×3	hit	−0.184 [−0.216, −0.153]	5e-29		R	
S vs collapsed	+8	3×3	AP	−0.137 [−0.146, −0.127]	7.5e-144		R	
ISIC (designated in advance; hit, AP, Dice; n = 735)								
N vs collapsed	+8	1×1	hit	−0.044 [−0.065, −0.022]	1.0e-04	**0.0017**	O	−0.042

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Table S7 (continued)

Contrast	T	Unit	Metric	diff [95% CI]	P	P_{Holm}	Src	Replicate
N vs collapsed	+8	1×1	AP	−0.168 [−0.182, −0.154]	4.9e-80	**2.2e-78**	O	−0.154
N vs collapsed	+8	1×1	Dice	−0.136 [−0.154, −0.117]	3.3e-36	**1.2e-34**	O	−0.120
N vs collapsed	+8	3×3	hit	−0.095 [−0.124, −0.067]	1.9e-10	**4.5e-09**	O	−0.088
N vs collapsed	+8	3×3	AP	−0.081 [−0.092, −0.070]	3.7e-57	**1.5e-55**	O	−0.091
N vs collapsed	+8	3×3	Dice	−0.049 [−0.057, −0.040]	2.6e-23	**8.7e-22**	O	−0.058
N vs collapsed	−8	1×1	hit	−0.219 [−0.257, −0.182]	6.2e-29	**2.2e-27**	O	−0.322
N vs collapsed	−8	1×1	AP	−0.093 [−0.113, −0.073]	1.5e-16	**4.2e-15**	O	−0.186
N vs collapsed	−8	1×1	Dice	+0.002 [−0.008, +0.013]	0.21	1	O	−0.038
N vs collapsed	−8	3×3	hit	+0.026 [−0.007, +0.059]	0.14	1	O	−0.073
N vs collapsed	−8	3×3	AP	−0.014 [−0.031, +0.004]†	0.027	0.33	O	−0.096
N vs collapsed	−8	3×3	Dice	−0.012 [−0.022, −0.002]	5.7e-06	**1.0e-04**	O	−0.053
N vs S	+8	1×1	hit	−0.012 [−0.024, +0.000]	0.078		O	−0.024
N vs S	+8	1×1	AP	−0.157 [−0.169, −0.144]	3e-81		O	−0.161
N vs S	+8	1×1	Dice	−0.134 [−0.154, −0.114]	3.4e-32		O	−0.130
N vs S	+8	3×3	hit	+0.124 [+0.088, +0.159]	1.9e-11		O	+0.188
N vs S	+8	3×3	AP	+0.017 [+0.006, +0.028]†	0.47		O	+0.017
N vs S	+8	3×3	Dice	−0.021 [−0.028, −0.015]	4.9e-10		O	−0.022
N vs S	−8	1×1	hit	+0.042 [+0.015, +0.069]	0.0032		O	−0.031
N vs S	−8	1×1	AP	+0.217 [+0.206, +0.229]	6.2e-116		O	+0.119
N vs S	−8	1×1	Dice	+0.104 [+0.094, +0.114]	1.9e-62		O	+0.108
N vs S	−8	3×3	hit	+0.019 [−0.008, +0.045]	0.2		O	+0.188
N vs S	−8	3×3	AP	+0.018 [+0.009, +0.027]	1.8e-10		O	+0.099
N vs S	−8	3×3	Dice	+0.019 [+0.013, +0.025]	3.4e-05		O	+0.006
S vs collapsed	+8	1×1	hit	−0.018 [−0.039, +0.003]	0.14		P	
S vs collapsed	+8	1×1	AP	+0.006 [−0.002, +0.014]†	8e-08		P	
S vs collapsed	+8	1×1	Dice	+0.010 [+0.003, +0.016]	3.1e-05		P	
S vs collapsed	+8	3×3	hit	−0.276 [−0.312, −0.241]	3.6e-46		P	
S vs collapsed	+8	3×3	AP	−0.108 [−0.121, −0.095]	5.3e-57		P	
S vs collapsed	+8	3×3	Dice	−0.036 [−0.044, −0.029]	1.3e-23		P	
NIH, refitted head (post hoc; hit, AP; n = 984)								
N vs collapsed	+8	1×1	hit	−0.065 [−0.090, −0.041]	3.8e-07	**8.3e-06**	O	−0.064
N vs collapsed	+8	1×1	AP	−0.070 [−0.083, −0.057]	1.2e-23	**4e-22**	O	−0.070
N vs collapsed	+8	3×3	hit	−0.087 [−0.114, −0.062]	4.7e-11	**1.2e-09**	O	−0.088
N vs collapsed	+8	3×3	AP	−0.061 [−0.073, −0.050]	5.2e-20	**1.6e-18**	O	−0.061
N vs collapsed	−8	1×1	hit	−0.003 [−0.027, +0.021]	0.87	1	O	−0.003
N vs collapsed	−8	1×1	AP	+0.030 [+0.018, +0.041]	3.7e-06	**7.3e-05**	O	+0.030
N vs collapsed	−8	3×3	hit	+0.063 [+0.038, +0.089]	2.3e-06	**4.7e-05**	O	+0.064
N vs collapsed	−8	3×3	AP	+0.045 [+0.032, +0.058]	4.2e-06	**7.9e-05**	O	+0.045
N vs S	+8	1×1	hit	−0.030 [−0.043, −0.019]	6.9e-08		O	−0.030
N vs S	+8	1×1	AP	−0.042 [−0.048, −0.035]	9.5e-36		O	−0.042
N vs S	+8	3×3	hit	−0.009 [−0.028, +0.010]	0.42		O	−0.010
N vs S	+8	3×3	AP	−0.018 [−0.023, −0.013]	1.2e-21		O	−0.018
N vs S	−8	1×1	hit	−0.013 [−0.028, +0.002]	0.12		O	−0.013
N vs S	−8	1×1	AP	+0.056 [+0.048, +0.064]	9.2e-60		O	+0.056
N vs S	−8	3×3	hit	+0.013 [−0.004, +0.030]	0.17		O	+0.014
N vs S	−8	3×3	AP	+0.022 [+0.016, +0.027]	7.9e-25		O	+0.022
S vs collapsed	+8	1×1	hit	−0.034 [−0.057, −0.010]	0.0064		P	
S vs collapsed	+8	1×1	AP	−0.028 [−0.039, −0.018]†	0.0504		P	
S vs collapsed	+8	3×3	hit	−0.078 [−0.105, −0.052]	2.1e-08		P	
S vs collapsed	+8	3×3	AP	−0.043 [−0.055, −0.032]	4.2e-08		P	
NIH, model trained in this study (post hoc; hit, AP; n = 984)								
N vs collapsed	+8	1×1	hit	−0.055 [−0.075, −0.036]	7.7e-08	**1.8e-06**	O	−0.057
N vs collapsed	+8	1×1	AP	−0.089 [−0.102, −0.077]	1.1e-50	**4.3e-49**	O	−0.075
N vs collapsed	+8	3×3	hit	−0.098 [−0.119, −0.076]	2.8e-19	**8.4e-18**	O	−0.051
N vs collapsed	+8	3×3	AP	−0.082 [−0.094, −0.071]	8e-52	**3.3e-50**	O	−0.049
N vs collapsed	−8	1×1	hit	+0.005 [−0.019, +0.029]	0.75	1	O	−0.043
N vs collapsed	−8	1×1	AP	+0.000 [−0.012, +0.012]	0.17	1	O	−0.022
N vs collapsed	−8	3×3	hit	−0.001 [−0.027, +0.025]	1	1	O	−0.095
N vs collapsed	−8	3×3	AP	−0.003 [−0.016, +0.010]†	0.0076	0.11	O	−0.022
N vs S	+8	1×1	hit	−0.047 [−0.062, −0.033]	1.2e-10		O	−0.032
N vs S	+8	1×1	AP	−0.078 [−0.090, −0.066]	1.8e-49		O	−0.057
N vs S	+8	3×3	hit	+0.035 [+0.014, +0.056]	0.0015		O	+0.051
N vs S	+8	3×3	AP	−0.017 [−0.022, −0.011]	1.5e-21		O	−0.007
N vs S	−8	1×1	hit	+0.098 [+0.072, +0.123]	6.5e-14		O	+0.017
N vs S	−8	1×1	AP	+0.040 [+0.032, +0.049]	4.2e-15		O	+0.034
N vs S	−8	3×3	hit	−0.026 [−0.047, −0.006]	0.014		O	−0.032
N vs S	−8	3×3	AP	+0.006 [−0.001, +0.014]†	2e-09		O	+0.013
S vs collapsed	+8	1×1	hit	−0.025 [−0.044, −0.007]	0.0088		P	
S vs collapsed	+8	1×1	AP	−0.019 [−0.026, −0.012]	2.7e-08		P	
S vs collapsed	+8	3×3	hit	−0.102 [−0.128, −0.075]	2.5e-13		P	
S vs collapsed	+8	3×3	AP	−0.042 [−0.052, −0.032]	1e-19		P	
RSNA, disjoint models (post hoc; hit, AP; n = 1,385)								
N vs collapsed	+8	1×1	hit	−0.091 [−0.113, −0.069]	3.4e-16		D	
N vs collapsed	+8	1×1	AP	−0.073 [−0.080, −0.064]	1.5e-82		D	
N vs collapsed	+8	3×3	hit	−0.144 [−0.170, −0.120]	3.3e-29		D	
N vs collapsed	+8	3×3	AP	−0.119 [−0.129, −0.109]	6e-119		D	
N vs collapsed	−8	1×1	hit	−0.012 [−0.037, +0.014]	0.4		D	
N vs collapsed	−8	1×1	AP	+0.048 [+0.039, +0.056]	6.5e-42		D	
N vs collapsed	−8	3×3	hit	−0.008 [−0.035, +0.019]	0.59		D	
N vs collapsed	−8	3×3	AP	+0.020 [+0.013, +0.028]	2.5e-14		D	
N vs S	+8	1×1	hit	+0.007 [+0.001, +0.014]	0.031		D	
N vs S	+8	1×1	AP	+0.019 [+0.015, +0.023]	2.4e-35		D	
N vs S	+8	3×3	hit	+0.023 [+0.008, +0.038]	0.0048		D	
N vs S	+8	3×3	AP	+0.006 [+0.003, +0.009]	2.1e-06		D	
N vs S	−8	1×1	hit	−0.093 [−0.116, −0.070]	7.5e-15		D	
N vs S	−8	1×1	AP	+0.047 [+0.041, +0.053]	4.4e-80		D	
N vs S	−8	3×3	hit	+0.141 [+0.118, +0.164]	1.5e-31		D	
N vs S	−8	3×3	AP	+0.035 [+0.030, +0.040]	4.7e-67		D	
S vs collapsed	+8	1×1	hit	−0.098 [−0.119, −0.077]	2.9e-19		D	

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Table S7 (continued)

Contrast	T	Unit	Metric	diff [95% CI]	P	P_{Holm}	Src	Replicate
S vs collapsed	+8	1×1	AP	−0.091 [−0.099, −0.084]	1.9e-136		D	
S vs collapsed	+8	3×3	hit	−0.168 [−0.193, −0.143]	9.5e-39		D	
S vs collapsed	+8	3×3	AP	−0.125 [−0.134, −0.116]	7e-142		D	

Table S8. Pixel average precision (PxAP) on the benchmarks with masks, a secondary outcome named in the localisation protocol. The localisation protocol names PxAP for VOC masks and COCO; it is not among the 192 primary contrasts (Table S6, deviation (i)) and is reported here, one line per dataset and backbone (identifiers as in Table S2). PxAP pools every pixel of the evaluation split over the threshold sweep, so each value is a single number for the split, without a paired interval or test. The first three columns give the PxAP of the collapsed map, plain CAM and the centre prior; the last four give each separated map minus the collapsed map (N, necessity; S, sufficiency; aggregator, the replaced aggregator; relocation, the relocated normalisation, ViT-B/16 only). Every separated map scores below the collapsed map except the sufficiency map of the replaced aggregator on VOC masks with ViT-B/16 (+0.030), and this includes both maps of the relocated-normalisation arm.

		PxAP			Separated map − collapsed			
Dataset	Backbone	Collapsed	Plain CAM	Centre prior	N, aggregator	S, aggregator	N, relocation	S, relocation
voc12seg	resnet50	0.464	0.487	0.321	−0.287	−0.013		
voc12seg	vgg16gap	0.375	0.387	0.321	−0.191	−0.054		
voc12seg	vitb16	0.425	0.443	0.321	−0.262	+0.030	−0.026	−0.027
coco17	resnet50	0.241	0.249	0.156	−0.131	−0.068		
coco17	vgg16gap	0.195	0.201	0.156	−0.086	−0.059		
coco17	vitb16	0.238	0.246	0.156	−0.134	−0.023	−0.011	−0.020

Table S9. Within-model necessity versus sufficiency contrasts on the localisation benchmarks (post hoc).

Dataset	Backbone	Arm	Metric	N	S	N − S [95% CI]	P_{Holm}	Verdict
CUB	ResNet-50	aggregator	MaxBoxAccV2	0.2509	0.5645	−0.3136 [−0.3232, −0.3054]	0.005	S better
CUB	ResNet-50	aggregator	best IoU	0.3516	0.6466	−0.2950 [−0.2995, −0.2906]	$< 10^{-300}$	S better
CUB	ResNet-50	aggregator	pointing, strict	0.9581	0.9600	−0.0019 [−0.0035, −0.0003]	0.036	S better
CUB	ResNet-50	aggregator	pointing, 15 px	0.9924	0.9934	−0.0010 [−0.0019, −0.0003]	0.062	no difference†
CUB	VGG-16	aggregator	MaxBoxAccV2	0.2509	0.3786	−0.1277 [−0.1374, −0.1182]	0.005	S better
CUB	VGG-16	aggregator	best IoU	0.3496	0.5650	−0.2154 [−0.2204, −0.2105]	$< 10^{-300}$	S better
CUB	VGG-16	aggregator	pointing, strict	0.9767	0.9786	−0.0019 [−0.0031, −0.0007]	0.022	S better
CUB	VGG-16	aggregator	pointing, 15 px	0.9865	0.9877	−0.0012 [−0.0022, −0.0003]	0.117	no difference†
CUB	ViT-B/16	aggregator	MaxBoxAccV2	0.2691	0.3235	−0.0544 [−0.0621, −0.0495]	0.010	S better
CUB	ViT-B/16	relocation	MaxBoxAccV2	0.3996	0.4769	−0.0774 [−0.0855, −0.0706]	0.010	S better
CUB	ViT-B/16	aggregator	best IoU	0.4511	0.5476	−0.0965 [−0.0995, −0.0934]	$< 10^{-300}$	S better
CUB	ViT-B/16	relocation	best IoU	0.5970	0.6376	−0.0406 [−0.0437, −0.0373]	4.4×10^{-118}	S better
CUB	ViT-B/16	aggregator	pointing, strict	0.4537	0.4534	+0.0003 [−0.0035, +0.0041]	0.933	no difference
CUB	ViT-B/16	relocation	pointing, strict	0.6545	0.7478	−0.0934 [−0.1061, −0.0809]	4.6×10^{-45}	S better
CUB	ViT-B/16	aggregator	pointing, 15 px	0.5853	0.5851	+0.0002 [−0.0040, +0.0041]	1.000	no difference
CUB	ViT-B/16	relocation	pointing, 15 px	0.7654	0.8386	−0.0732 [−0.0842, −0.0623]	1.2×10^{-35}	S better
VOC boxes	ResNet-50	aggregator	pointing, strict	0.8398	0.8434	−0.0036 [−0.0056, −0.0016]	0.002	S better
VOC boxes	ResNet-50	aggregator	pointing, 15 px	0.9163	0.9194	−0.0030 [−0.0051, −0.0010]	0.015	S better
VOC boxes	VGG-16	aggregator	pointing, strict	0.8093	0.8145	−0.0052 [−0.0073, −0.0032]	3.7×10^{-6}	S better
VOC boxes	VGG-16	aggregator	pointing, 15 px	0.8945	0.8967	−0.0021 [−0.0040, −0.0004]	0.088	no difference†
VOC boxes	ViT-B/16	aggregator	pointing, strict	0.6990	0.7093	−0.0104 [−0.0138, −0.0071]	3.4×10^{-9}	S better
VOC boxes	ViT-B/16	relocation	pointing, strict	0.7072	0.7311	−0.0239 [−0.0305, −0.0178]	3.8×10^{-13}	S better
VOC boxes	ViT-B/16	aggregator	pointing, 15 px	0.8234	0.8385	−0.0151 [−0.0184, −0.0121]	8.8×10^{-21}	S better
VOC boxes	ViT-B/16	relocation	pointing, 15 px	0.8285	0.8476	−0.0191 [−0.0244, −0.0141]	5.7×10^{-13}	S better
VOC masks	ResNet-50	aggregator	pointing, strict	0.6670	0.6712	−0.0042 [−0.0102, +0.0023]	0.291	no difference
VOC masks	ResNet-50	aggregator	pointing, 15 px	0.9082	0.9106	−0.0023 [−0.0070, +0.0019]	0.810	no difference
VOC masks	ResNet-50	aggregator	peak-mask IoU	0.2622	0.4785	−0.2162 [−0.2236, −0.2091]	$< 10^{-300}$	S better
VOC masks	VGG-16	aggregator	pointing, strict	0.6642	0.6702	−0.0061 [−0.0130, +0.0005]	0.314	no difference
VOC masks	VGG-16	aggregator	pointing, 15 px	0.8887	0.8915	−0.0028 [−0.0065, +0.0009]	0.630	no difference
VOC masks	VGG-16	aggregator	peak-mask IoU	0.2603	0.4478	−0.1875 [−0.1946, −0.1806]	1.7×10^{-299}	S better
VOC masks	ViT-B/16	aggregator	pointing, strict	0.5207	0.5291	−0.0084 [−0.0158, −0.0014]	0.107	no difference†
VOC masks	ViT-B/16	relocation	pointing, strict	0.5268	0.5571	−0.0303 [−0.0447, −0.0154]	2.6×10^{-4}	S better
VOC masks	ViT-B/16	aggregator	pointing, 15 px	0.7885	0.8062	−0.0177 [−0.0252, −0.0107]	3.0×10^{-6}	S better
VOC masks	ViT-B/16	relocation	pointing, 15 px	0.7857	0.8272	−0.0415 [−0.0536, −0.0303]	3.3×10^{-11}	S better
VOC masks	ViT-B/16	aggregator	peak-mask IoU	0.2777	0.4317	−0.1540 [−0.1613, −0.1468]	8.6×10^{-277}	S better
VOC masks	ViT-B/16	relocation	peak-mask IoU	0.4261	0.4375	−0.0114 [−0.0145, −0.0083]	5.8×10^{-4}	S better
COCO	ResNet-50	aggregator	pointing, strict	0.4368	0.4338	+0.0030 [+0.0004, +0.0056]	0.095	no difference†
COCO	ResNet-50	aggregator	pointing, 15 px	0.7085	0.7100	−0.0015 [−0.0033, +0.0003]	0.432	no difference
COCO	ResNet-50	aggregator	peak-mask IoU	0.1956	0.2903	−0.0947 [−0.0979, −0.0918]	$< 10^{-300}$	S better
COCO	VGG-16	aggregator	pointing, strict	0.4175	0.4219	−0.0044 [−0.0069, −0.0018]	0.005	S better
COCO	VGG-16	aggregator	pointing, 15 px	0.6495	0.6498	−0.0003 [−0.0018, +0.0012]	1.000	no difference
COCO	VGG-16	aggregator	peak-mask IoU	0.1847	0.2611	−0.0765 [−0.0794, −0.0735]	$< 10^{-300}$	S better
COCO	ViT-B/16	aggregator	pointing, strict	0.3656	0.3687	−0.0031 [−0.0064, +0.0003]	0.169	no difference
COCO	ViT-B/16	relocation	pointing, strict	0.3728	0.3603	+0.0125 [+0.0068, +0.0185]	1.8×10^{-4}	N better
COCO	ViT-B/16	aggregator	pointing, 15 px	0.5953	0.5990	−0.0037 [−0.0068, −0.0005]	0.052	no difference†
COCO	ViT-B/16	relocation	pointing, 15 px	0.5775	0.5687	+0.0087 [+0.0026, +0.0149]	0.018	N better
COCO	ViT-B/16	aggregator	peak-mask IoU	0.2101	0.2666	−0.0565 [−0.0592, −0.0541]	$< 10^{-300}$	S better
COCO	ViT-B/16	relocation	peak-mask IoU	0.2647	0.2634	+0.0012 [+0.0003, +0.0022]	2.6×10^{-17}	N better

N and S are the necessity and sufficiency maps of the same model (aggregator arm: log-sum-exp pooling at $T = +1$ with the reused head; relocation arm: ViT-B/16 with the layer normalisation after pooling and a refitted head). The difference is N − S with its unadjusted 95% percentile bootstrap interval (2000 resamples; 1000 for MaxBoxAccV2). Tests as for the primary contrasts: paired image bootstrap (MaxBoxAccV2), exact McNemar (pointing), Wilcoxon signed-rank (best IoU, peak-mask IoU). Holm within each dataset, backbone and metric family, here enlarged from the four (convolutional backbones) or eight (ViT-B/16) primary contrasts by the one or two N-vs-S contrasts. †Holm verdict and interval disagree.

Tally of the within-model necessity versus sufficiency contrasts (Holm, $\alpha = 0.05$).

Arm	Metric class	n	N better	S better	No difference
aggregator	maximum location	24	0	9	15
aggregator	extent	12	0	12	0
aggregator	all	36	0	21	15
relocation	maximum location	8	2	6	0
relocation	extent	4	1	3	0
relocation	all	12	3	9	0
all	maximum location	32	2	15	15
all	extent	16	1	15	0
all	all	48	3	30	15

Maximum location: the two pointing games; extent: MaxBoxAccV2, best IoU, peak-mask IoU. Counts over dependent contrasts, descriptive only; "no difference" means not significant after Holm, not equivalence.

Table S10. Comparisons named by the localisation protocol and not among the primary contrasts (analysed post hoc). Unless marked 3×3, rows use the 1×1 unit. Controls: the separated maps against the activation-magnitude and random maps, Holm within each dataset, backbone and metric family over these rows, on all twelve dataset–backbone cells. Context: the collapsed map against the same controls. Mean reference: the primary analogues and the within-model contrast recomputed with the dataset-mean cell as the reference, run for ResNet-50 on CUB and VOC masks only; 22 of the 28 primary-analogue verdicts equal those obtained with the eight-image reference bank. Better and worse: the first map minus the second is positive or negative with $P_{\text{Holm}} < 0.05$ (for necessity against sufficiency, better means necessity higher); counts over dependent contrasts, descriptive only. The rows marked 3×3 repeat the comparisons at the 3×3 perturbation unit; those with a log-sum-exp map carry the caveat of Note S1.11 and are indicative. Per-row values are in the code-and-data archive.

Set	Contrast	n	Better	Worse	No difference
Controls	necessity (LN after pool) vs activation magnitude	12	12	0	0
Controls	necessity (LN after pool) vs random map	12	12	0	0
Controls	necessity (LSE) vs activation magnitude	36	26	10	0
Controls	necessity (LSE) vs random map	36	30	6	0
Controls	sufficiency (LN after pool) vs activation magnitude	12	12	0	0
Controls	sufficiency (LN after pool) vs random map	12	12	0	0
Controls	sufficiency (LSE) vs activation magnitude	36	30	6	0
Controls	sufficiency (LSE) vs random map	36	36	0	0
Context	collapsed map vs activation magnitude	36	30	6	0
Context	collapsed map vs random map	36	36	0	0
Mean reference	necessity (LSE) vs sufficiency (LSE)	7	0	3	4
Mean reference	necessity (LSE) vs centre prior	7	4	3	0
Mean reference	necessity (LSE) vs collapsed map	7	0	3	4
Mean reference	sufficiency (LSE) vs centre prior	7	7	0	0
Mean reference	sufficiency (LSE) vs collapsed map	7	2	1	4
3×3 aggregator, maximum location	necessity (LSE) vs collapsed map	24	0	21	3
3×3 aggregator, maximum location	sufficiency (LSE) vs collapsed map	24	2	19	3
3×3 aggregator, maximum location	necessity (LSE) vs centre prior	24	19	3	2
3×3 aggregator, maximum location	sufficiency (LSE) vs centre prior	24	18	4	2
3×3 aggregator, maximum location	necessity (LSE) vs sufficiency (LSE)	24	9	10	5
3×3 aggregator, maximum location	necessity (LSE) vs activation magnitude	24	22	2	0
3×3 aggregator, maximum location	sufficiency (LSE) vs activation magnitude	24	21	2	1
3×3 aggregator, maximum location	necessity (LSE) vs random map	24	24	0	0
3×3 aggregator, maximum location	sufficiency (LSE) vs random map	24	24	0	0
3×3 aggregator, extent	necessity (LSE) vs collapsed map	12	0	12	0
3×3 aggregator, extent	sufficiency (LSE) vs collapsed map	12	0	12	0
3×3 aggregator, extent	necessity (LSE) vs centre prior	12	3	8	1
3×3 aggregator, extent	sufficiency (LSE) vs centre prior	12	6	6	0
3×3 aggregator, extent	necessity (LSE) vs sufficiency (LSE)	12	0	11	1
3×3 aggregator, extent	necessity (LSE) vs activation magnitude	12	5	7	0
3×3 aggregator, extent	sufficiency (LSE) vs activation magnitude	12	7	4	1
3×3 aggregator, extent	necessity (LSE) vs random map	12	7	4	1
3×3 aggregator, extent	sufficiency (LSE) vs random map	12	10	1	1
3×3 relocation, maximum location	necessity (LN after pool) vs collapsed map	8	8	0	0
3×3 relocation, maximum location	sufficiency (LN after pool) vs collapsed map	8	8	0	0
3×3 relocation, maximum location	necessity (LN after pool) vs centre prior	8	6	2	0
3×3 relocation, maximum location	sufficiency (LN after pool) vs centre prior	8	6	2	0
3×3 relocation, maximum location	necessity (LN after pool) vs sufficiency (LN after pool)	8	0	5	3
3×3 relocation, maximum location	necessity (LN after pool) vs activation magnitude	8	8	0	0
3×3 relocation, maximum location	sufficiency (LN after pool) vs activation magnitude	8	8	0	0
3×3 relocation, maximum location	necessity (LN after pool) vs random map	8	8	0	0
3×3 relocation, maximum location	sufficiency (LN after pool) vs random map	8	8	0	0
3×3 relocation, extent	necessity (LN after pool) vs collapsed map	4	4	0	0
3×3 relocation, extent	sufficiency (LN after pool) vs collapsed map	4	4	0	0
3×3 relocation, extent	necessity (LN after pool) vs centre prior	4	3	1	0
3×3 relocation, extent	sufficiency (LN after pool) vs centre prior	4	3	0	1
3×3 relocation, extent	necessity (LN after pool) vs sufficiency (LN after pool)	4	0	3	1
3×3 relocation, extent	necessity (LN after pool) vs activation magnitude	4	4	0	0
3×3 relocation, extent	sufficiency (LN after pool) vs activation magnitude	4	4	0	0
3×3 relocation, extent	necessity (LN after pool) vs random map	4	4	0	0
3×3 relocation, extent	sufficiency (LN after pool) vs random map	4	4	0	0
3×3 context, maximum location	collapsed map vs activation magnitude	24	22	2	0
3×3 context, maximum location	collapsed map vs random map	24	24	0	0
3×3 context, extent	collapsed map vs activation magnitude	12	8	4	0
3×3 context, extent	collapsed map vs random map	12	12	0	0

Table S11. The structural check on 32 constructed heads (post hoc). Each head acts on a random $16 \times 7 \times 7$ feature map with random weights; the measured label is the float64 certificate of Note S1.2. †Correct label, wrong blocking operation named. With the pattern rules of the archived check function, derived from the audit and not tested beyond it, the verdict agrees with the certificate on 22 of the 32 heads (Note S1.5).

ID	Head (on a $16 \times 7 \times 7$ map)	Check	ε_1	ε_2	Measured	Agree
P01	mean as a module (AdaptiveAvgPool2d) → Flatten → Linear	degenerate	1.9×10^{-15}	4.9×10^{-15}	degenerate	yes
P02	mean as a tensor op x.mean((2,3)) → Linear, no declared pool	degenerate	1.9×10^{-15}	4.9×10^{-15}	degenerate	yes
P03	max pool as a module (AdaptiveMaxPool2d) → Linear	aggregator-blocked	1.0×10^{0}	1.1×10^{0}	blocked (hard)	yes
P04	max pool as timm SelectAdaptivePool2d(pool_type='max') → Linear	aggregator-blocked	1.0×10^{0}	1.1×10^{0}	blocked (hard)	yes
P05	max as a tensor op x.amax((2,3)) → Linear, no declared pool	degenerate	1.0×10^{0}	1.1×10^{0}	blocked (hard)	**no**
P06	max as a tensor op, model declares global_pool='max'	aggregator-blocked	1.0×10^{0}	1.1×10^{0}	blocked (hard)	yes
P07	attention pool module without MLP (timm AttentionPoolLatent, mlp=None) → Linear; global_pool='map'	degenerate	3.4×10^{-2}	1.0×10^{-2}	blocked (hard)	**no**
P08	attention pool + LayerNorm + MLP (timm AttentionPoolLatent as in SigLIP 'map') → Linear	nonlinearity-blocked	3.1×10^{-1}	8.9×10^{-2}	blocked (hard)	yes†
P09	attention pool written as tensor ops (Linear k, v; softmax over cells) → Linear, no declared pool	degenerate	2.5×10^{-1}	7.5×10^{-2}	blocked (hard)	**no**
P10	attention pool (tensor ops) followed by an MLP (Linear-GELU-Linear)	nonlinearity-blocked	2.2×10^{-1}	1.0×10^{-1}	blocked (hard)	yes†
P11	stack of 3 class tokens reduced by Conv1d(3,1,1) → Linear (CoaT head)	degenerate	3.7×10^{-16}	2.7×10^{-16}	class-token	**no**
P12	stack of 3 class tokens reduced by Linear(3,1) over the token axis → Linear	degenerate	7.4×10^{-16}	2.4×10^{-16}	class-token	**no**
P13	single class token x[:, 0] → Linear, global_pool='token'	class-token	0	0	class-token	yes
P14	LayerNorm after the mean (AdaptiveAvgPool2d → LayerNorm → Linear)	normalisation-blocked	6.1×10^{-1}	3.8×10^{-1}	normalisation-blocked	yes
P15	LayerNorm before the mean, per cell (LayerNorm2d → AdaptiveAvgPool2d → Linear)	degenerate	2.0×10^{-15}	2.0×10^{-15}	degenerate	yes
P16	LayerNorm before the mean, per token, mean as a tensor op (LayerNorm → x.mean(1) → Linear)	normalisation-blocked	1.8×10^{-15}	7.8×10^{-15}	degenerate	**no**
P17	RMSNorm after the mean (timm RmsNorm)	normalisation-blocked	6.5×10^{-1}	3.7×10^{-1}	normalisation-blocked	yes
P18	RMSNorm after the mean (torch nn.RMSNorm)	normalisation-blocked	6.5×10^{-1}	3.7×10^{-1}	normalisation-blocked	yes
P19	SimpleNorm2d after the mean (ConvNeXt-Zepto-RMS head: pool → SimpleNorm2d → Linear)	nonlinearity-blocked	5.2×10^{-1}	3.7×10^{-1}	normalisation-blocked	**no**
P20	per-cell MLP before the mean (Conv1x1-GELU-Conv1x1 → AdaptiveAvgPool2d → Linear)	degenerate	1.2×10^{-14}	6.0×10^{-14}	degenerate	yes
P21	per-token MLP before a tensor-op mean (Linear-GELU-Linear → x.mean(1) → Linear)	nonlinearity-blocked	1.2×10^{-14}	5.5×10^{-14}	degenerate	**no**
P22	GeM pooling as a module (p = 3) → Linear	aggregator-blocked	5.3×10^{-1}	2.5×10^{-1}	blocked (hard)	yes
P23	GeM pooling as tensor ops → Linear, no declared pool	degenerate	5.3×10^{-1}	2.5×10^{-1}	blocked (hard)	**no**
P24	mixed avg+max concat as timm SelectAdaptivePool2d('catavgmax') → Linear	aggregator-blocked	6.8×10^{-1}	6.2×10^{-1}	blocked (hard)	yes
P25	mixed avg+max concat as tensor ops → Linear, no declared pool	degenerate	6.8×10^{-1}	6.2×10^{-1}	blocked (hard)	**no**
P26	per-cell class scores (Conv1x1) pooled by log-sum-exp over cells (tensor op), T = 1	degenerate	1.3×10^{-1}	6.8×10^{-2}	blocked (hard)	**no**
P27	linear cell mixing before the mean (Conv3x3, no activation → AdaptiveAvgPool2d → Linear)	degenerate	5.1×10^{-15}	1.3×10^{-14}	degenerate	yes
P28	nonlinear cell mixing before the mean (Conv3x3 → ReLU → AdaptiveAvgPool2d → Linear)	degenerate	1.1×10^{0}	8.8×10^{-1}	blocked (hard)	**no**
P29	self-attention over cells before the mean (nn.MultiheadAttention → AdaptiveAvgPool2d → Linear)	degenerate	2.0×10^{-1}	9.0×10^{-2}	blocked (hard)	**no**
P30	MLP tail (AdaptiveAvgPool2d → Linear → GELU → Linear)	nonlinearity-blocked	2.0×10^{-1}	6.6×10^{-2}	blocked (hard)	yes
P31	BatchNorm1d (inference) after the mean → Linear	degenerate	1.5×10^{-14}	5.6×10^{-14}	degenerate	yes
P32	ReLU in the tail that is never crossed on the reachable set (Linear bias +50 → ReLU → Linear)	nonlinearity-blocked	1.1×10^{-12}	2.0×10^{-12}	degenerate	**no**

Table S12. Audit of the 30 census architectures whose head trace was never measured (post hoc). Each architecture was measured with random weights in float64 (Note S1.6) and its head read in the timm 1.0.27 source. Trace: index of the distinct head trace. The three CoaT models reduce a stack of class tokens with a one-dimensional convolution, and no spatial cell moves their logit (measured label vacuous: a class-token readout); the two ConvNeXt-Zepto-RMS models carry an RMS normalisation that the lookup tables label nonlinear. For mobilenetv4_hybrid_large the measurement on unit-scale features decides, because its randomly initialised body returns near-zero features.

Trace	Architecture	Verdict of the check	Measured label	Agree
1	vit_medium_patch16_gap_256	degenerate	additive	yes
1	vit_medium_patch16_gap_384	degenerate	additive	yes
1	vit_relpos_base_patch32_plus_rpn_256	degenerate	additive	yes
1	vit_relpos_medium_patch16_rpn_224	degenerate	additive	yes
1	vit_so150m_patch16_reg4_gap_256	degenerate	additive	yes
1	vit_so150m_patch16_reg4_gap_384	degenerate	additive	yes
1	vit_srelpos_medium_patch16_224	degenerate	additive	yes
1	vit_srelpos_small_patch16_224	degenerate	additive	yes
2	mambaout_base_plus_rw	nonlinearity-blocked	hard-blocked	yes
2	mambaout_base_short_rw	nonlinearity-blocked	hard-blocked	yes
2	mambaout_base_tall_rw	nonlinearity-blocked	hard-blocked	yes
2	mambaout_base_wide_rw	nonlinearity-blocked	hard-blocked	yes
2	mambaout_small_rw	nonlinearity-blocked	hard-blocked	yes
3	deit_base_distilled_patch16_224	class-token	vacuous	yes
3	deit_base_distilled_patch16_384	class-token	vacuous	yes
3	deit_small_distilled_patch16_224	class-token	vacuous	yes
3	deit_tiny_distilled_patch16_224	class-token	vacuous	yes
4	coat_mini	degenerate	vacuous	**no**
4	coat_small	degenerate	vacuous	**no**
4	coat_tiny	degenerate	vacuous	**no**
5	efficientvit_l1	nonlinearity-blocked	hard-blocked	yes
5	efficientvit_l2	nonlinearity-blocked	hard-blocked	yes
5	efficientvit_l3	nonlinearity-blocked	hard-blocked	yes
6	convnext_zepto_rms	nonlinearity-blocked	normalisation-only	**no**
6	convnext_zepto_rms_ols	nonlinearity-blocked	normalisation-only	**no**
7	tf_mobilenetv3_large_minimal_100	nonlinearity-blocked	hard-blocked	yes
7	tf_mobilenetv3_small_minimal_100	nonlinearity-blocked	hard-blocked	yes
8	convnext_large_mlp	nonlinearity-blocked	hard-blocked	yes
9	mobilenetv4_hybrid_large	nonlinearity-blocked	hard-blocked (unit-scale features)	yes
10	test_vit3	nonlinearity-blocked	hard-blocked	yes

Table S13. Conjunctive sufficiency against the collapsed map in every medical arm (post hoc). S(−8) is the sufficiency map of the log-sum-exp head at $T = -8$ (the conjunctive sufficiency map), collapsed the collapsed map, both at the same unit (the cell block perturbed); the columns give each map's mean and their paired difference S(−8) − collapsed with a 95% paired bootstrap interval as in Table S7. *P* is exact McNemar for hit and Wilcoxon signed-rank otherwise, uncorrected. Metrics are those of each arm's grid (Table S7): hit and AP everywhere, IoU for NIH with the distributed head, Dice on ISIC. *Src* gives the run each value comes from, with the codes of Table S7: M, the stored maps of the first run; R, maps regenerated from the same distributed RSNA weights; P, the second, independent training run (refitted NIH head, retrained NIH model of this study, retrained ISIC model); D, the cross-fitted disjoint RSNA models; C, a further retrained ISIC model whose maps were computed in closed form (its forward-difference values differ from these by at most 8×10^{-9}). *First run* is the first run's mean difference for the same cell where it was stored (no per-item values, hence no interval); for Src M it equals the value shown and is omitted. The held-out rows restrict the distributed models to items outside their training data; there the stored and the closed-form maps give identical results. For NIH with the distributed head and the distributed RSNA model, the closed form reproduces every difference shown to within 6×10^{-5}. †Interval and test disagree.

Unit	Metric	S(−8)	Collapsed	S(−8) − collapsed [95% CI]	*P*	Src	First run
NIH, distributed head (designated in advance; *n* = 984; Src M)							
1×1	hit	0.184	0.227	−0.043 [−0.063, −0.022]	6.6×10^{-5}	M	
1×1	AP	0.156	0.220	−0.064 [−0.074, −0.056]	1.3×10^{-51}	M	
1×1	IoU	0.093	0.140	−0.047 [−0.053, −0.041]	1.5×10^{-52}	M	
3×3	hit	0.214	0.227	−0.012 [−0.033, +0.007]	0.26	M	
3×3	AP	0.196	0.224	−0.028 [−0.037, −0.019]	1.1×10^{-12}	M	
3×3	IoU	0.126	0.145	−0.018 [−0.024, −0.013]	4.3×10^{-9}	M	
NIH, refitted head (post hoc; *n* = 984; Src P)							
1×1	hit	0.226	0.215	+0.010 [−0.013, +0.034]	0.44	P	+0.010
1×1	AP	0.169	0.195	−0.026 [−0.036, −0.017]	5.9×10^{-7}	P	−0.026
3×3	hit	0.260	0.210	+0.050 [+0.025, +0.075]	0.00013	P	+0.050
3×3	AP	0.228	0.205	+0.023 [+0.012, +0.035]†	0.24	P	+0.023
NIH, own model (post hoc; *n* = 984; Src P)							
1×1	hit	0.185	0.245	−0.060 [−0.086, −0.034]	9.7×10^{-6}	P	−0.092
1×1	AP	0.169	0.225	−0.056 [−0.068, −0.045]	6.3×10^{-25}	P	−0.040
3×3	hit	0.177	0.240	−0.063 [−0.087, −0.039]	3×10^{-7}	P	+0.025
3×3	AP	0.193	0.227	−0.034 [−0.044, −0.024]	1.9×10^{-14}	P	−0.009
RSNA, distributed model (designated in advance; *n* = 1,385; Src R)							
1×1	hit	0.523	0.479	+0.045 [+0.022, +0.067]	0.00014	R	+0.044
1×1	AP	0.345	0.410	−0.065 [−0.073, −0.056]	9.5×10^{-63}	R	−0.065
3×3	hit	0.168	0.389	−0.222 [−0.248, −0.196]	9.3×10^{-62}	R	−0.222
3×3	AP	0.304	0.388	−0.085 [−0.093, −0.076]	1.2×10^{-90}	R	−0.085
RSNA, cross-fitted disjoint models (post hoc; *n* = 1,385; Src D)							
1×1	hit	0.258	0.176	+0.082 [+0.057, +0.107]	4×10^{-10}	D	
1×1	AP	0.237	0.236	+0.001 [−0.007, +0.008]	0.76	D	
3×3	hit	0.126	0.274	−0.149 [−0.173, −0.126]	5.2×10^{-35}	D	
3×3	AP	0.263	0.277	−0.014 [−0.022, −0.007]	0.0064	D	
ISIC (designated in advance; *n* = 735; Src P)							
1×1	hit	0.498	0.789	−0.291 [−0.331, −0.253]	4.1×10^{-44}	P	−0.261
1×1	AP	0.338	0.643	−0.305 [−0.325, −0.285]	1.1×10^{-99}	P	−0.310
1×1	Dice	0.298	0.444	−0.147 [−0.165, −0.129]	2×10^{-43}	P	−0.102
3×3	hit	0.448	0.709	−0.261 [−0.301, −0.222]	5.8×10^{-36}	P	+0.007
3×3	AP	0.449	0.644	−0.195 [−0.214, −0.176]	3.4×10^{-67}	P	−0.032
3×3	Dice	0.409	0.468	−0.059 [−0.069, −0.049]	6.3×10^{-30}	P	−0.031
RSNA, distributed model, held-out items outside the model's training data (post hoc; *n* = 283; Src R)							
1×1	hit	0.516	0.473	+0.042 [−0.011, +0.095]	0.16	R	
3×3	hit	0.170	0.360	−0.191 [−0.244, −0.141]	2.7×10^{-12}	R	
NIH, distributed head, held-out items outside the model's training data (post hoc; *n* = 182; Src M)							
1×1	hit	0.154	0.225	−0.071 [−0.115, −0.033]	0.00098	M	
3×3	hit	0.209	0.209	+0.000 [−0.044, +0.044]	1	M	
ISIC, retrained model, closed form (post hoc; *n* = 735; Src C)							
1×1	hit	0.073	0.363	−0.290 [−0.325, −0.254]	2.3×10^{-51}	C	
1×1	AP	0.200	0.385	−0.185 [−0.200, −0.169]	6.6×10^{-82}	C	
1×1	Dice	0.229	0.285	−0.056 [−0.067, −0.045]	8.7×10^{-28}	C	
3×3	hit	0.223	0.438	−0.215 [−0.253, −0.177]	1.3×10^{-25}	C	
3×3	AP	0.235	0.422	−0.187 [−0.203, −0.172]	4.8×10^{-87}	C	
3×3	Dice	0.247	0.337	−0.090 [−0.100, −0.079]	5.2×10^{-48}	C	

S1 Extended methods

This note gives the details needed to reproduce the study that the main text does not state. The readout, the two quantities, the identity, the structural check and the certificate are defined in Section 2 of the main text and are not repeated. Fig. S1 extends Figure 1 of the main text, and Table S14 lists the design of every experiment.

S1.1 Two properties of the identity

Affinity on the segment is not enough, as the following example shows. Let f be the identity, the aggregator a sum over two cells and $g(x, y) = (x + y - 1)^2$. The activations $a = [(1, 0), (0, 0)]$ and $b = [(0, 0), (0, 1)]$ give pooled points $(1, 0)$ and $(0, 1)$, and g is zero on the segment between them, yet the necessity of both cells is -1 and their sufficiency $+1$. The identity requires g to be affine on the convex hull of the $2N + 2$ pooled points that the single-cell substitutions reach, and its extension to regions of cells requires affinity on the zonotope spanned by all N displacements, placed at each of the two evaluation points.

Quasi-arithmetic means order the two maps identically. To see this, let $z = \varphi^{-1}\big(N^{-1}\sum_u \varphi(s_u)\big)$ with φ strictly monotone, and write $\Phi_x = \sum_u \varphi(s_u(x))$ and $d_t = \varphi(s_t(a)) - \varphi(s_t(b))$. Then $\mathrm{nec}(t) = \varphi^{-1}(\Phi_a/N) - \varphi^{-1}((\Phi_a - d_t)/N)$ and $\mathrm{suf}(t) = \varphi^{-1}((\Phi_b + d_t)/N) - \varphi^{-1}(\Phi_b/N)$. Because φ^{-1} is strictly monotone, both are strictly monotone functions of d_t in the same direction, so against one reference the two maps rank the cells identically. Log-sum-exp pooling is the case $\varphi(s) = e^{Ts}$, which gives Eq. (2) of the main text. Max and k-of-N pooling score each cell but are not of this form, and attention and bilinear pooling have no single per-cell score. With these operators the two maps can order the cells differently against a single reference. Under a quasi-arithmetic mean, by contrast, the orderings can differ only when the maps are averaged over several references, because the references enter the two maps differently: for log-sum-exp, necessity is normalised by Z_a and sufficiency by each reference's own Z_b.

S1.2 Certification and numerical precision

The set-level residual ε_2 is the mean absolute difference between the logit change of a random half-map substitution and the sum of its cells' single-cell changes over 64 random half-subsets, divided by the mean absolute logit change, and ε_3 is the corresponding pairwise interaction residual [15] over 256 random cell pairs. Each residual is the worst value over three probe pairs. The certificate establishes additivity on the set that the substitutions reach, which is what the identity needs, rather than additivity as a global property. Two readouts constructed to be non-additive as functions, a sinusoid along a direction orthogonal to every displacement and a rectifier whose threshold is never crossed within the reachable set, pass it, and both have necessity identical to sufficiency. Because the certificate samples subsets of size one, two and $N/2$, it is a strong screen rather than a proof. A readout was labelled vacuous when no other cell moved the logit by more than 10^{-12} of the largest, or when its spatial mass was below 10^{-9}.

Experiments run in float32, namely the operator sweeps and the normalisation edit, report their degenerate points as correlations together with their float32-level residuals, and the degenerate orderings of the normalisation edit were re-certified in float64. Forward differences of the logit cancel when a cell's share of the pooled sum lies below working precision, which affects log-sum-exp necessity at $T > 0$ and sufficiency at $T < 0$ in float32 and float64 alike. All log-sum-exp necessity and sufficiency maps that are compared by rank or shown cell by cell (Fig. S2(b), Fig. S3(b,c), Note S2.4 and the faithfulness comparison of Note S2.3) were therefore computed from Eq. (2) of the main text in float64 with the numerically stable logarithm $\log(1+x)$. The localisation estimator already evaluates the closed form, and it agrees with an independent implementation of it to 6×10^{-14} on real ResNet-50 activations. The medical maps were computed by float64 forward differences, and the primary contrasts were recomputed from the closed form where the models were available (Note S2.9). The reported grid values are those of the forward-difference maps.

In the census every entry was certified in float64 with random weights twice: on the features its own random-weight trunk produces, and on a unit-scale Gaussian feature map fed to its head. Entries whose classifier is initialised to zero, so that the logit does not depend on the cells, were certified after re-drawing those weights. The four PiT-distilled models return only their class and distillation tokens from the trunk. Their certificate, which read additive over those two tokens, was replaced by the vacuous label by inspection. The four entries with more than 10^9 parameters, the three EVA-giant/14 models at 224, 336 and 560 pixels and RegNetY-2560, were neither checked nor certified. The resulting counts and the entries whose two labels differ are given in Note S2.6.

S1.3 Reference activations

The operator sweeps drew one reference per image from a bank of real validation activations (32 for Fig. S2(b) and Fig. S3(a–c), 16 for Fig. S3(d)) and did not average. The medical references were eight images labelled "No Finding". The evaluation of the structural check used another validation image's activation as the reference, over three image pairs. A reference whose pooled vector equals that of the probed input, such as a spatial rearrangement of it, was excluded because it lies off the data manifold. This reference and ten other distinct references, twelve settings in all, form the sensitivity analysis of Table S5 (Note S2.10).

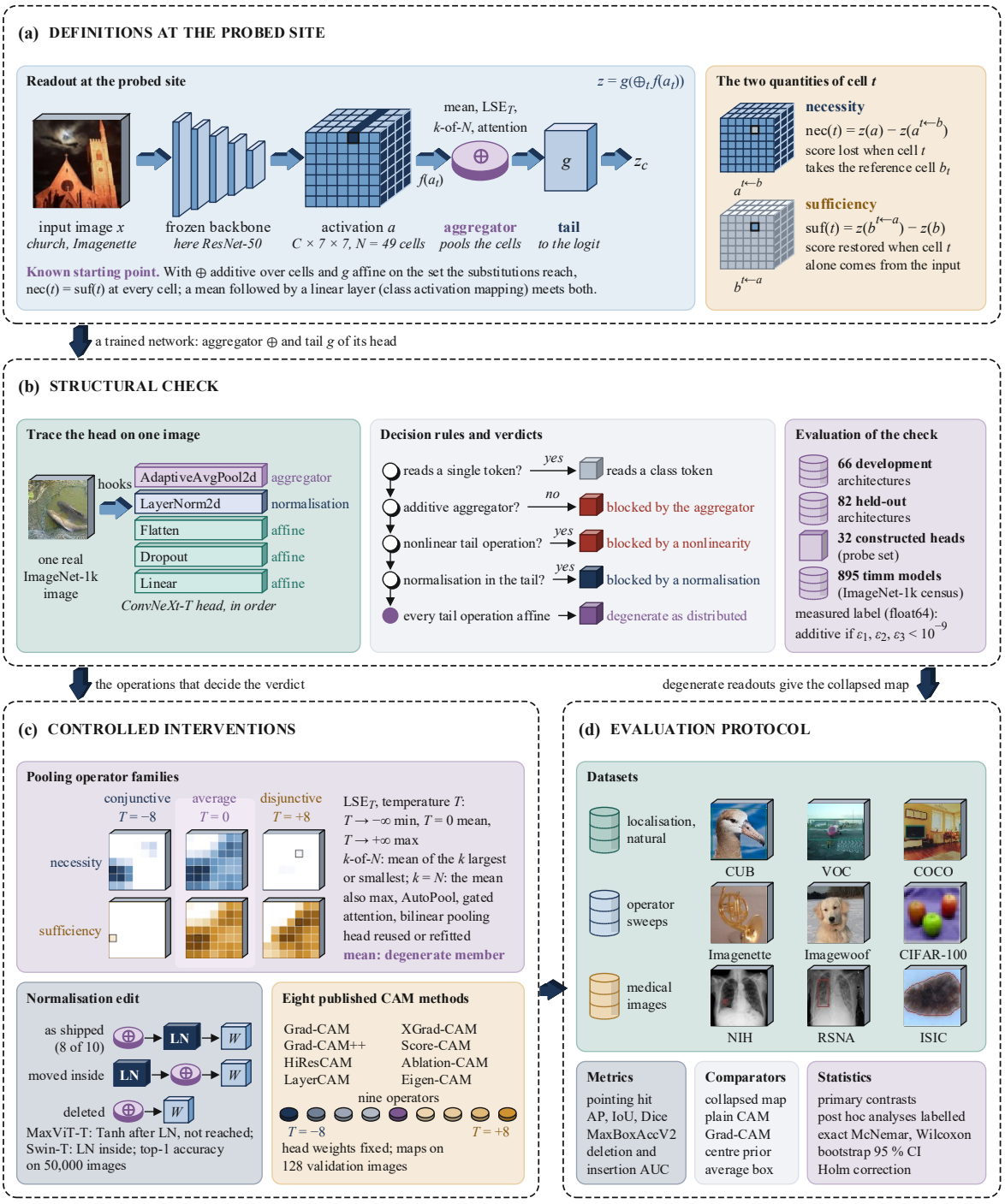


Figure S1. Overall design of the study. (a) Definitions at the probed site. A frozen backbone (here ResNet-50) encodes the input image x (a church from Imagenette) into an activation a of C channels on a 7×7 grid of $N = 49$ cells. The head scores each cell with f, an aggregator $\oplus$ pools the cells (mean, log-sum-exp with temperature T (LSE$_T$), k-of-N or attention) and a tail g maps the pooled vector to the class logit z_c (Section 2.1 of the main text). Necessity and sufficiency of cell t substitute that cell against a reference activation b (Eq. (1) of the main text), where blue cells belong to a and slate cells to b. With $\oplus$ additive and g affine, as in class activation mapping (CAM), the two coincide at every cell. (b) Structural check (Table 1 of the main text). Forward hooks record the modules the head executes on one input (shown for ConvNeXt-T on a real image, whereas the census used random weights and input), each is labelled, and four rules return one of five verdicts. The check was written on 66 development architectures, evaluated on 82 held-out architectures and on a probe set of 32 constructed heads against a double-precision (float64) measurement of the residuals ε_1–ε_3 (Note S1.2), and run as a census over 895 ImageNet-1k classifiers of the timm library. (c) Controlled interventions. Pooling operators are varied along the conjunctive–disjunctive axis with the mean as the degenerate member. The maps are the closed-form necessity (blue) and sufficiency (ochre) maps of Eq. (2) of the main text at $T = -8$, 0 and +8, on one scale per operator, so a map much smaller than its partner looks nearly blank. The layer normalisation (LN) of ten distributed models, after the mean as shipped by nine of the ten models, is moved inside the mean or deleted, and eight published CAM methods are evaluated under nine operators with the head weights fixed. (d) Evaluation protocol. Localisation benchmarks CUB-200-2011 (CUB), PASCAL VOC 2012 (VOC) and MS-COCO (COCO), operator-sweep datasets Imagenette, Imagewoof and CIFAR-100, and medical datasets NIH ChestX-ray14 (NIH), RSNA Pneumonia (RSNA) and ISIC 2018 (ISIC), each shown with one real image and its ground truth, together with the metrics, comparators and statistics of Notes S1.9 and S1.10. AP, average precision; AUC, area under the deletion or insertion curve; CI, confidence interval; IoU, intersection over union.

S1.4 Pooling families and heads

Log-sum-exp pooling [5] reduces to the mean at $T = 0$ and approaches a maximum (disjunctive pooling) as $T \to +\infty$ and a minimum (conjunctive pooling) as $T \to -\infty$. The k-of-N family averages the k largest or the k smallest per-cell scores [6, 7], and AutoPool [8] learns its position on the axis. Gated attention pooling [2], max pooling and a projected second-order (bilinear) pooling [16] complete the set. The bilinear head keeps the mean but applies a quadratic per-cell map and a non-affine tail (signed square root and L2 normalisation). Heads were fitted with AdamW [17] and a cross-entropy objective on frozen features, and within a comparison every member of a family was fitted identically. The settings of each experiment are given in Table S14.

For pooling of each channel before the linear layer (Note S2.7, Table S17), log-sum-exp with temperature T or a generalised mean with exponent p was applied to each channel of the last ResNet-50 feature map before the distributed linear layer, and compared with log-sum-exp of the per-cell class scores. Maps were computed by exact algebra evaluated stably (closed form) and by a literal substitution loop over the 49 cells of 64 ImageNet validation images, against one reference each.

S1.5 Implementation of the structural check

The reference implementation is specific to the timm library. Labels are assigned from the module's class name through fixed lookup tables written on the development set: for example, linear, dropout and batch-normalisation modules are affine, a class whose name ends in "Norm" is a normalisation, and any unknown class is nonlinear. When no hooked module reduces the cells, which happens when timm pools with a tensor operation, the check reads the pool type the model declares and defaults to a mean. Of the 148 checked architectures, 46 are decided on this branch. The real input serves only to trigger the hooks. A degenerate verdict is correct when every cell-mixing operation is a hooked module, and an operation written as a tensor expression in the head's forward pass is invisible to it. A blocked verdict can be a false negative when a nonlinearity is never active on the reachable set, and it does not say how far apart the two quantities are, so the measured separation of blocked architectures is reported as well (Note S2.6). For models outside timm, the same predicates can be read from an operator-level trace, for example one produced by the symbolic tracer of PyTorch, with "unknown" rather than a mean as the default. The check assumes, without testing it, that every operation the head executes before the aggregator acts on each cell separately and that the axis the aggregator removes is the spatial one. In the census the trace was recorded on random weights and a random input of the model's input size. A class-token readout rests on a single token at its head, and at the first site where the spatial cells reach the logit the cells are mixed by self-attention, which is not additive over cells.

The conservative check reads condition (i) of Section 2.3 of the main text as requiring that the spatial reduction is the only one: in both readings the operator trace must contain exactly one operator that removes the spatial axes, and in the module reading it must run inside the aggregator module. A non-leaf module that completes before the aggregator is accepted only when it runs no operator of its own (a plain sequential container), and its leaf children are checked against the per-cell whitelist. The whitelists were taken from the module classes of the 66 development architectures and filtered to the per-cell, additive-pool and affine categories of the protocol. Where the conditions do not hold, the conservative check returns the module-level verdict if it is blocked or class-token, and that verdict is the audited one, with three pattern rules (a maximum over tokens, a stack of class tokens, an unlisted normalisation) derived from the census audit and the held-out VOLO-D1 disagreement. The protocol numbers five conditions: condition (i) of the main text combines its first and fourth, and conditions (ii)–(iv) are its second, third and fifth. A post hoc variant with two operator-level conditions (only per-cell operators before the reduction, only affine operators after it) was added after the protocol check had been evaluated and is not part of the protocol. The archive provides the check as a function that returns the verdicts, names the blocking operation and flags, by the pattern rules, the timm heads known to defeat it, together with a function that computes both maps for any head by literal substitution.

The torchvision classifiers expose no split into trunk and head, so the split at the last trunk feature map was transcribed by hand for each of 19 families from the torchvision source, and splitting changed no output (largest relative difference 0.0 over 80 models). In 27 of the 80 models no hooked module reduces the cells (DenseNet, MNASNet, MobileNetV2, ShuffleNetV2, AlexNet, VGG and the five vision transformers, which reduce with tensor operations), so the verdict rests on the default-mean branch. The five vision transformers ship a classifier initialised to zero, and they were also labelled with those weights re-drawn (80 labelled models).

S1.6 Evaluation of the check

The development architecture that could not be measured is CrossViT-9. The 66 development and 82 held-out architectures were chosen family by family from the same library in two disjoint splits, so the 148 are a purposive sample. The normalisations deleted for the normalisation-only label were identified both from the check's own trace and independently of it, as every post-pool module whose class is a normalisation, and the two give the same labels. The code blocks of the archived check are byte-identical, by their recorded hashes, to blocks of the development code, whose files were last modified before the held-out lists were drawn. The archived held-out evaluation itself was run after the VOLO-D1 disagreement had been diagnosed, and it also records the verdict of a post hoc operator-level variant of the original check (Note S2.6). These are local file times, not an external commitment. A head trace is the aggregator followed by the ordered tail modules and their labels. Every census architecture whose head trace

Table S14. Experimental design. Head regime: refitted means that a head is fitted for every operator, reused means that the head fitted on the averaged readout is kept and only the operator changes, and shipped means that the distributed classifier is used unchanged. Maximum-location metrics: pointing (strict and with 15-pixel tolerance). Extent metrics: MaxBoxAccV2, best IoU and peak-mask IoU. AUC, area under the deletion or insertion curve; CAM, class activation mapping; GAP, global average pooling; IoU, intersection over union; LSE, log-sum-exp pooling.

Experiment	Backbone	Data	Head	Reference	Outcome and comparators
Operator sweeps (Notes S2.2 and S2.3)	ResNet-50 (also VGG-16, ConvNeXt-T) and an end-to-end convolutional network	Imagenette, Imagewoof, CIFAR-100	refitted	one per image from a bank	separation, insertion and deletion AUC
1,000 classes (Note S2.4)	ResNet-50	ImageNet-1k val., 224 images	shipped	one per image, bank of 32	separation, rank agreement, deletion AUC
Normalisation edit (Note S2.5)	ten distributed transformer and convolutional models	ImageNet-1k val., 50,000 images	shipped, one layer moved	bank	separation (256 images), top-1
Check evaluation (Note S2.6)	148 timm architectures, census of 895, 32 constructed heads	ImageNet-1k val. (census: random input)	shipped (census, constructed heads: random weights)	3 image pairs (measured set)	verdict against measured label
Localisation (Note S2.8)	ResNet-50, VGG-16 with GAP head (448 px), ViT-B/16 patch tokens (224 px)	CUB-200-2011, PASCAL VOC 2012 (boxes, masks), MS-COCO val2017	GAP head, LSE ($T = +1$) reused, normalisation relocated and refitted	8 fitting images	5 metrics; collapsed map, plain CAM, centre prior
End-to-end localisation (Note S2.8)	ResNet-50 (448 px), three seeds per head	CUB-200-2011	four heads trained end to end	8 training images	MaxBoxAccV2, best IoU, pointing, top-1; collapsed map, plain CAM
Medical (Note S2.9)	DenseNet-121 (TorchXRayVision and own), ResNet-50 (ISIC, three further seeds, Note S2.9)	NIH ChestX-ray14, RSNA Pneumonia, ISIC 2018	distributed head reused, refitted arm	8 "No Finding" images	hit rate, Dice; collapsed map, CAM, Grad-CAM, centre prior, activation magnitude, average box

never occurs among the measured architectures (30 architectures, ten traces) was measured with random weights in float64 and its verdict audited against the head's source code (Table S12). The 32 constructed heads (Table S11) probe each rule of the check: pools and attention written as tensor operations, stacks of class tokens, normalisations after and before the mean, cell-mixing operations before the pool and a nonlinearity that is never active. Each was certified in float64 as in Note S1.2 and its verdict compared with the certificate.

S1.7 Models, datasets and training

Beyond the networks named in the main text, the hybrid designs include [19]. The operator sweeps use Imagenette and Imagewoof [20], ten-class subsets of ImageNet, and CIFAR-100 [21]. ImageNet-1k [22] images were taken from a re-encoded public mirror, and Table S15 reports mirror accuracy. To reconcile the cost of the normalisation edit with a published ablation (Note S2.5), a linear probe was fitted on six frozen timm transformers under each placement of the normalisation, on 40 ImageNet validation images per class, and evaluated on ten others per class, and a ViT-Tiny/16 without a class token was trained from scratch on Imagenette under each placement with five paired seeds. All runs used PyTorch [23] 2.6.0 with CUDA 12.4.

S1.7.1 Natural-image localisation

The localisation backbones were frozen ResNet-50 and VGG-16 with the torchvision ImageNet weights (pool5 features, with the fully connected classifier replaced by global average pooling and a linear head) and a ViT-B/16 pre-trained on ImageNet-21k and fine-tuned on ImageNet-1k (patch tokens, class token discarded), each giving a 14×14 grid. On each benchmark a linear head was fitted to the average-pooled cells of a fitting split disjoint from the evaluation split, with per-channel standardisation, zero initialisation, AdamW with a learning rate of 3×10^{-3} and a weight decay of 10^{-4}, a cosine schedule and 400 full-batch epochs, under cross-entropy on CUB and binary cross-entropy elsewhere. This head is the degenerate readout, certified additive. The temperature of the aggregator arm was fixed before any localisation outcome (Note S1.11). The splits were CUB-200-2011 [24] train and test (5,994 and 5,794 images), PASCAL VOC 2012 [25] train and val (5,717 and 5,823 images, 8,856 image–class records) and train and segmentation-val (1,449 images, 2,147 records), and the 4,952 annotated MS-COCO val2017 [26] images split in half with a fixed seed (2,476 and 2,476 images, 7,339 records). Maps were upsampled bilinearly to 224×224 pixels and min–max scaled. Because the backbones are frozen and only a linear head is fitted, rather than fine-tuned as in the weakly supervised localisation protocol [10], absolute MaxBoxAccV2 values are not comparable with published results of that protocol.

For the heads trained end to end, a torchvision ResNet-50 initialised on ImageNet-1k was fine-tuned on the 5,994 training images

of CUB-200-2011 at 448×448 pixels (14×14 cells) with four heads and three seeds each, under one recipe: SGD with momentum 0.9, weight decay 10^{-4}, learning rates 0.01 for the head and 0.001 for the backbone with a cosine schedule to zero over 30 epochs, batch 32, mixed precision for the backbone and float32 for the head, random resized crops (scale 0.5 to 1) and horizontal flips. Head H0 is global average pooling and a linear layer (degenerate), and H1 is log-sum-exp pooling of the per-cell class scores at $T = +1$. H2 is gated attention pooling with a hidden size of 256, followed by a linear layer, and H3 is global average pooling, layer normalisation and a linear layer (normalisation-blocked). Test images were resized whole to 448×448 pixels without cropping. Necessity and sufficiency maps of the true class were computed in float64 for every cell against eight reference activations from the training split and averaged. The H1 maps were computed in closed form, the H2 maps by substituting the cell-local attention logit and class projection and the H3 maps by substitution in the pooled vector, each checked against the literal substitution loop through the head module on the first three test images and two references of every model. The primary family of six contrasts compares each map of H1–H3 with the collapsed map of H0 in MaxBoxAccV2, seed-matched and averaged over seeds, with a 95% percentile bootstrap over images (1,000 resamples) and Holm correction. The maps each head yields by its own construction (the per-cell class scores s_t that H1 pools, the attention weights a_t of H2 and the attention-weighted class evidence $a_t\langle w_c, x_t\rangle$ that H2 sums, Grad-CAM of each head, and CAM of H3 with its layer normalisation ignored) were scored post hoc with the same images, boxes, scaling, metrics and bootstrap, with Holm correction over the seven maps of H1–H3, and every map was also scored after replacing its 14×14 values by their ranks within the image.

S1.7.2 Medical images

The NIH ChestX-ray14 [27] evaluation used the 880 images with radiologist-drawn boxes (984 boxes) within a 5,720-image sample of the official test list (the 880 boxed images and 4,840 others). The distributed TorchXRayVision [14] DenseNet-121 classifier for NIH, whose readout is a rectified linear unit, global average pooling and a linear layer, reaches a mean area under the receiver operating characteristic curve (ROC AUC) of 0.668 over the fourteen findings on this sample with the preprocessing of this study. The DenseNet-121 trained in this study was trained on a 7,000-image sample of the official training list for 8 epochs and reaches 0.693. RSNA Pneumonia [28] provided 1,385 boxed studies from the challenge's labelled training set. ISIC 2018 Task 1 [29] lesion masks with ISIC 2017 Part 3 diagnosis labels [30] provided 735 evaluation images. The first ResNet-50 (ImageNet initialisation, class-weighted cross-entropy, 12 epochs) was trained on the other 1,715 of the 2,450 Task 1 images that carry a diagnosis (macro ROC AUC 0.834). On ISIC, Dice was computed on the top-q fraction of the upsampled map, with q chosen from nine values by two-fold cross-tuning, and for every map this came within 0.003 of the in-sample optimum. The disjoint RSNA models are five DenseNet-121 classifiers (ImageNet initialisation, mean pooling and one linear logit) trained with the recipe of the NIH model of this study on four of five patient-level stratified folds of the 3,884 RSNA studies available locally, namely the 1,385 boxed evaluation studies and 2,499 negative studies of the challenge's labelled training set, with box coordinates for the other positive patients taken from the challenge's label file, and each model explains only its held-out fold. The post hoc NIH arm refitted the linear head on the frozen features under each operator. The three ISIC classifiers with seeds 0, 1 and 2 were trained under the ISIC recipe above, initialised from the torchvision ImageNet-1k weights with the classifier re-initialised, and evaluated on the same 735 images. Their necessity and collapsed maps were computed both in closed form and by forward difference, and the per-cell class scores of every image are archived, from which the closed-form maps can be recomputed. ISIC models 1 and 2 (Table 3 of the main text) started from the timm ImageNet-1k weights of ResNet-50, and model 3 and seeds 0–2 from the torchvision weights. Models 2 and 3 reach a macro ROC AUC of 0.851 and 0.868. Model 2 repeats the configuration of model 1, and model 3 that of seed 0 (split, recipe, seed and initial weights), so each pair differs only through non-deterministic GPU kernels.

The metrics of the grids differ by arm. NIH with the distributed head is scored by hit rate, average precision (AP) and IoU, RSNA and the post hoc NIH arms by hit rate and AP, and ISIC by hit rate, AP and Dice. Cells other than the primary carry an uncorrected per-cell 95% bootstrap interval. The protocol chose $T = +8$ because it was the temperature of the largest measured separation in the float32 operator sweeps, with $T = -8$ secondary, and it motivated the hypothesis by the observation, in those sweeps, that the CAM family tracks the sufficiency side once the readout leaves the degenerate point. That observation was a cancellation artefact of float32 forward differences: in closed form the two maps order the cells identically against one reference (Note S2.4). Under disjunctive pooling necessity concentrates its magnitude on the maximum (Note S2.3), which disfavours it on metrics that read extent but not on the pointing hit, which reads only the location of the maximum. The conjunctive necessity map and the disjunctive sufficiency map, which spread their magnitude, are reported beside it, and the conjunctive sufficiency map is reported for every arm as a post hoc contrast. Averaged over eight references, the necessity and sufficiency maps of NIH have a rank correlation of 0.33 to 0.54 across the four operator–resolution cells (closed form). Every cell is listed in Table S7.

S1.8 Mapping methods

Score-CAM was applied at the activation site, where each channel's normalised map masks the activation and the channel scores are turned into weights by a softmax across channels, which differs from its input-space original. Every method was evaluated for the class the degenerate model predicts.

S1.9 Localisation measures and baselines

MaxBoxAccV2 [11] sweeps 100 map thresholds, draws a box around every 8-connected component and averages over the overlap thresholds $\{0.3, 0.5, 0.7\}$. Best IoU is the per-image maximum box IoU over thresholds, and peak-mask IoU is the per-record maximum mask IoU over thresholds, with VOC void pixels excluded. The pointing game [12] counts a hit when the map's maximum falls inside a box (CUB, VOC boxes) or a mask (VOC masks, COCO) of the class, and its tolerant variant enlarges boxes by 15 pixels or dilates masks. The tolerance is applied at the 224×224 evaluation resolution rather than at the original image resolution, which makes the tolerant game more lenient than in its original definition [12]. CUB used MaxBoxAccV2, best IoU and both pointing games, VOC boxes used the pointing games, and VOC masks and COCO used the pointing games and peak-mask IoU. The centre prior, the image- and class-independent baseline of the pointing game [12] and of the weakly supervised localisation protocol [10, 11], is an isotropic Gaussian at the grid centre with $\sigma = 3.5$ cells (in the medical experiments the negative squared distance to the centre, which ranks the cells identically). Unlike the class-agnostic average mask of [13], scored by precision and recall, the average-annotation box is the mean box of the same finding, built from box annotations, cross-fitted over patients of the evaluation set on NIH and built from non-evaluation patients on RSNA, and scored at the centroid of its plateau. A map that is identically zero has its maximum at the first pixel and is scored as a miss. The effect of this rule on Grad-CAM is reported in Note S2.9. ChestX-ray14 also defines its own localisation measure, the accuracy at thresholds of the intersection over the detected bounding box [27], but the pointing hit is used here because it applies to every map without a threshold. Faithfulness was summarised with the deletion and insertion curves of [9].

S1.10 Statistical analysis

Tests were two-sided: the Wilcoxon signed-rank test [31], the exact McNemar test [32], and a paired image bootstrap [33] for MaxBoxAccV2, which maximises over thresholds, with the P value taken as twice the smaller tail of the bootstrap distribution (floored at 1/1,000). Fisher's exact test compared two independent proportions. Percentile bootstrap intervals used 2,000 resamples for localisation (1,000 for MaxBoxAccV2 and for the end-to-end experiment) and 10,000 for the medical comparisons. Where the unit is one of five training seeds, t intervals were used, and proportions from small samples carry Wilson intervals [34]. Because NIH carries 984 boxes on 880 images, the NIH primary contrast was also re-estimated with an image-level cluster bootstrap. The Holm step-down correction [35] was applied at $\alpha = 0.05$ within each of the 36 localisation families of four or eight contrasts, and no correction runs across families. A localisation record is one image–class pair, and the verdicts were also recomputed with the image as the exchangeable unit by an image-level paired sign-flip permutation test. The verdict of a contrast comes from the test, while its interval is unadjusted, so the two can disagree, and they do in 3 of the 192 primary contrasts. Tallies count dependent contrasts: the two separated maps share a model, several metrics share records, and the two VOC benchmarks share heads, fitting split and 1,122 of the 1,449 segmentation images. The faithfulness intervals of Note S2.3 are unadjusted. The accuracy cost of each normalisation edit was tested with an exact McNemar test on the 50,000 per-image outcomes.

S1.11 Designated-in-advance and post hoc analyses

Table S6 lists every analysis with its status and the four deviations (i)–(iv) of the localisation and medical analyses from their protocol documents. The localisation protocol fixed $T = +1$ for the aggregator arm without giving a reason, and the medical protocol fixed $T = +8$ for the reason given in Note S1.7.2. The protocol documents carry no time stamp, so their timing cannot be verified independently. They are included, with the code that produced every verdict, in the code-and-data archive. The 192 localisation contrasts are called primary rather than designated in advance, because 48 of them were not in the written protocol. The 3×3 log-sum-exp maps were computed with a block summation of exponentials whose accuracy was characterised only on a stand-in (Table S6, deviation (iv), and Note S2.8), so those rows are indicative. The 3×3 maps of the relocation arm do not use that summation.

The conservative check, the measured census, the channel-wise pooling probe, the end-to-end experiment and the three retrained ISIC classifiers were fixed in protocol documents before they were run. The documents and their SHA-256 digests are part of the archive, a record kept by the author rather than by a third party. Their departures from the protocols are as follows. The unit-scale label of the census (Note S2.6) is a rule taken from the census audit and not written in the census protocol, and the post hoc variant of the conservative check is not part of the protocol. The protocols promised to archive trained checkpoints, but the end-to-end and ISIC checkpoints are kept by the author and available on request because of their size. The protocol also asked for a test on heads of the HuggingFace transformers library. It was not run, because the library was not available in the offline environment. In the end-to-end experiment the four seed-2 runs were stopped by a shutdown of the host, and H0 with seed 1 twice by the operator, once after 17 epochs, before any of their outputs existed. All were trained again from the start with the same seeds, and the readings of details the protocol leaves open are listed in the code-and-data archive. For the retrained ISIC classifiers, the timm ResNet-50 weights used for ISIC models 1 and 2 (Table 3 of the main text) were not available offline, so the torchvision weights were used, as for model 3. In all ISIC models, cuDNN kernels were not forced to be deterministic. The scoring of each end-to-end head's own maps and of within-image ranks is post hoc.

S2 Extended results

This note lists, as numbered paragraphs, the results that the main text cites and that no table carries, in the order of the methods. Each subsection names the tables and figures that report the same analysis in full.

S2.1 The identity at double precision

1. On the DenseNet-121 [18] chest-radiograph classifier of Note S1.7.2, a literal double-precision loop over cells gave a worst relative difference between the two quantities of 3.56×10^{-14} at single-cell resolution and 9.77×10^{-15} at 3×3 resolution.
2. Over the 89 architectures of the checked set certified additive (pretrained weights), the largest residuals were $\varepsilon_1 = 4.43 \times 10^{-14}$, $\varepsilon_2 = 8.09 \times 10^{-14}$ and $\varepsilon_3 = 4.17 \times 10^{-14}$. The 48 measured readouts that are not additive have ε_1 between 0.38 and 1.29 (Note S2.6).
3. No measured distributed readout had a residual between 10^{-12} and 10^{-4}, so the threshold of 10^{-9} is not delicate.
4. At the degenerate point Grad-CAM, HiResCAM and XGrad-CAM produced the same map, as the published equivalences predict [1, 3, 4].

S2.2 Operations that separate the two quantities

Fig. S2 gives every panel of this analysis.

1. A per-cell layer normalisation and a per-cell multilayer perceptron before pooling both left the separation at zero, with a float64 set-level residual of at most 4.2×10^{-15}.
2. Replacing the mean by log-sum-exp, max or gated attention pooling separated the quantities by between 0.067 (log-sum-exp at $T = 0.25$) and 0.628 (max pooling). Keeping the mean but adding a non-affine tail separated them by 0.626 for the bilinear head and by 0.155 for a layer normalisation after the mean (set-level residual 0.68). A two-layer perceptron after the mean separated them by 0.049 for single cells, 0.022 for 3×3 regions and 0.0023 for half-maps.
3. Against one reference, the rank separation was exactly zero for log-sum-exp pooling at every temperature. It was 0.725 for max pooling (where 91.5% of the necessity values are exactly zero, so the value largely reflects ties), 0.907 for gated attention pooling, 0.633 for the bilinear head and 0.071 for the perceptron after the mean.
4. These small separations depend on the reference scale, falling to 3.3×10^{-5} as the reference approaches the input, and the set-level residual was larger for the perceptron after the mean (0.278) than for the mildest aggregator replacement (0.264), whose separation does not decrease with region size (0.067 for single cells, 0.119 for 3×3 regions). All of these heads stayed within 0.56 top-1 points of the averaged readout on Imagenette.
5. For the same average-pooling head, necessity and sufficiency of the predicted-class softmax probability had a correlation of 0.853 with a refitted 10-class head and 0.688 with the distributed 1,000-class ResNet-50. Every statement about degenerate readouts concerns logit-level maps.

S2.3 Operator families with average pooling as the degenerate member

1. When the aggregator was made learnable, its parameter drifted from the mean toward the disjunctive side in all 15 runs (three learning rates by five seeds), and from ±2 further outward in all 30 runs, none of which returned toward the mean. Only the five runs initialised at the mean with the smallest learning rate gained accuracy (0.148 points, t 95% CI $[0.121, 0.174]$). Over all 45 runs accuracy fell by 1.03 points on average, and by 0.26 points over the 15 initialised at the mean.
2. Deletion and insertion curves [9] were computed within one model (ResNet-50 on Imagenette, a head refitted for each temperature, 32 images with one reference each, curves on the softmax probability), with the head's own reference-subtracted per-cell score $s_t(a) - s_t(b)$ as comparator and necessity, which equals sufficiency in rank here, from the closed form. On deletion the comparator was lower, and therefore better, under conjunctive pooling, by 0.022 to 0.035, and it could not be distinguished from necessity under disjunctive pooling (at most 0.006). On insertion necessity was better under strongly conjunctive pooling, by 0.020 (95% CI $[0.008, 0.033]$) at $T = -8$ and by 0.018 ($[0.006, 0.029]$) at $T = -4$, and indistinguishable elsewhere. The random-ordering baseline of these curves shifted with the operator, from 0.47 to 0.65 across temperatures.

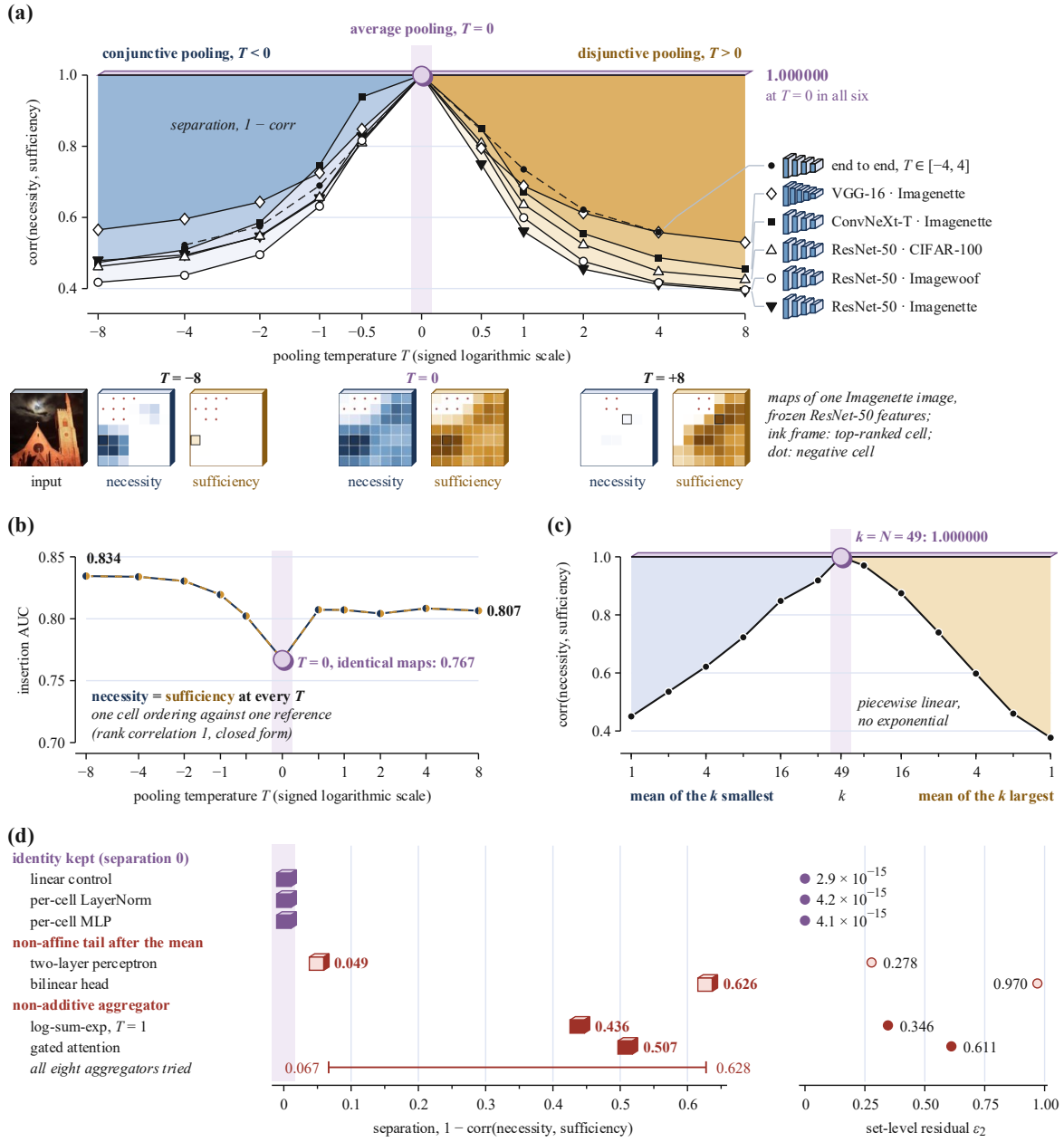


Figure S2. Operations that separate necessity from sufficiency. (a) Correlation between the per-cell necessity and sufficiency estimates (float32, 32 images) against the log-sum-exp pooling temperature T (signed logarithmic axis, eleven temperatures) for five frozen-backbone settings (VGG-16 and ConvNeXt-T on Imagenette, and ResNet-50 on Imagenette, Imagewoof and CIFAR-100) and one network trained end to end, each with its own marker and named at its right end beside a backbone glyph. ResNet-50 on Imagenette (filled inverted triangles) is the setting of panels (b) and (d) and of the maps below. At $T \geq 4$ it nearly coincides with ResNet-50 on Imagewoof. The shaded area is the separation $1 - \text{corr}$ of the frozen-backbone settings, navy on the conjunctive side ($T < 0$) and ochre on the disjunctive side ($T > 0$), deeper where more settings overlap. The end-to-end network, sampled on $T \in [-4, 4]$, is dashed. All six reach 1.000000 at $T = 0$ (average pooling, purple disc). Below the axis, the necessity (navy) and sufficiency (ochre) maps of one Imagenette validation image on its 7×7 grid at $T = -8$, 0 and +8 (log-sum-exp heads on frozen ResNet-50 features, closed form in float64). Lightness is the square root of the value relative to the largest absolute value of the pair, the ink frame marks the top-ranked cell and a red dot a negative cell. Along this axis the separation is a difference of magnitude: the two maps of a pair share their top-ranked cell, and against one reference the cell ordering is shared (panel (b)). (b) Insertion area under the curve (AUC) across the same axis (ResNet-50, Imagenette, one reference per image). Against one reference the necessity and sufficiency maps of a log-sum-exp head order the cells identically (Eq. (2) of the main text), so their insertion curves coincide at every temperature and are drawn as one two-tone curve with split markers: 0.767 at $T = 0$, 0.834 at $T = -8$ and 0.807 at $T = +8$ (closed form, float64). (c) k-of-N pooling, the mean of the k smallest (left) or k largest (right) of the $N = 49$ cell scores, reparameterised so that $k = N$ falls at $T = 0$. It closes there, at 1.000000, and the curve is piecewise linear. (d) Separation (left, cube markers) and set-level residual ε_2 (right) of seven readouts of the same features (ResNet-50, Imagenette), grouped by where they break the identity: a linear control and two per-cell nonlinear maps before pooling (per-cell layer normalisation and a per-cell multilayer perceptron, MLP) keep it, with separation 0 in float64 and ε_2 between 2.9×10^{-15} and 4.2×10^{-15}. A non-affine tail after the mean (a two-layer perceptron, 0.049, $\varepsilon_2 = 0.278$; the bilinear head, 0.626, $\varepsilon_2 = 0.970$, open markers) and a non-additive aggregator (log-sum-exp at $T = 1$, 0.436, $\varepsilon_2 = 0.346$; gated attention, 0.507, $\varepsilon_2 = 0.611$, filled markers) separate. The bar spans the separation of all eight aggregators tried, 0.067 to 0.628.

Table S15. Placement of the pool-adjacent normalisation in ten distributed ImageNet models. Corr: correlation between the necessity and sufficiency maps over 256 images, as shipped and after the normalisation is moved to the other side of the spatial mean. Top-1: shipped model, on the 50,000-image validation set of the ImageNet mirror. Relocate and Delete: change in top-1 points under relocation and under deletion of the normalisation. The cheaper available edit is in bold, and * marks an exact McNemar $P < 0.05$. Deletion leaves Swin-T degenerate, so it is not an edit that changes that model, and MaxViT-T is reached by neither edit. Corr and the rank correlations of Note S2.5 are float32 measurements. Re-measured in float64 with the literal per-cell loop on 96 images, the shipped EVA-02-Ti gives 0.545 (95% CI $[0.484, 0.605]$) against 0.41 ($[0.375, 0.449]$, 256 images), whereas the float64 intervals of the other eight separable configurations overlap their float32 intervals. RoPE, rotary position embedding.

Model	Normalisation	Corr shipped	Corr edited	Reached	Top-1	Relocate	Delete
EVA-02-Ti/14 (336 px)	after the mean	0.41	1.00	yes	77.45	−0.71*	**−0.16***
NaFlexViT-B/16 (GAP)	after the mean	0.49	1.00	yes	79.88	**−0.09***	−0.81*
ViT-Betwixt/16 (4 registers)	after the mean	0.49	1.00	yes	80.92	−0.35*	**+0.05**
BEiT-v2-B/16	after the mean	0.62	1.00	yes	83.12	−0.35*	**−0.12**
ViT-B/16 (RoPE, 1 register)	after the mean	0.63	1.00	yes	80.71	−0.12*	**−0.04**
ViT-M/16 (4 registers)	after the mean	0.65	1.00	yes	81.06	−0.45*	**−0.10**
ViT-Little/16 (4 registers)	after the mean	0.66	1.00	yes	79.91	−0.80*	**−0.43***
ConvNeXt-T	after the mean	0.86	1.00	yes	78.91	−3.90*	**−0.59***
Swin-T	inside the mean	1.00	0.87	yes	77.58	**−0.08**	n/a
MaxViT-T	after the mean	0.51	0.93	no	79.98	n/a	n/a

timm weight tags, in row order: eva02_tiny_patch14_336.mim_in22k_ft_in1k, naflexvit_base_patch16_gap.e300_s576_in1k, vit_betwixt_patch16_reg4_gap_256.sbb_in1k, beitv2_base_patch16_224.in1k_ft_in1k, vit_base_patch16_rope_reg1_gap_256.sbb_in1k, vit_medium_patch16_reg4_gap_256.sbb_in1k, vit_little_patch16_reg4_gap_256.sbb_in1k, convnext_tiny.fb_in1k, swin_tiny_patch4_window7_224.ms_in1k and maxvit_tiny_tf_224.in1k.

S2.4 The coincidence at 1,000 classes under the distributed classifier

Fig. S3 gives every panel of this analysis.

1. With the distributed ResNet-50 weight matrix inserted unmodified, average pooling reproduced the shipped model (73.28% top-1 on the evaluation split). Over 224 validation images the correlation between the two quantities was 1 to ten decimals at $T = 0$ in float32, 0.374 (95% CI $[0.363, 0.386]$) at $T = +8$ and 0.411 ($[0.395, 0.429]$) at $T = -8$. Top-1 fell to 64.63% at $T = +8$ and to 18.12% at $T = -8$.
2. Grad-CAM agreed equally with both quantities, with a Spearman correlation of 0.941 at $T = 0$, 0.976 ($[0.965, 0.985]$) at $T = +8$ and 0.030 ($[-0.010, 0.072]$) at $T = -8$.
3. Computed by float32 forward differences instead of the closed form, the small values cancel and the two maps appear to rank cells differently (Spearman 0.29 at $T = +8$), a difference that is an artefact of the arithmetic.

S2.5 Relocating the pool-adjacent normalisation

Table S15 and Fig. S4 report this analysis per model.

1. The ten models are eight from the measured set and two others. The rank correlation of the two maps was between 0.36 (EVA-02-Ti) and 0.75 (ConvNeXt-T) as shipped, and 0.80 for Swin-T with the normalisation moved out of the mean.
2. Making the eight separable models degenerate by relocation cost between 0.09 and 3.90 top-1 points (mean 0.85, or 0.76 over the nine relocations, including that of Swin-T in the reverse direction), and eight of the nine relocations were significant by the exact McNemar test (Swin-T, $P = 0.11$). Deletion cost at most 0.81 points (mean 0.27) and was significant on four of eight models. The cost of the cheaper edit was at most 0.59 points, for deletion on ConvNeXt-T (−0.594, 95% CI $[-0.778, -0.410]$, exact McNemar $P = 2.5 \times 10^{-10}$), and it was not significant on five of the nine models by the exact McNemar test, where the widest interval reaches a loss of 0.26 points. The one degenerate model became separable for 0.08 points.
3. Every edit that made a model degenerate was re-certified in float64, and the worst residual was $\varepsilon_1 = 2.4 \times 10^{-13}$ (EVA-02-Ti with the normalisation moved inside the mean).
4. A published ablation of self-supervised transformers [36] reports 76.3% for a mean-pooled readout against 69.7% with a normalisation inside the mean. On six frozen timm transformers (two supervised average-pooling models, two supervised class-token models, DINO [37] and MAE [38]), the linear probe absorbed the normalisation's trained affine parameters to within 0.0005, and the inside-the-mean ordering cost nothing on the self-supervised bodies (+0.14 and −0.39 points). For the ViT-Tiny/16 trained from scratch, the inside-the-mean ordering cost 1.08 points (t 95% CI $[-1.53, -0.62]$) and the outside ordering 0.42 points ($[-0.94, +0.10]$, not significant).

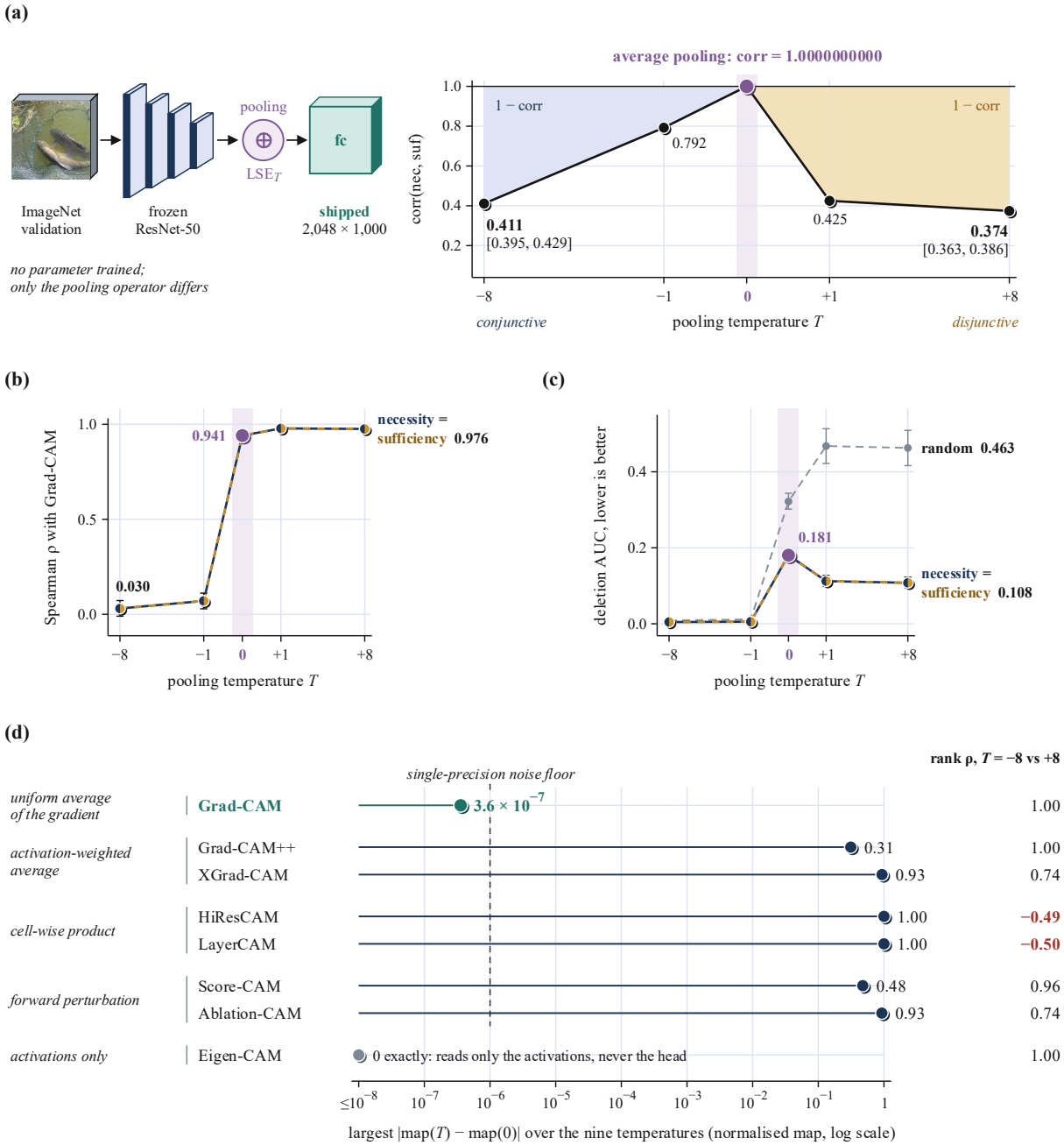


Figure S3. The coincidence at 1,000 classes, and the response of eight mapping methods to the pooling operator at fixed head weights. (a) Set-up and result. An ImageNet validation image passes through the frozen ResNet-50 backbone, a log-sum-exp pooling operator of temperature T (LSE$_T$) and the distributed fully connected (fc) layer, whose 2,048 × 1,000 weight matrix is inserted unmodified into every operator, so that no parameter is trained and only the pooling operator differs. The curve gives corr(nec, suf), the Pearson correlation between the necessity and the sufficiency map, over 224 validation images at five temperatures with bootstrap 95% intervals. The side $T < 0$ is conjunctive (towards minimum pooling) and $T > 0$ disjunctive (towards maximum pooling), and the gap to 1 is shaded navy on the conjunctive side and ochre on the disjunctive side. The correlation is 1.0000000000 at $T = 0$ (average pooling), 0.792 at $T = -1$ and 0.425 at $T = +1$, and falls to 0.411 ([0.395, 0.429]) at $T = -8$ and 0.374 ([0.363, 0.386]) at $T = +8$. (b) Spearman rank correlation (ρ) between Grad-CAM and the two quantities. Against one reference the two maps share one cell ordering, so the agreement is the same for both and is drawn as one curve, navy with ochre dashes and split navy/ochre markers: 0.941 at $T = 0$, 0.976 at $T = +8$ and 0.030 at $T = -8$. At $T = 0$ the agreement is below 1 because the two quantities subtract the reference's per-cell scores, whereas Grad-CAM, which equals plain CAM there, does not. (c) Deletion area under the curve (AUC, lower is better) of the necessity and sufficiency maps, which coincide (one curve, as in (b): 0.181 at $T = 0$ and 0.108 at $T = +8$), with a random ordering for reference (dashed, 0.463 at $T = +8$). At $T < 0$ a min-pooled logit collapses after any deletion, so both curves, including the random one, are near zero there. Panels (b) and (c) use the closed form in float64, with bootstrap 95% intervals. In (a)–(c) the purple band and marker denote the degenerate point $T = 0$. (d) Largest movement of each normalised map across nine temperatures relative to its value at $T = 0$ (128 images, logarithmic scale, with the dashed line marking the single-precision noise floor), grouped by how the map weights the gradient: a uniform average of the gradient over cells (Grad-CAM, 3.6×10^{-7}, below the noise floor), an activation-weighted average (Grad-CAM++, XGrad-CAM), a cell-wise product (HiResCAM, LayerCAM), forward perturbation (Score-CAM, Ablation-CAM), or activations only (Eigen-CAM, which reads only the activations and does not move at all). The rank correlation (ρ) of each map between $T = -8$ and $T = +8$ is given at the right, and the two negative values (HiResCAM −0.49, LayerCAM −0.50) are set in red.

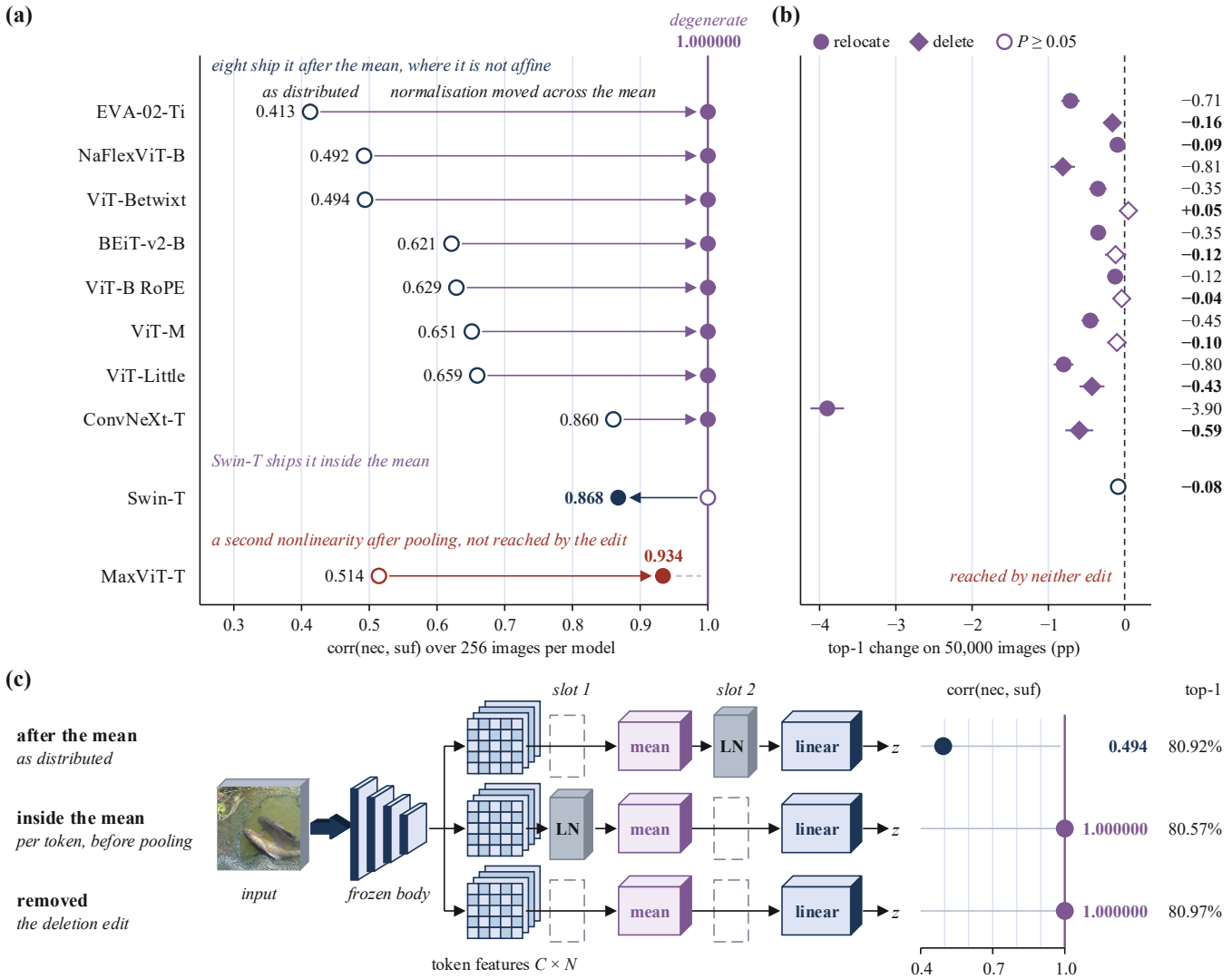


Figure S4. Relocating the pool-adjacent normalisation in ten distributed ImageNet models (model names as in Table S15). (a) Correlation between the necessity and sufficiency estimates, corr(nec, suf), over 256 images per model, as distributed (ring) and after the normalisation is moved to the other side of the spatial mean (dot). The purple line marks the degenerate value 1.000000. Eight models ship the normalisation after the mean, where it is not affine, and go from 0.41–0.86 to 1.000000 (degenerate). Swin-T ships it inside the mean and goes from 1.000000 to 0.868 (separable). MaxViT-T, which keeps a second nonlinearity after pooling, is not reached by the edit and moves only from 0.514 to 0.934 (dashes: the remaining distance to 1). (b) Top-1 change in percentage points (pp) on the full 50,000-image validation set, on the same rows: relocation (circle) and deletion (diamond) of the normalisation, each with its paired 95% confidence interval. A filled mark is significant by the exact McNemar test ($P < 0.05$) and an open mark is not. Purple marks an edit that makes the model degenerate, navy one that makes it separable, and the dashed line is no change. All nine relocations are below zero (eight significant, mean −0.76, largest −3.90 for ConvNeXt-T). Deletion is cheaper than relocation on seven of eight models and significant on four. The right-hand column gives the relocation (upper) and deletion (lower) change, the cheaper available edit in bold. Deletion leaves Swin-T degenerate, so only its relocation is shown. MaxViT-T is reached by neither edit. (c) The three orderings on one model (ViT-Betwixt), each drawn as the head it produces from the same input, the same frozen body and the same weights (input: a real ImageNet validation image; token features $C \times N$; LN, layer normalisation; slots 1 and 2, the positions before and after the mean; z, the class logit), with corr(nec, suf) and top-1 accuracy of each: only the normalisation after the mean is non-degenerate (0.494, with top-1 80.92%, against 80.57% inside the mean and 80.97% removed). Before any edit the head decomposition reproduces the distributed outputs of all ten models at zero relative error.

S2.6 Census, held-out agreement and versions of the structural check

Tables S1, S11, S12 and Table S16, Table 2 of the main text and Fig. S5 report this analysis per architecture or per version of the check.

1. Over the 895 census entries the original check returned 532 degenerate verdicts, 103 blocked by a normalisation, 118 nonlinearity-blocked and 142 class-token readouts, with no failures and no verdict blocked by the aggregator. The verdicts recomputed from the stored census record were identical on all 895 entries.
2. The audit of the 30 census architectures whose head trace had never been measured found five wrong verdicts (Table S12). Three CoaT models reduce a stack of three class tokens with a one-dimensional convolution, which the check reads as an additive aggregator over cells. When measured, no spatial cell moves their logit. Two ConvNeXt-Zepto models use a root-mean-square (RMS) normalisation that the lookup tables label nonlinear. When measured, they are blocked only by that normalisation. The eleven VOLO checkpoints share the head of VOLO-D1, whose hidden maximum over cells the check reads as a class-token readout. With all of these corrected, the census counts 529 degenerate architectures (59.1%), 105 normalisation-blocked (11.7%), 116 nonlinearity-blocked (13.0%), 134 class-token readouts (15.0%) and 11 blocked by the aggregator (1.2%).
3. The 895 entries realise 43 distinct head traces, of which 14 are degenerate, 5 normalisation-blocked, 18 nonlinearity-blocked, 5 class-token readouts and 1 blocked by the aggregator. The 33 traces of the 148 checked architectures cover 865 entries. A broader population, every architecture with any ImageNet-1k classifier weights (901 classified), gives 58.8% degenerate after the same corrections.
4. Of the 267 verdicts decided on the declared-pool branch, 62 are degenerate verdicts among the 529, all of which declare average pooling, which an operator-level trace confirms, and 142 are the class-token verdicts of the check. The population includes the 13 small test models that timm registers with ImageNet-1k weights, 7 of them degenerate. Without them the audited degenerate share is 522 of 882 (59.2%). It excludes the four entries with more than 10^9 parameters (Note S1.2).
5. The operator-level variant of the original check (added after the VOLO-D1 disagreement and not part of the check) changes 23 of the 895 verdicts: all 11 VOLO checkpoints from class-token to aggregator-blocked, all 11 CrossViT checkpoints from class-token to degenerate, and one attention-pooled test model from nonlinearity-blocked to class-token. The CrossViT change is itself a mislabel, since the mean the variant finds is taken over the two branches' class-token logits rather than over cells. The census reported uses the module-level check.
6. Measured census. Of the 895 entries, 877 were certified directly (518 additive, 98 normalisation-only, 127 hard-blocked, 134 vacuous). The other 18 have a classifier initialised to zero and were certified after re-drawing those weights: the gMixer, gMLP, MLP-Mixer, ResMLP and Sequencer2D entries are additive and the seven Hiera entries normalisation-only. Agreement of the five-class verdict with these labels was 879 of 895 for the original check, 883 for the post hoc variant of the conservative check and 895 for the audited and conservative checks. All 30 entries of the traces audited by hand had the label the audit gave them. Split by head trace (post hoc), the original check agreed with all 790 entries whose head trace a development architecture shares (772 certified directly and the 18 re-drawn) and with 89 of the 105 whose trace none shares, against 56 of 105 for the majority-verdict baseline (degenerate). The 16 errors are the eleven VOLO, three CoaT and two ConvNeXt-Zepto entries, and its binary answer was right on 94 of the 105. On the 147 measured development and held-out architectures, the label at random weights equalled the label at pretrained weights on 139, and the eight others are zero-initialised classifiers. For the 144 checked architectures that belong to the census population, the verdict on random weights equalled the verdict on trained weights.
7. The own-feature and unit-scale labels differed on 22 entries. The unit-scale label is reported, a rule taken from the census audit and not written in the census protocol (Note S1.11). Twenty of the 22 are structurally additive heads (eleven RepVGG, three ConvMixer, three pruned EfficientNet, Inception-v4, NASNet-A-Large and RepViT-M2.3) whose own-feature certificates fail numerically: the readouts of ConvMixer, pruned EfficientNet and Inception-v4 nearly vanish (largest necessity values 3×10^{-15} to 8×10^{-14}), the per-cell differences of RepVGG are zero or at rounding level ($\leq 3\times10^{-17}$), so that the normalised residuals measure rounding or overflow, and NASNet and RepViT (largest necessity values 1.2×10^{-8} and 4.0×10^{-6}, and ε_2 of 1.1 to 1.3×10^{-9}) lie just above the 10^{-9} threshold. On their own features they read hard-blocked (one RepVGG model vacuous), on unit-scale features additive. The other two, GhostNetV2-130 and MobileNetV4-hybrid-L, read additive on their own features although a nonlinearity follows the pool, because at random weights that nonlinearity stays in one regime. On unit-scale features they read hard-blocked.
8. Scored against the own-feature labels, under which the census is 57.1% additive (511 entries), the original, audited and conservative checks would each return 19 false degenerate verdicts, all on the first group of the previous paragraph, and the conservative check would agree with 851 of the 877 directly certified labels instead of all 877 (855 without the four PiT-distilled labels of Note S1.2), the original check with 835, and the original check would be right on 34 instead of 35 of the 41 entries with head designs it had not seen. At random weights the audit also met heads whose residuals fall inside the gap of Note S2.1 (EfficientViT-L, ε_1 about 3×10^{-5}), because small activations keep a nonlinearity near its linear regime. On unit-scale features they are clearly blocked (ε_1 0.37 to 0.56).

9. Of the 148 development and held-out architectures, the check found 89 degenerate as distributed, 24 normalisation-blocked, 23 nonlinearity-blocked and 12 class-token readouts. On the 82 held-out architectures it agreed with the measured label on 81 (98.8%). The nearest-trace baseline reached 77 of 82 and the majority verdict 57 of 82 (69.5%). On the 11 with a new head trace it agreed on 10 (Wilson 95% interval 0.62–0.98), against 6 for both baselines (exact McNemar $P = 0.125$ against the nearest-trace baseline, four discordant pairs). These new traces are new combinations of module classes that all occur in development traces. The one disagreement, VOLO-D1, adds a maximum over cells in a tensor expression that no module records. Its verdict is class-token, whereas its measured readout is not additive ($\varepsilon_1 = 0.469$).

10. Identifying the post-pool normalisations independently of the lookup tables changed none of the labels, and all 24 normalisation-only architectures became additive when those normalisations were deleted (worst residual 1.0×10^{-12}), whereas none of the 24 hard-blocked ones did.

11. On the 32 constructed heads (Table S11) the original check agreed with the float64 certificate on 18. It returned degenerate on 10, eight of them non-additive and two vacuous stacks of class tokens. These heads were pools and attention written as tensor operations without a declared pool type (among them generalised-mean, log-sum-exp and concatenated average and max pools), an attention-pooling module that the model declares as a pool type the lookup tables do not know, stacks of class tokens, and nonlinear cell mixing before the pool. It missed three degenerate heads, a per-token normalisation or perceptron before a mean written as a tensor operation and a nonlinearity that is never active, and it mislabelled one normalisation. Each error follows from one of six properties of the implementation. A reduction that no module performs is taken to be a mean unless the model declares otherwise. With a declared pool, every recorded operation is treated as tail. Operations before the aggregator are not inspected, and the reduced axis is not verified to be spatial. The lookup tables are incomplete, and a nonlinearity counts as blocking even where it is inactive. Where the aggregator is a pooling module, or a declared pool over the spatial axis, and only per-cell operations precede it, the check erred only through its lookup tables (an unknown normalisation, and an attention pool declared under a name it does not list) and through a nonlinearity that is never active. The operator-level variant agreed on 20 of the 32.

12. Versions of the check (Table S16). On the constructed heads the original check returned a false degenerate verdict on P05, P07, P09, P23, P25, P26, P28 and P29, the audited variant on P07, P09, P23, P28 and P29, and the conservative check on P23 alone, a generalised mean written as tensor operations. The protocol target of no false degenerate verdict is therefore not met on the constructed heads. The conservative check left P07, P09, P11, P12, P27, P28 and P29 unknown. On the census the conservative conditions never fired (no entry unknown). On torchvision it left the six Swin models and the five vision transformers unknown. With the zero-initialised classifiers of the five vision transformers re-drawn, the original and audited checks called all five degenerate although they read a class token, whereas the conservative and post hoc versions did not.

13. Over the 48 measured architectures whose readout is not additive, the 47 with a blocked verdict and VOLO, ε_1 ranged from 0.38 to 1.29 (median 0.92). On the 24 measured normalisation-only architectures it ranged from 0.38 to 1.25 (median 0.84), with a separation from 0.015 to 0.79.

14. For the 21 nonlinearity-blocked architectures whose edit was costed, the cheapest edit that makes the readout degenerate cost between 0.06 and 18.52 top-1 points (median 3.80, Fig. S5(c)). For two held-out models of the 23, EfficientViT-B1 and VGG-19, no cost was measured, the first because it fails with a shape error under the edit and the second because it was not run. Deleting the activation alone sufficed on all twelve tails without a normalisation and on none of the nine with one, but it was the cheapest edit on only four of the twelve. The cheapest edit was the replacement of the activation by its fitted affine part on four architectures, a move of the linear head inside the mean on eight (four with a normalisation and four without), and the deletion of both normalisation and activation on the remaining five. For MaxViT-T, deleting the hyperbolic tangent alone left the readout non-additive ($\varepsilon_1 = 0.90$), whereas deleting the normalisation and the hyperbolic tangent together made it additive at a cost of 0.06 points.

15. Cost per model on a CPU (median, mean): the original check 0.24 s and 0.56 s, the conservative check 0.25 s and 0.57 s, and the certificate 3.41 s and 7.30 s on the census. On torchvision the three costs were 0.27 s, 0.33 s and 3.81 s.

S2.7 Channel-wise pooling and gradient-weighted maps

Table S17 reports the channel-wise pooling probe of Section 4.1 of the main text per head. The paragraphs below also give the gradient-weighted maps of its Section 4.2.

1. Under channel-wise pooling the closed form and the substitution loop give the same orderings, and these differ between the two maps (Table S17). Under score-then-pool log-sum-exp the closed form orders the two maps identically. The substitution loop agrees with the closed form to a relative 2×10^{-15} there too, but at float64 resolution many of its cell values are tied (905 to 3,285 tied values over the 64 images, against at most one in the closed form), because a cell far below the maximum moves the logit by less than one unit in the last place. It is the ties, not a difference in ordering, that lower its rank correlation.

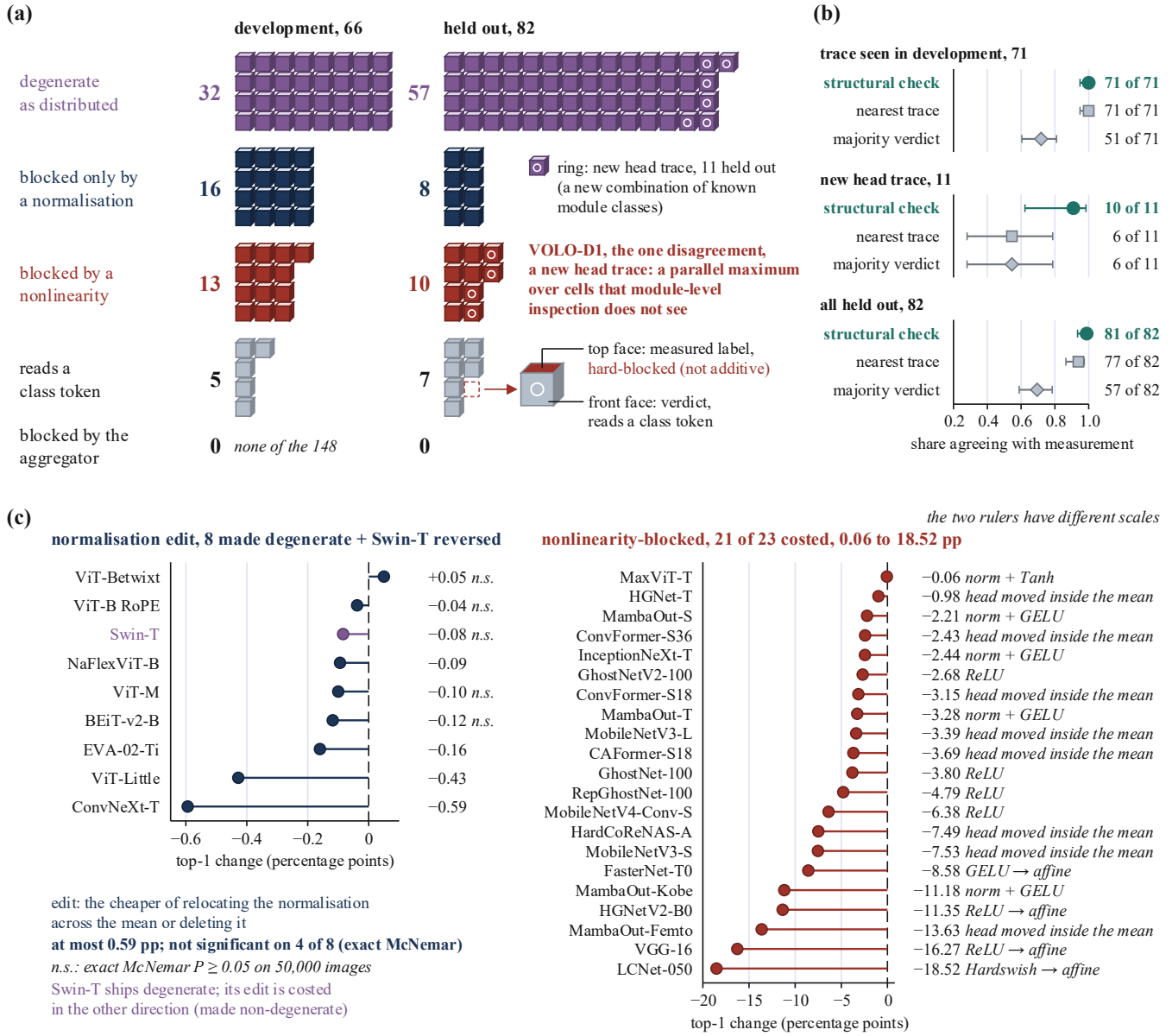


Figure S5. Verdicts of the structural check, their agreement with direct double-precision measurement, and the cost of making each readout degenerate. (a) One cube per architecture, coloured by the verdict of the check. Rows are the five verdicts and columns the 66 development and 82 held-out architectures. Development: 32 degenerate as distributed, 16 blocked only by a normalisation, 13 blocked by a nonlinearity and five class-token readouts. Held out: 57, 8, 10 and 7. No architecture received the fifth verdict, blocked by the aggregator (bottom row, 0 of 148). A white ring marks the 11 held-out architectures whose head trace does not occur in the development set (a new combination of known module classes). The one held-out disagreement, VOLO-D1, is lifted out of its slot and enlarged: its front face shows the verdict (class-token readout) and its top face the measured label (hard-blocked, that is, not additive, in red as for the verdict scored against that label). Its head contains a parallel maximum over cells, a non-additive aggregator that module-level inspection does not see. (b) Share of held-out architectures on which a rule agrees with measurement, with Wilson 95% intervals, for the structural check, a nearest-trace baseline that copies the verdict of the development architecture with the same head trace, and the majority verdict, shown separately for the 71 architectures whose trace occurs in development (71, 71 and 51 agree), the 11 with a new trace (10, 6 and 6, exact McNemar $P = 0.125$ for the check against nearest trace) and all 82 (81, 77 and 57). (c) Top-1 change, in percentage points (pp), of the cheapest edit that moves each readout across the degenerate point, one lollipop per costed architecture, on two rulers with different scales. Left: the normalisation edit, the cheaper of relocating the normalisation across the mean or deleting it, for the eight architectures it makes degenerate (at most 0.59 pp, not significant on four of the eight by the exact McNemar test) and for the Swin transformer (Swin-T), which ships degenerate and is costed in the other direction. A change with exact McNemar $P \geq 0.05$ on the 50,000 validation images is marked n.s. Right: the 21 costed nonlinearity-blocked architectures of the 23 (0.06 to 18.52 pp), each labelled with the operation the edit removes: the activation (ReLU, rectified linear unit), normalisation and activation together (GELU, Gaussian error linear unit; Tanh, hyperbolic tangent), the activation replaced by its fitted affine part (→ affine; Hardswish, hard swish activation), or the head moved inside the mean. Colours: purple, degenerate as distributed; navy, blocked only by a normalisation; red, blocked by a nonlinearity (verdict) or hard-blocked (measured label); slate, class-token readout; green, the structural check.

Table S16. False degenerate verdicts, coverage and binary accuracy of four versions of the check. False degenerate: degenerate where the float64 certificate is normalisation-only or hard-blocked. Coverage: share of entries not left unknown. Binary accuracy: whether the two maps can differ, over decided entries and over all entries (unknown counted wrong). Census: the 877 entries certified directly, with the labels of Note S2.6 (unit-scale label where the own-feature and unit-scale labels differ). Post hoc variant: the variant with two operator-level conditions (Note S1.5).

Set	Check	False degen.	Coverage	Binary (decided)	Binary (all)
32 constructed heads	original	8	100.0%	65.6%	65.6%
	audited	5	100.0%	75.0%	75.0%
	conservative	1	78.1%	84.0%	65.6%
	post hoc variant	0	75.0%	87.5%	65.6%
147 architectures	original	0	100.0%	99.3%	99.3%
	audited	0	100.0%	100.0%	100.0%
	conservative	0	100.0%	100.0%	100.0%
	post hoc variant	0	99.3%	100.0%	99.3%
Census (877)	original	0	100.0%	98.7%	98.7%
	audited	0	100.0%	100.0%	100.0%
	conservative	0	100.0%	100.0%	100.0%
	post hoc variant	0	98.6%	100.0%	98.6%
torchvision (75)	original	0	100.0%	100.0%	100.0%
	audited	0	100.0%	100.0%	100.0%
	conservative	0	92.0%	100.0%	92.0%
	post hoc variant	0	86.7%	100.0%	86.7%

Table S17. Spearman correlation between the necessity and the sufficiency map of one image (mean and minimum over 64 images). ch-LSE and ch-GeM: log-sum-exp with temperature T and generalised mean with exponent p applied to each channel, followed by the linear layer; stp-LSE: log-sum-exp of the per-cell class scores.

Head	T or p	Closed form		Substitution loop	
		Mean	Min.	Mean	Min.
ch-LSE	1	0.649	0.036	0.649	0.036
ch-LSE	4	0.342	−0.362	0.342	−0.362
ch-LSE	8	0.326	−0.396	0.326	−0.396
ch-GeM	1	1.000	1.000	1.000	1.000
ch-GeM	3	0.654	−0.086	0.654	−0.086
ch-GeM	4	0.552	−0.233	0.552	−0.233
ch-GeM	8	0.400	−0.318	0.400	−0.318
stp-LSE	1	1.000	1.000	0.937	0.525
stp-LSE	4	1.000	1.000	0.514	0.245
stp-LSE	8	1.000	1.000	0.366	0.245

2. At fixed head weights the Grad-CAM map equalled the class activation map to within 7×10^{-14} for the mean, log-sum-exp at any temperature, top-k and bottom-k means, max, min, median and linear-gate attention. Gated attention and generalised-mean pooling ($p = 2$ and 3), which are not shift-equivariant, moved the normalised Grad-CAM map away from the class activation map by 0.20, 0.05 and 0.04 on average over the predicted classes of 48 images (largest 0.89, 0.13 and 0.09). Heads refitted for each operator gave different Grad-CAM maps (Spearman 0.60–0.93), as did the refitted medical heads.
3. With the maps averaged over eight references, Grad-CAM agreed at $T = +8$ with averaged sufficiency at 0.99 and with averaged necessity at 0.92 (0.99 and 0.81 with 32 references), and at the localisation temperature $T = +1$ at 0.99 and 0.94 (closed form).

S2.8 Localisation on natural images

Tables S2, S3 and S8–S10, Table S18 and Fig. S6 report these analyses contrast by contrast.

1. On CUB the centre prior scores 0.945 and 0.987 on the two pointing games by itself.
2. Replacing the Wilcoxon and McNemar tests by a paired sign-flip permutation test changed two of the 192 primary verdicts (39, 25 and 32 against the collapsed map). Making the image rather than the image–class record the exchangeable unit of that test changed none. Restricted to the metrics of the protocol, the tally against the collapsed map is 28, 16 and 28 of 72.

3. Aggregator arm. The separated maps won 20 of 72 contrasts, lost 25 and tied 27: 3, 10 and 11 on ResNet-50, 1, 10 and 13 on VGG-16, and 16, 5 and 3 on ViT-B/16. Sixteen of its 20 wins were on the two maximum-location metrics, and of its 24 extent contrasts it won 4 and lost 20. On ResNet-50, CUB top-1 fell from 62.2% to 41.4% and VOC and COCO mean average precision by 1.5 to 3.2 points under the reused head, and at the scale of the heads' per-cell scores $T = +1$ pools over about 1.3 to 2.7 effective cells. The separation of this arm was 0.19 to 0.78 across the twelve dataset–backbone cells, and larger separation went with fewer net wins (Spearman -0.76, $n = 12$, post hoc). The cells are not independent, since the two VOC benchmarks share heads and fitting split. The localisation features were not stored. On ImageNet ResNet-50 activations with eight references at $T = +1$, the rank correlation of the averaged maps is 0.95. Runs at other temperatures (ResNet-50 on CUB and VOC masks), made after the primary outcome, are not interpreted, because their estimator returned non-finite values for some maps at $|T| = 8$.

4. Relocation arm. The separated maps won 19 of 24 contrasts and lost 1, including all 8 extent contrasts of the reported set. This arm separates the quantities by 0.02 to 0.22. In the post hoc control on the same pre-normalisation tokens, the additive readout (a mean and a linear map, certified with worst residual 1.4×10^{-13}) was better than the separated pair in 15, worse in 1 and indistinguishable in 8 of 24 paired comparisons (Holm within each family of 24), and plain CAM better in 22, worse in 1 and indistinguishable in 1.

5. Comparator. The collapsed map was significantly weaker than plain CAM in 20 of 36 dataset–backbone–metric comparisons (all 12 on ViT-B/16, with Holm over the metrics of each dataset–backbone cell), better in 3 and indistinguishable in 13. Against plain CAM the separated maps won 13, lost 6 and tied 5 in the relocation arm and 5, 27 and 40 in the aggregator arm, and 16 of the 39 wins over the collapsed map survived.

6. Refitted log-sum-exp head ($T = +1$, ResNet-50, post hoc, VOC with boxes and masks). Its maps won 11, lost 11 and tied 2 of 24 contrasts against the collapsed map: 10 of 16 maximum-location contrasts were wins (pointing on VOC and COCO), 1 of 8 extent contrasts, and all 8 CUB contrasts were losses. The median per-record separation of its two maps was 0.097 on VOC boxes, 0.067 on VOC masks and 0.003 on COCO, against 0.504 on CUB, and the refitted head was the better classifier on VOC and COCO (COCO mean average precision 0.659 against 0.620 for the averaged head).

7. On CUB under MaxBoxAccV2 the ResNet-50 necessity map of the aggregator arm scored 0.251, the score of a box covering the whole image, against 0.564 for the centre prior and 0.565 for the sufficiency map. On the other backbones necessity scored 0.251 and 0.269 and sufficiency 0.379 and 0.324.

8. Within-model contrasts (Table S9). In the aggregator arm the sufficiency map was better than the necessity map in 21 of 36 contrasts and the necessity map in none, with 15 ties. All 12 extent contrasts favoured sufficiency, and 15 of the 24 maximum-location contrasts were ties. These counts use the Holm family of the primary set enlarged by the within-model contrasts. With those contrasts in a family of their own, or uncorrected, the maximum-location counts are 1 for necessity, 14 for sufficiency and 9 ties. In the relocation arm sufficiency was better in 9 of 12 contrasts and necessity in 3, all on COCO, among them the extent metric of COCO (peak-mask IoU).

9. Protocol controls (Table S10). Against the activation-magnitude and random controls the separated maps won 170 of 192 contrasts, all 22 losses being on CUB or on peak-mask IoU. With the dataset-mean reference, 22 of 28 verdicts matched those obtained with the reference bank.

10. 3×3 unit. Against the collapsed map the separated maps of the aggregator arm were better in 2, worse in 64 and tied in 6 of 72 contrasts (2, 40 and 6 on the pointing games, and 0, 24 and 0 on extent), and those of the relocation arm better in all 24. Against the readout of the same tokens that cannot represent the distinction, the separated 3×3 maps were worse in 15 of 24 paired comparisons and better in none (post hoc). Within each model, the 3×3 pointing contrasts of the aggregator arm split 9 for necessity, 10 for sufficiency and 5 ties, with necessity better on strict pointing for every convolutional backbone. On the stand-in (the shipped ResNet-50 head on 7×7 ImageNet activations at $T = +1$) the block summation located the maximum as the closed form does in 223 of 224 necessity maps and all 224 sufficiency maps, with largest normalised errors of 0.30 and 0.94, and at $T = +2$ the necessity agreement fell to 0.94.

11. The participation ratio of a map, $(\sum_t |m_t|)^2 / \sum_t m_t^2$, counts the grid cells over which its magnitude is spread (post hoc, means over images). In all 16 primary 1×1 rows (12 dataset–backbone combinations in the aggregator arm and 4 in the relocation arm) the averaged sufficiency map was the more spread of the two: over 11 to 161 of the 196 grid cells against 1.6 to 23 for the necessity map in the aggregator arm, and over 136 to 177 against 122 to 173 in the relocation arm.

12. Heads trained end to end: estimator checks. Every map agreed with the literal substitution loop to within 1.2×10^{-13} of the largest map value, and the maps of H0 were identical to within 2×10^{-14}. Against a single reference the two maps of H1 had a rank separation of exactly 0 in each seed with the cancellation-free closed form. The plain double-precision evaluation gives 0.01, through ties among cells whose share of the partition function is below its resolution, and the localisation maps of the two evaluations differ by at most 1.7×10^{-14}.

13. Heads trained end to end: primary family (Table S18). The sufficiency map of H3 exceeded the collapsed map in each seed (0.0085, 0.0142 and 0.0074, with a t interval over the three seeds of +0.001 to +0.019), and also exceeded plain CAM of H0 in MaxBoxAccV2 and best IoU, but not in strict pointing. Every necessity map scored below the sufficiency map of the same model in all three metrics, and the necessity map of H1 scored 0.2509 in each seed, the MaxBoxAccV2 of a box

Table S18. Heads trained end to end on CUB-200-2011 (ResNet-50, three seeds each, 5,794 test images). Top-1 and MaxBoxAccV2 are means over seeds. Separation is 1− Pearson correlation of the necessity and sufficiency maps, and rank separation 1− Spearman correlation, against one reference and after averaging over eight, per image and averaged. Differences are each map minus the collapsed map of H0, seed-matched and averaged over seeds, with a 95% bootstrap interval over images for MaxBoxAccV2. Each of the six primary contrasts (H1–H3) has a Holm-adjusted P of 0.006, the smallest value 1,000 resamples allow. Best IoU and strict pointing are secondary outcomes (all differences with an absolute value of at least 0.005 have $P_{\text{Holm}} \leq 0.006$, and those of 0.001 or less are not significant). N, necessity; S, sufficiency; LN, layer normalisation; LSE, log-sum-exp. [a]Cancellation-free closed form. Plain double-precision evaluation gives 0.01 / 0.18.

Head	Map	Top-1	Separation 1 / 8 ref.	Rank sep. 1 / 8 ref.	MaxBoxAccV2	Difference [95% interval]	Best IoU difference	Pointing difference
H0, mean	collapsed	0.824	0 / 0	0 / 0	0.605	–	–	–
	plain CAM				0.609	+0.005 [+0.002, +0.007]	+0.002	0.000
H1, LSE	N	0.765	0.43 / 0.47	0 / 0.17[a]	0.251	−0.354 [−0.362, −0.347]	−0.276	−0.006
	S				0.538	−0.067 [−0.072, −0.062]	−0.048	−0.005
H2, attention	N	0.818	0.52 / 0.44	0.99 / 0.81	0.424	−0.181 [−0.188, −0.173]	−0.093	−0.130
	S				0.444	−0.161 [−0.168, −0.155]	−0.042	+0.001
H3, mean + LN	N	0.819	0.24 / 0.19	0.35 / 0.37	0.423	−0.182 [−0.188, −0.175]	−0.062	−0.013
	S				0.615	+0.010 [+0.006, +0.015]	+0.019	+0.001

covering the whole image. Layer normalisation cost 0.5 top-1 points (−0.0051, 95% t interval −0.0081 to −0.0021 over the three seed-matched differences), log-sum-exp pooling 5.8 points, and gated attention a difference that was not significant (−0.0063, −0.0159 to +0.0034).

14. Heads trained end to end: each head's own maps (post hoc, Holm over the seven maps). The pipeline reproduced the stored CAM of H0 exactly in every seed. The attention weights of H2 were better than the collapsed map of H0 (MaxBoxAccV2 0.632, +0.027 [+0.022, +0.032], or +0.014, +0.045 and +0.023 by seed with a t interval over seeds of −0.013 to +0.067), as was CAM of H3 with the normalisation ignored (0.620, +0.015 [+0.011, +0.020]). The class scores of H1 (−0.094), Grad-CAM of H1 (−0.052), the attention-weighted evidence of H2 (−0.026), Grad-CAM of H2 (−0.175) and Grad-CAM of H3 (−0.094) were worse, all with Holm-adjusted $P = 0.007$. Grad-CAM of H0 scored 0.611. In strict pointing the attention weights and CAM of H3 did not differ from the collapsed map. In best IoU the attention weights were worse (−0.023 [−0.026, −0.021], Holm-adjusted $P = 0.007$). Paired with the sufficiency map of the same head (a single contrast), CAM of H3 with the normalisation ignored was the better map: +0.005 [+0.002, +0.008] in MaxBoxAccV2 ($P = 0.001$, the smallest value 1,000 resamples allow, and +0.007, +0.002 and +0.007 by seed), +0.004 [+0.003, +0.005] in best IoU and +0.024 [+0.021, +0.026] on within-image ranks. In strict pointing the two did not differ.

15. Heads trained end to end: within-image ranks (post hoc, with the same bootstrap and Holm families). Of the 15 contrasts with the collapsed map, two changed sign, both maps of H2: the attention weights (+0.027 on values, −0.001 [−0.006, +0.004] on ranks) and the attention-weighted evidence (−0.026 on values, +0.020 [+0.015, +0.025] on ranks, positive in every seed). Most losses shrank, most of all for H1 necessity (−0.354 to −0.075), whereas Grad-CAM of H3 lost more (−0.094 to −0.112). The gains grew: H3 sufficiency +0.017 [+0.013, +0.021], CAM of H3 +0.041, plain CAM of H0 +0.012.

S2.9 Medical images

Tables S4, S7, S13 and Table S19, Table 3 of the main text and Fig. S6(c,d) report these analyses cell by cell.

1. Training overlap. Reconstructing the TorchXRayVision patient-level 80/20 split from its published recipe (a patient-grouped shuffle split with seed 0, as in the library's training script) places, by this estimate, 716 of the 880 NIH box-annotated images and 1,102 of the 1,385 RSNA evaluation studies in the training part. The original split files cannot be checked. On the held-out remainder (164 NIH images and 283 RSNA studies) the collapsed-map hit rate was 0.473 against 0.480 on the RSNA training studies (Fisher $P = 0.89$) and 0.225 against 0.227 on NIH, and the primary contrast was negative in both subsets.

2. Classifier quality. The mean ROC AUC of the NIH classifier (0.668) was the same on the images outside its training split and under the library's own preprocessing. Its Emphysema output was below chance (0.487), whereas the library's multi-dataset model reaches 0.580 on Emphysema and the NIH DenseNet-121 trained in this study 0.743. The refitted NIH head had a mean ROC AUC of 0.692 under the mean and 0.650 under the operator. The cross-fitted RSNA models reach a pooled out-of-fold ROC AUC of 0.828 (folds 0.813–0.844). ISIC models 2 and 3 reach a macro AUC of 0.851 and 0.868 (model 1, 0.834), and the three retrained ISIC classifiers (seeds 0–2) 0.878, 0.881 and 0.870. Models 1 and 2 started from timm weights and the other four from torchvision weights (Note S1.7.2).

3. Primary contrasts. They were NIH −0.090 (exact McNemar $P = 3.3 \times 10^{-13}$), RSNA −0.226 ($P = 5.5 \times 10^{-59}$) and ISIC Dice −0.136 (Wilcoxon $P = 3.3 \times 10^{-36}$). On the 182 NIH boxes of the 164 held-out images the NIH contrast was −0.088 ([−0.137, −0.038]), and with the NIH head refitted $P = 3.8 \times 10^{-7}$. With the image rather than the box as the resampling

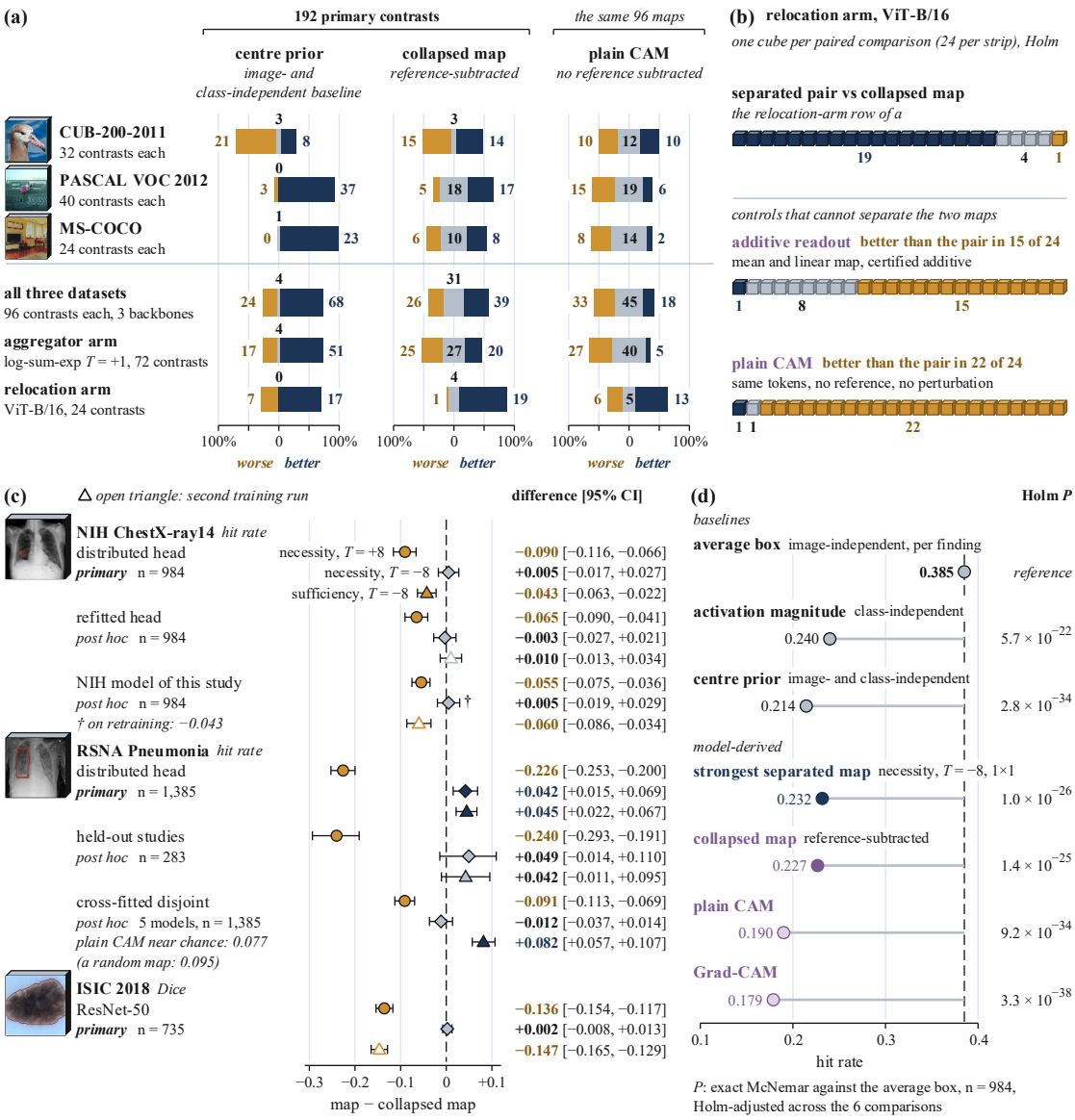


Figure S6. Localisation against human annotation. (a) Outcome of the 192 primary localisation contrasts on CUB-200-2011, PASCAL VOC 2012 and MS-COCO (three backbones, five metrics): the two separated maps against the centre prior, an image- and class-independent baseline (96 contrasts), and against the collapsed, reference-subtracted map (96), Holm-corrected within families, with the same 96 separated maps scored against plain class activation mapping (CAM) of each arm's own readout beside them. Each row is a diverging bar of shares: worse (orange) to the left of 0, no difference after Holm correction (slate) centred on 0, better (navy blue) to the right, with the counts printed. Rows: each dataset with one image and its ground truth (32, 40 and 24 contrasts per comparator), all three datasets (96), the aggregator arm (log-sum-exp pooling at $T = +1$, 72) and the relocation arm (ViT-B/16, 24). (b) The post hoc control on the relocation arm, one cube per paired comparison (24 per strip), coloured from the side of the separated pair. The first strip repeats that arm's result against the collapsed map, and the other two score, against the separated pair, two readouts on the same pre-normalisation tokens that cannot represent the distinction: an additive readout (a mean and a linear map) and plain CAM on these tokens, which uses no reference and no perturbation. (c) Medical contrasts at 1×1 resolution, map minus collapsed map (hit rate, or Dice on ISIC 2018), with 95% bootstrap confidence intervals (CI) printed on the right: the designated in advance necessity map at $T = +8$ (circles), the necessity map at $T = -8$ (diamonds) and, post hoc, the sufficiency map at $T = -8$ (triangles). A mark is orange when its interval lies below 0, navy blue when above 0 and slate when it includes 0. The designated-in-advance arms are NIH ChestX-ray14 and RSNA Pneumonia with the distributed heads and ISIC 2018 with ResNet-50. The other rows are post hoc: NIH with the head refitted and with the NIH model of this study, the 283 RSNA studies outside the distributed classifier's training split, and five cross-fitted disjoint RSNA models, whose plain CAM localises near chance (0.077 against 0.095 for a random map). Filled triangles come from the same training run as the circles and diamonds. Open triangles (outline in the same colour code), for the refitted NIH head, the NIH model of this study and ISIC, come from a second, independent training run, because the first run stored no per-item maps (first-run means +0.010, −0.092 and −0.102, Table S7). On RSNA the triangles' advantage is confined to this 1×1 hit. At 3×3 resolution the same map is worse than the collapsed map, and on average precision it is worse or no better (Table S13). The dagger marks a cell that changes sign on retraining (second run −0.043). Each row gives its status and the number of images n. (d) Hit rate against radiologist boxes on NIH ChestX-ray14 ($n = 984$): baselines (the average annotation box of the same finding, cross-fitted over patients; activation magnitude; the centre prior) and model-derived maps (the strongest separated map, necessity at $T = -8$ and 1×1; the collapsed map; plain CAM; gradient-weighted CAM, Grad-CAM), each joined to the average box (dashed line, 0.385) by its gap, with the exact McNemar test against the average box, Holm-adjusted across the six comparisons.

unit, the NIH primary is −0.090 ([−0.115, −0.066]). In the cross-fitted RSNA models plain CAM and Grad-CAM scored 0.077 and 0.078, a random map 0.095, the centre prior 0.108, the collapsed map 0.176 and the necessity map at $T = +8$ 0.085.

4. Closed form. Recomputed from Eq. (2) of the main text, the NIH primary contrast is unchanged item for item and the RSNA primary is −0.226 ([−0.253, −0.199]). The primary of the first ISIC model could not be recomputed, because its per-item maps and trained model were not retained. The second training run supplies the replicate cells of Table S7 (primary −0.120). On model 3, trained for this recomputation, the primary Dice contrast is −0.050 ([−0.059, −0.041]) with the forward-difference estimator and −0.025 ([−0.035, −0.015], $P = 0.0024$) in closed form, whereas the same contrast by hit rate is +0.022 ([−0.001, 0.045]) and by average precision near zero. The conjunctive necessity map beats the collapsed map on Dice (+0.050, [0.038, 0.063]), as does the disjunctive sufficiency map at 1×1 on hit rate (+0.026, [0.001, 0.050]) and on AP (+0.021, [0.014, 0.028]).

5. Retrained ISIC seeds 0–2 (Table S19). At the primary cell the forward difference gave −0.016 ([−0.030, −0.003], $P = 0.021$), −0.049 and −0.044, and the two estimators differed only where the forward difference returns an exact zero, in 2.7–4.3% of the necessity cells at $T = +8$. Elsewhere they agreed to a relative 5×10^{-13} and chose the same top cell in at least 98.9% of the images. Against the first model (−0.136, [−0.154, −0.117]), the effect was smaller in all three and not distinguishable from zero in seed 0, where the hit contrast at the same cell was positive (+0.038, [0.012, 0.064]). Model 3 repeats the configuration of seed 0 and differs from it only through non-deterministic GPU kernels, so the gap between their primaries (−0.025 against −0.005 in closed form, −0.050 against −0.016 by forward difference) is run-to-run variation. Within each seed the conjunctive necessity map ($T = -8$, 1×1) had a higher hit rate than the conjunctive sufficiency map, by +0.298 ([0.263, 0.335]), +0.207 ([0.173, 0.241]) and +0.216 ([0.186, 0.246]).

6. H2. Of the 32 designated-in-advance cells, 16 favoured necessity, 10 sufficiency and 6 spanned zero: 12, 0 and 4 at $T = -8$, and 4, 10 and 2 at $T = +8$. On the 20 cells that read average precision, IoU or Dice, the map that the mechanism expects to spread its magnitude (sufficiency at $T = +8$, necessity at $T = -8$) was better in 19, whereas the 12 pointing-hit cells do not follow it consistently. At the primary cell H2 is −0.050 on NIH, +0.001 on RSNA (spanning zero) and −0.134 on ISIC (Dice).

7. H1 cells favouring necessity (counts in Table S7): two IoU cells of the distributed NIH head (+0.008 and +0.007, the first with Wilcoxon $P = 0.087$, so that its interval and test disagree), the RSNA 1×1 hit (+0.042, 95% CI [0.015, 0.069]) and, in the post hoc refitted NIH arm, AP 1×1 (+0.030), hit 3×3 (+0.063) and AP 3×3 (+0.045). Among the conjunctive-necessity losses are the RSNA 3×3 hit (−0.211) and the ISIC 1×1 hit (−0.219). In the disjoint RSNA models the conjunctive hit cells tied (−0.012 and −0.008) and the AP cells improved (+0.048 and +0.020). One of the three exceptions to a disjunctive sufficiency map below the collapsed map is ISIC Dice at 1×1 (+0.010, [0.003, 0.016]). In the second training run the primary contrasts of the NIH model of this study and of ISIC stayed negative (−0.057 and −0.120).

8. Conjunctive sufficiency (Table S13). At the 1×1 pointing hit it improved on the collapsed map on RSNA, by +0.045 ([0.022, 0.067]) with the distributed model and by +0.082 ([0.057, 0.107]) with the cross-fitted models, while on the 283 held-out studies the difference spanned zero (+0.042, [−0.011, 0.095]). It did not improve on NIH (−0.043 with the distributed head, +0.010 spanning zero with the refitted head and −0.060 with the NIH model of this study) or on ISIC (−0.291, Dice −0.147). On RSNA the same map was worse than the collapsed map at 3×3 resolution (−0.222 with the distributed model, −0.149 with the cross-fitted models and −0.191 on the held-out studies) and on average precision (−0.065 at 1×1 with the distributed model). The other cells whose interval lies above zero are the 3×3 hit (+0.050, [0.025, 0.075]) and the 3×3 AP (+0.023, [0.012, 0.035], with Wilcoxon $P = 0.24$, so interval and test disagree) of the refitted NIH head.

9. Comparators. In the chest-radiograph arms the collapsed map localised better than plain CAM and Grad-CAM (0.227 against 0.190 and 0.179 on NIH and 0.479 against 0.252 and 0.252 on RSNA, exact McNemar $P \leq 7 \times 10^{-4}$). Grad-CAM and plain CAM coincided image for image except where the Grad-CAM map was identically zero (193 of 984 NIH items). Scoring those maps as no prediction, or with the centroid rule of the average box, reduces the advantage of the collapsed map over Grad-CAM to +0.035 ($P = 0.004$, 791 items) and +0.030 ($P = 0.004$). A map of activation magnitude tied the best model-derived map on NIH (0.240 against 0.232, $P = 0.57$).

10. Average-annotation box. On NIH it scored 0.385 against 0.232 for the strongest model-derived map (exact McNemar $P = 3.4 \times 10^{-27}$ unadjusted, 1.0×10^{-26} with Holm adjustment across the six comparisons of Fig. S6(d)). Pooled to the maps' own 7×7 resolution, the leave-one-patient-out box scored 0.371 ($P = 1.9 \times 10^{-25}$), and the cross-fitted box had the highest hit rate of all maps on each of the eight findings (Table S4). On RSNA, built from the 4,627 non-evaluation positive patients, it scored 0.560 against 0.523 for the strongest model-derived map, sufficiency under conjunctive pooling ($P = 0.008$), and it tied at the maps' own 7×7 resolution (0.529, $P = 0.60$). No average box was built for ISIC, where the centre prior is the best map (hit rate 0.917, and Dice 0.530 against at most 0.488 for any other map). The centre prior scored 0.214 on NIH, driven by cardiomegaly, and 0.108 on RSNA.

11. Resolution. At 7×7 resolution only 372 pixel positions are reachable by the pointing protocol, and 99% of nodule boxes are smaller than one cell in both dimensions, so nodule hit rates, at most 0.04 for every model-derived map, are bounded by resolution. The overall ceiling is 0.816 on NIH and 0.969 on RSNA.

Table S19. Necessity map minus collapsed map on ISIC for three retrained classifiers (closed form, 95% bootstrap interval over 735 images, 10,000 resamples). Mean Dice of the necessity and collapsed maps at the primary cell ($T = +8$, 1×1): 0.280 and 0.285, 0.237 and 0.289, 0.239 and 0.280. AP, average precision.

T	Grid	Outcome	Seed 0	Seed 1	Seed 2
+8	1×1	Dice	−0.005 [−0.017, 0.007]	−0.052 [−0.068, −0.035]	−0.042 [−0.055, −0.029]
		AP	0.012 [0.001, 0.023]	−0.030 [−0.042, −0.018]	−0.011 [−0.023, 0.001]
		Hit	0.038 [0.012, 0.064]	0.005 [−0.019, 0.030]	0.022 [−0.008, 0.052]
	3×3	Dice	−0.048 [−0.058, −0.038]	−0.064 [−0.075, −0.053]	−0.074 [−0.085, −0.063]
		AP	−0.018 [−0.029, −0.006]	−0.051 [−0.064, −0.039]	−0.054 [−0.065, −0.042]
		Hit	−0.056 [−0.088, −0.024]	−0.069 [−0.099, −0.041]	−0.044 [−0.073, −0.014]
−8	1×1	Dice	0.000 [−0.012, 0.012]	0.026 [0.019, 0.034]	0.045 [0.036, 0.053]
		AP	−0.111 [−0.128, −0.095]	−0.106 [−0.128, −0.084]	−0.047 [−0.065, −0.029]
		Hit	−0.039 [−0.084, 0.007]	−0.061 [−0.109, −0.012]	−0.114 [−0.152, −0.076]
	3×3	Dice	−0.023 [−0.037, −0.010]	0.004 [−0.008, 0.016]	−0.024 [−0.035, −0.012]
		AP	−0.119 [−0.134, −0.104]	−0.147 [−0.166, −0.128]	−0.116 [−0.133, −0.099]
		Hit	−0.176 [−0.216, −0.136]	−0.205 [−0.248, −0.162]	−0.162 [−0.201, −0.122]

12. Spread on NIH (distributed head, participation ratio as in Note S2.8). The sufficiency map at $T = +8$ spread over 8.3 of the 49 grid cells against 7.4 for necessity at 1×1 on average (23.2 against 18.5 at 3×3), and necessity at $T = -8$ over 15.1 against 6.0 at 1×1.

S2.10 Dependence on the reference

1. Along the interpolation from the reference toward the input (Table S5), the correlation between the two quantities rose almost monotonically toward 1 on all three models swept, with one reversal of 0.015 on ConvNeXt-T. Across the different kinds of reference (zeros, the dataset mean, noise, a spatial shuffle and the bank) the ordering by distance did not hold, with 2 to 11 violations out of 55 pairs per model.
2. Six of the twelve settings put the distributed transformer above 0.92, and the spatial-shuffle reference gives 0.984 to 0.9998 on the three models. Degenerate orderings return 1 at every reference evaluated.

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